

THE TATE AND STANDARD CONJECTURES FOR ABELIAN VARIETIES OF DIMENSION AT MOST FIVE OVER FINITE FIELDS

NINGYI LI

ABSTRACT. We prove the Tate conjecture over finite fields and the numerical Hodge standard conjecture over arbitrary fields of positive characteristic for abelian varieties of dimension at most five. The positivity statement holds for every ample rational divisor class. Consequently, abelian varieties of dimension at most five satisfy all the ℓ -adic standard conjectures over arbitrary fields of positive characteristic, for every prime ℓ invertible in the base field. Over finite fields and their algebraic closures, these abelian varieties satisfy all the standard conjectures with respect to every Weil cohomology theory. The proof constructs a positively polarized abelian motive from Gross–Schoen heights and monodromy, and identifies the residual intersection pairing with a Rosati trace form. The same construction establishes the Tate and standard conjectures for certain families of mixed powers of unbounded dimension over finite fields, and yields a Beilinson–Bloch rank formula for a class of constant abelian threefolds over global function fields.

CONTENTS

1. Introduction	1
2. Motivic and specialization preliminaries	5
3. Ordinary abelian varieties of dimension at most five	8
4. The tensor block with four odd factors	9
5. A decomposable fivefold	11
6. Reduction to the residual block	12
7. The residual correspondence	21
8. The Tate conjecture	31
9. The Hodge standard conjecture over finite fields	32
10. Standard conjecture D and further consequences	40
11. Specialization of the standard conjectures	41
12. Powers and products with curves	42
Acknowledgements	47
References	47

1. INTRODUCTION

We prove the Tate conjecture over finite fields and the numerical Hodge standard conjecture in positive characteristic for abelian varieties of dimension at most five. We also obtain results for certain products of unbounded dimension and for constant threefolds over function fields.

Tate conjecture. Let $\mathbf{F}_q$ have characteristic p . The Tate conjecture asserts that Frobenius invariant classes in even cohomology, with the corresponding Tate twist, are spanned by algebraic cycles. Tate proves this for divisor classes [Tat66, Theorem 4]. Broe proves algebraicity in codimension two

2020 *Mathematics Subject Classification.* Primary 14C25; Secondary 14K15, 14F20, 11G10.

Key words and phrases. Tate conjecture, standard conjectures, Hodge standard conjecture, abelian fivefolds, algebraic cycles, Gross–Schoen cycles, Honda–Tate theory.

for abelian fourfolds [Bro26b, Theorem 1.1]; together with the divisor theorem and hard Lefschetz, this gives all codimensions. Our first theorem extends the dimension bound to five.

Theorem 1.1. *Let $A/\mathbf{F}_q$ be an abelian variety of dimension $d \leq 5$. For every $0 \leq r \leq d$ and $\ell \neq p$, the cycle class map*

$$\mathrm{CH}^r(A)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell} \longrightarrow H_{\mathrm{\acute{e}t}}^{2r}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(r))^{\mathrm{Frob}_q=1}$$

is surjective.

Together with Kahn's theorem [Kah03, Theorem 1.10], this gives equality of rational and numerical equivalence with rational coefficients, over finite fields and their algebraic closures. In particular, the cycle class maps in Theorem 1.1 are isomorphisms. Kahn's Corollaries 2.1–2.2 [Kah03] also identify the orders of the poles of the zeta function with the ranks of the numerical cycle groups and give $K_i(A) \otimes \mathbf{Q} = 0$ for $i > 0$.

Standard conjectures. For an abelian variety A of dimension d , let $N^r(A)_{\mathbf{Q}}$ denote codimension- r cycles modulo numerical equivalence. For an ample rational divisor class θ and $0 \leq 2r \leq d$, let

$$P_{\mathrm{num}}^r(A, \theta) = \ker(\theta^{d-2r+1} : N^r(A)_{\mathbf{Q}} \longrightarrow N^{d-r+1}(A)_{\mathbf{Q}}).$$

The numerical Hodge standard conjecture I asserts that

$$(x, y) \longmapsto (-1)^r \deg_A(xy\theta^{d-2r}) \tag{1.1}$$

is positive definite on this space after extension to $\mathbf{R}$. Its homological formulation uses the rational image of the cycle class map and the cohomologically primitive subspace. Standard conjecture D asserts that numerical and homological equivalence agree. Subscripts ℓ specify ℓ -adic cohomology.

Ancona proves numerical Hodge positivity in dimension at most four [Anc21, Theorem 3.18]. Combined with Broe's Tate theorem, this also gives D_{ℓ} in dimension at most four over arbitrary fields of positive characteristic [Bro26b, Corollary 1.2]. We prove the following extension.

Theorem 1.2. *Let A/k be an abelian variety of dimension at most five, where k has characteristic $p > 0$.*

- (i) *For every ample rational divisor class, A satisfies the numerical Hodge standard conjecture I and its ℓ -adic homological formulation I_{ℓ} for every prime $\ell \neq p$. Standard conjecture D_{ℓ} holds for every such prime.*
- (ii) *If k is finite or an algebraic closure of a finite field, A satisfies the homological Hodge standard conjecture for every ample rational divisor class and satisfies standard conjecture D, with respect to every Weil cohomology theory.*

Algebraic Lefschetz and Künneth decompositions give standard conjectures A, B, and C for abelian varieties [Anc21, Theorem 4.1(1),(4)]. Thus Theorem 1.2 completes the standard conjectures in the stated settings.

Correspondences from curves. The geometric construction underlying both theorems gives the following description of the third cohomology of abelian threefolds with specified Newton slopes.

Theorem 1.3. *Let $Y/\overline{\mathbf{F}}_p$ be an abelian threefold with Newton slopes*

$$0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1.$$

There are a finite disjoint union T of smooth projective curves and a correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$, both defined over a finite field, such that, for every $\ell \neq p$,

$$\Gamma_* : H^1(T, \mathbf{Q}_{\ell})(-1) \longrightarrow H^3(Y, \mathbf{Q}_{\ell})$$

is surjective.

Idea of the proof. For ordinary fivefolds, canonical lifting converts Frobenius invariant classes into Hodge classes, whose algebraicity is known in dimension five. Supersingular varieties have Tate classes generated by divisors, and Tankeev's theorem treats geometrically simple fivefolds [Tan84, Theorem 1.2]. For the remaining products, the arithmetic reduction isolates divisor products and classes supported on subproducts of dimension at most four. The additional Tate classes occur in summands of the form

$$T^2(B_{\text{ao}}) \otimes h^1(B_{\text{ord}}) \otimes h^1(E_{\text{ss}})(2). \quad (1.2)$$

Here B_{ao} and B_{ord} are geometrically simple abelian surfaces, respectively almost ordinary and ordinary, E_{ss} is a supersingular elliptic curve, and $T^2(B)$ is the orthogonal complement of the divisor motive in $h^2(B)$. The three factors are selected by a residual Tate relation. The characteristic polynomial of Frobenius on $T^2(B_{\text{ao}}) \otimes h^1(E_{\text{ss}})(1)$ is the square of the characteristic polynomial on $h^1(B_{\text{ord}})$, after a finite extension. We seek an algebraic correspondence realizing this identity.

To construct it, we first prove Theorem 1.3. A Moret–Bailly family through the generic threefold with these Newton slopes yields a family of curves of genus three and compact type. Its positive Gross–Schoen height gives a primitive correspondence Γ with $\deg(\Gamma^2) > 0$. The curve and correspondence descend to a finite extension of the generic field, over which the cylinder map is Galois equivariant. The corresponding subgroup has finite index and preserves the connected algebraic monodromy group Sp_6 . The image of the cylinder map is therefore stable under this group. Since the primitive third cohomology is an irreducible representation of dimension fourteen, the cylinder map is surjective onto this subspace. Spreading the correspondence and using density of isogeny classes carries this surjection to every threefold with the prescribed slopes. An ample curve supplies the Lefschetz summand.

Apply this construction to $B_{\text{ao}} \times E_{\text{ss}}$ and project onto $T^2(B_{\text{ao}}) \otimes h^1(E_{\text{ss}})(1)$. Poincaré complete reducibility produces an actual factor of the Jacobian; the restricted ample polarization gives a nondegenerate pairing on this factor. Nilpotence for finite-dimensional motives [AK02, Proposition 9.1.14] lifts the resulting realization isomorphism to an isomorphism of Chow motives. Thus the projected motive is the degree-one motive of an abelian fourfold. Comparing Frobenius polynomials and applying Tate's isogeny criterion [Tat66, Theorem 1(c)] gives, over a finite extension, an isomorphism of Chow motives

$$T^2(B_{\text{ao}}) \otimes h^1(E_{\text{ss}})(1) \simeq h^1(B_{\text{ord}})^{\oplus 2}. \quad (1.3)$$

Divisor correspondences now give algebraic representatives for the Tate classes in (1.2), together with numerical duals. Descent of coefficients, averaging over a finite extension, and hard Lefschetz complete the proof of Theorem 1.1.

For positivity, monodromy makes the generic cylinder composed with its adjoint a scalar endomorphism. The trace identity computes its scalar:

$$\lambda = \frac{\deg(\Gamma^2)}{14} > 0.$$

The primitive motive is thereby identified with the degree-one motive of an abelian sevenfold, with its polarization form multiplied by λ . Spreading this polarized correspondence and transporting it along isogenies preserves the positive rational scaling of the form. A residual cycle then corresponds to a rational homomorphism u of polarized abelian varieties, and its primitive self-intersection is a positive rational multiple of $\text{Tr}_\ell(u^\dagger u)$. Here $\dagger$ is the Rosati adjoint and Tr_ℓ is the ordinary cohomological trace. Rosati positivity therefore proves positivity on the residual classes.

To treat a prescribed ample rational class, we decompose the polarized variety up to isogeny into orthogonal simple factors and choose maximal CM fields stable under the Rosati involution. The resulting rational CM orbit decomposition is orthogonal for the tensor pairing in degree four, whose restriction to primitive classes is a positive rational multiple of the intersection form. Divisor

products, classes supported in dimension at most four, and residual classes exhaust its numerical summands, so their positive forms give numerical Hodge positivity over finite fields.

Finally, over an arbitrary field of positive characteristic, we spread the variety, polarization, and cycles over a scheme of finite type over $\mathbf{F}_p$. Smooth proper base change preserves a nonzero primitive ℓ -adic algebraic class on a finite-field fibre. Standard conjecture D and numerical positivity on that fibre give strict positivity on the original class. Algebraic Lefschetz correspondences then give D_ℓ and identify the homological and numerical primitive forms. This proves Theorem 1.2(i).

Mixed powers. For the triples $B_{\text{ao}}, B_{\text{ord}}, E_{\text{ss}}$ selected by the arithmetic reduction, the Chow isomorphism (1.3) also controls the products

$$B_{\text{ao}}^a \times B_{\text{ord}}^b \times E_{\text{ss}}^c, \quad a, b, c \geq 0.$$

The Frobenius conditions defining these triples are stated in Proposition 6.10; they give the identity of characteristic polynomials used in (1.3) and persist under finite extension. The dimensions $2a+2b+c$ are unbounded. We prove the Tate conjecture for these products in every codimension, for every $\ell \neq p$, over every finite extension of the field of definition of the residual data (Corollary 12.1). We also prove all the standard conjectures over finite fields and their algebraic closures, for every Weil cohomology theory and every ample rational divisor class (Corollary 12.5).

The Chow isomorphism (1.3) makes their motives, after a finite extension, summands of finite direct sums of tensor powers of $h(B_{\text{ao}} \times E_{\text{ss}})$ and their Tate twists. This reduces algebraicity to divisor classes on powers of $B_{\text{ao}} \times E_{\text{ss}}$. The polarized construction also identifies the induced motivic forms with the geometric forms.

Products with curves. Let Y/k be an abelian threefold over a finite field with the Newton slopes of Theorem 1.3. We construct an abelian variety J_3/k of dimension ten and an isomorphism of rational Chow motives over k ,

$$h^3(Y)(1) \simeq h^1(J_3)$$

(Proposition 12.6). For every smooth projective geometrically connected curve C/k , the cycle class maps for $C \times Y$ are then isomorphisms in every codimension, over every finite extension of k and for every $\ell \neq \text{char}(k)$ (Corollary 12.7).

Let $K = k(C)$, and let $\text{CH}^2(Y_K)_{\mathbf{Q}, \text{hom}}$ denote the kernel of the cycle class map to $H^4(Y_{\bar{K}}, \mathbf{Q}_\ell(2))$. Broe's criterion [Bro26a, Proposition 4.20] applies to the cycle class isomorphisms for $C \times Y$. The identification $h^3(Y)(1) \simeq h^1(J_3)$ then expresses the resulting Beilinson–Bloch rank formula as

$$\dim_{\mathbf{Q}} \text{CH}^2(Y_K)_{\mathbf{Q}, \text{hom}} = \text{rank } J_3(K) = \text{ord}_{s=2} L(H^3(Y_{\bar{K}}, \mathbf{Q}_\ell), s),$$

where L is the Hasse–Weil L -function over K (Corollary 12.8).

Related results. Koshikawa proves the numerical Hodge standard conjecture for the square of a simple abelian variety of prime dimension and for its product with an elliptic curve [Kos24, Theorem 1.1]. Agugliaro proves it for all powers of geometrically simple abelian varieties of prime dimension over $\overline{\mathbf{F}}_p$, and for all powers of abelian threefolds over arbitrary fields [Agu25, Theorem 1.2 and Corollary 1.3]. The products treated here also include two geometrically simple surfaces in distinct isogeny classes. Milne proves related results for powers of surfaces and for mixed powers of a neat simple threefold and an elliptic curve [Mil22, Section 4(b) and Theorem 4.2].

Over the complex numbers, Markman's algebraicity theorem for Weil classes [Mar25, Corollary 1.6.1], together with Moonen–Zarhin's classification [MZ99, Theorem 0.2], gives the Hodge conjecture for abelian varieties of dimension at most five; see also [Mar26, Corollary 1.3]. Broe's fourfold Tate theorem combines Markman's result with Ancona's method for the Hodge standard conjecture.

Dupuy, Kedlaya, and Zureick-Brown relate Frobenius angle rank to Newton slopes and the Galois action [DKZB24, Theorems 1.5–1.6]. Our arithmetic reduction uses Zarhin's criteria for multiplicative independence [Zar15, Section 2 and Corollary 5.2].

Overview of the contents. Section 2 fixes the motivic conventions and proves the specialization lemmas. Sections 3–6 give the arithmetic reduction. Sections 7–9 construct the curve correspondences, prove the Tate conjecture, and establish positivity over finite fields. Sections 10–11 deduce the consequences for cycles and extend the standard conjectures to arbitrary fields of positive characteristic. Section 12 treats mixed powers and products with curves.

2. MOTIVIC AND SPECIALIZATION PRELIMINARIES

We use contravariant rational Chow motives. Write $\mathrm{CHM}(k)_{\mathbf{Q}}$ for their pseudo-abelian category, $\mathbf{1}$ for the unit motive, and $\mathbf{L} = \mathbf{1}(-1)$ for the Lefschetz motive. Our realization convention is

$$R_{\ell}h^i(X) = H_{\mathrm{et}}^i(X, \mathbf{Q}_{\ell}), \quad R_{\ell}(M(n)) = R_{\ell}(M)(n).$$

For an abelian variety C , a polarization gives an algebraic isomorphism $h^1(C)^{\vee} \simeq h^1(C)(1)$. The required Lefschetz isomorphisms and their inverses are algebraic [Kün93, Theorem 5.2].

For an endomorphism u of an ℓ -adic realization, $\mathrm{Tr}_{\ell}(u)$ denotes the ordinary linear trace. For an endomorphism of a motive, write $\mathrm{Tr}_{\ell}(u) = \mathrm{Tr}_{\ell}(R_{\ell}u)$ and $\mathrm{tr}_{\mathrm{cat}}(u)$ for its categorical trace. If the realization is concentrated in degree d , then, in $\mathbf{Q}_{\ell}$,

$$\mathrm{tr}_{\mathrm{cat}}(u) = (-1)^d \mathrm{Tr}_{\ell}(u). \quad (2.1)$$

Field traces retain their field subscripts, as in $\mathrm{Tr}_{F/\mathbf{Q}}$.

For a smooth projective d -fold X/k , let

$$N^r(X)_{\mathbf{Q}} = \mathrm{CH}^r(X)_{\mathbf{Q}} / \equiv_{\mathrm{num}}.$$

Since homological equivalence implies numerical equivalence, $D_{\ell}(X)$ is equivalent to the assertion that every numerically trivial rational cycle has zero ℓ -adic class.

Lemma 2.1. *Let $i: M \rightarrow N$ and $p: N \rightarrow M$ be morphisms of rational Chow motives with $pi = 1_M$.*

(1) *For every rational Chow motive T , the split injections*

$$\begin{aligned} \mathrm{Hom}(T, M) &\longrightarrow \mathrm{Hom}(T, N), & u &\longmapsto iu, \\ \mathrm{Hom}(M, T) &\longrightarrow \mathrm{Hom}(N, T), & u &\longmapsto up \end{aligned}$$

preserve and reflect numerical triviality and vanishing under each realization. Equality of numerical and homological equivalence and independence of realization kernels therefore pass from either Hom group involving N to the corresponding Hom group involving M .

(2) *An isomorphism of rational Chow motives preserves numerical triviality and all ℓ -adic realizations. In particular, standard conjecture D_{ℓ} and independence of cycle class kernels are invariant under isogeny of abelian varieties.*

Proof. Numerically trivial morphisms form an ideal. The identity $pi = 1_M$ shows that each split injection preserves and reflects numerical triviality. For any realization R , the identity $R(p)R(i) = 1_{R(M)}$ gives the same assertion for vanishing under R . This proves (1) and the assertion about isomorphisms in (2). For an isogeny $f: A \rightarrow A'$, choose an isogeny in the opposite direction whose two composites with f are multiplication by an integer $n > 0$. Multiplication by n acts as n^j on h^j [Sch94, Theorem 5.2(i)], so f^* is an isomorphism of rational Chow motives. The identity $f_*f^* = (\deg f)1_{h(A')}$ identifies its inverse with $(\deg f)^{-1}f_*$, giving inverse rational Chow correspondences. $\square$

Lemma 2.2. *Let A be an abelian variety of dimension g over an algebraically closed field, and write $z = \sum_s z_s$ for the Beauville decomposition of $z \in \mathrm{CH}^r(A)_{\mathbf{Q}}$.*

(1) *If z is numerically trivial, every z_s is numerically trivial.*

(2) *For $s \neq 0$, the class of z_s vanishes in every ℓ -adic realization with ℓ invertible in the base field.*

(3) *The algebraic Fourier transform preserves numerical triviality.*

Proof. Choose an integer $n \geq 2$ invertible in the base field. On $\mathrm{CH}^r(A)_{(s)}$, the operator $[n]^*$ acts by n^{2r-s} [Sch94, §5.11]. If S_r is the finite set of possible grades, the projector onto grade s is the polynomial

$$p_s = \prod_{\substack{t \in S_r \\ t \neq s}} \frac{[n]^* - n^{2r-t}}{n^{2r-s} - n^{2r-t}}.$$

The projection formula shows that $[n]^*$ preserves numerical triviality: for a complementary cycle y , $\deg([n]^*z \cdot y) = \deg(z \cdot [n]_*y) = 0$. Thus $z_s = p_s z$ is numerically trivial. On H^{2r} , multiplication by n acts by n^{2r} , so compatibility with the cycle class map gives $(n^{2r-s} - n^{2r}) \mathrm{cl}_\ell(z_s) = 0$, hence $\mathrm{cl}_\ell(z_s) = 0$ for $s \neq 0$. Finally, the Fourier transform is an algebraic correspondence defined using the Poincaré bundle, and algebraic correspondences preserve numerical triviality by the projection formula. $\square$

Theorem 2.3. *Let A/k be an abelian variety of dimension at most four over a finite field of characteristic p . For every $0 \leq r \leq \dim A$ and $\ell \neq p$, the cycle class map*

$$\mathrm{CH}^r(A)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \longrightarrow H_{\mathrm{et}}^{2r}(A_{\bar{k}}, \mathbf{Q}_\ell(r))^{\mathrm{Frob}_k=1}$$

is surjective, where Frob_k denotes geometric Frobenius.

Proof. We first prove surjectivity when $\dim A = 4$. Broe [Bro26b, Theorem 1.1] proves it for $r = 2$. Tate's divisor theorem [Tat66, Theorem 4] gives surjectivity for $r = 1$. If L is the class of an ample divisor over k , hard Lefschetz gives a Frobenius-equivariant isomorphism

$$L^2: H_{\mathrm{et}}^2(A_{\bar{k}}, \mathbf{Q}_\ell(1)) \xrightarrow{\sim} H_{\mathrm{et}}^6(A_{\bar{k}}, \mathbf{Q}_\ell(3)).$$

Divisor classes span the source invariants, so intersection with L^2 gives surjectivity for $r = 3$. The fundamental class and the origin give surjectivity for $r = 0$ and $r = 4$, respectively.

For $d = \dim A < 4$, choose an elliptic curve E/k and let $X = A \times E^{4-d}$. The projection $\pi: X \rightarrow A$ and the inclusion $i: A \rightarrow X$ at the origin satisfy $i^* \pi^* = 1$ on Chow groups and cohomology. For a Frobenius-invariant class a on A , surjectivity in dimension four provides a cycle with $\mathbf{Q}_\ell$ -coefficients on X whose class is $\pi^* a$. Pullback by i gives a cycle with class a . $\square$

Lemma 2.4. *Let $A/\bar{\mathbf{F}}_p$ be an abelian variety. Any finite collection of cycles, homomorphisms, isogenies, polarizations, and Chow projectors on powers and products of A , together with finitely many rational equivalences and polynomial identities among these data, descends to a common finite subfield after a finite extension. Passing to that extension preserves geometric numerical equivalence, the prescribed Chow identities, and every geometric ℓ -adic realization.*

Proof. Each geometric datum is of finite presentation over $\bar{\mathbf{F}}_p$. A rational equivalence is witnessed by finitely many principal divisors on integral subvarieties. The subvarieties, rational functions, coefficients, and operations in the prescribed identities descend to a common finite subfield. The identities therefore hold in the Chow groups over that field. Base change to $\bar{\mathbf{F}}_p$ recovers the original data. $\square$

Lemma 2.5. *Let k be an algebraically closed field of characteristic p , let X/k be smooth and projective, and let $z \in \mathrm{CH}^r(X)_{\mathbf{Q}}$. Suppose that $\mathrm{cl}_{\ell_0}(z) = 0$ and $\mathrm{cl}_{\ell_1}(z) \neq 0$ for two primes $\ell_0, \ell_1 \neq p$. There exist an integral connected scheme S of finite type over $\mathbf{F}_p$, a smooth projective model $\mathcal{X} \rightarrow S$, a relative rational cycle $\mathcal{Z}$, and a closed point $s \in S$ such that*

$$\mathrm{cl}_{\ell_0}(\mathcal{Z}_{\bar{s}}) = 0, \quad \mathrm{cl}_{\ell_1}(\mathcal{Z}_{\bar{s}}) \neq 0.$$

The cycle $\mathcal{Z}_{\bar{s}}$ is numerically trivial.

Proof. Choose a rational cycle representing z , and descend X and the finitely many components of this representative to a finitely generated field. Spread these data over an integral $\mathbf{F}_p$ -scheme of finite type. After passing to a nonempty smooth open subscheme, $\mathcal{X} \rightarrow S$ is smooth and projective

and the closure of every chosen component is flat over S . Their rational linear combination defines $\mathcal{Z}$. Invariance of ℓ -adic cohomology under extensions of algebraically closed fields shows that the two classes at a geometric generic point are respectively zero and nonzero. For each $i \in \{0, 1\}$, the relative cycle class gives a global section of

$$R^{2r} f_* \mathbf{Q}_{\ell_i}(r), \quad f: \mathcal{X} \longrightarrow S,$$

which is lisse by proper smooth base change. A global section of a lisse sheaf on a connected scheme is determined by its value at any geometric point. Thus the ℓ_0 -adic section vanishes on every geometric fibre and the ℓ_1 -adic section is nonzero on every geometric fibre. Choose a closed point s . The vanishing ℓ_0 -adic class implies that $\mathcal{Z}_{\bar{s}}$ has zero intersection number with every complementary cycle. Hence $\mathcal{Z}_{\bar{s}}$ is numerically trivial. $\square$

Proposition 2.6. (1) *Every abelian variety of dimension at most four over $\overline{\mathbf{F}}_p$ satisfies standard conjecture D_ℓ in every codimension and for every prime $\ell \neq p$.*

(2) *For every abelian variety of dimension at most four over an algebraically closed field of characteristic p , the kernels of the rational ℓ -adic cycle class maps are independent of $\ell \neq p$.*

Proof. For (1), let $A/\overline{\mathbf{F}}_p$ have dimension d , and let $z \in \mathrm{CH}^r(A)_{\mathbf{Q}}$ be numerically trivial. By Lemma 2.4, A and z descend to an abelian variety $A_0/\mathbf{F}_q$ and a cycle $z_0 \in \mathrm{CH}^r(A_0)_{\mathbf{Q}}$. Write Φ for geometric Frobenius. The class $\mathrm{cl}_\ell(z_0)$ is fixed by Φ . Frobenius acts semisimply on H^1 [Zar15, Section 2], hence on its exterior powers, which are the cohomology groups of A_0 . Poincaré duality therefore restricts to a perfect pairing

$$H_{\mathrm{et}}^{2r}(A_{0,\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))^{\Phi=1} \times H_{\mathrm{et}}^{2d-2r}(A_{0,\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(d-r))^{\Phi=1} \longrightarrow \mathbf{Q}_\ell.$$

By Theorem 2.3, algebraic cycle classes span the second invariant space. Numerical triviality of z_0 therefore makes $\mathrm{cl}_\ell(z_0)$ orthogonal to that space. The perfect pairing gives $\mathrm{cl}_\ell(z_0) = 0$, hence $\mathrm{cl}_\ell(z) = 0$. Compatibility of cycle classes with products and the trace map gives the converse implication.

For (2), suppose that the kernels at two primes ℓ_0 and ℓ_1 differ. After interchanging the primes, choose a cycle whose class vanishes at ℓ_0 and is nonzero at ℓ_1 . Lemma 2.5 preserves these two properties on a closed geometric fibre; after shrinking the base, its model is an abelian scheme. The specialized cycle is numerically trivial, so (1) makes its ℓ_1 -adic class zero, a contradiction. $\square$

Lemma 2.7. *Let A/k be a supersingular abelian variety of dimension g over an algebraically closed field of characteristic $p > 0$.*

(1) *For every r and $\ell \neq p$, algebraic cycles span*

$$H_{\mathrm{et}}^{2r}(A, \mathbf{Q}_\ell(r)).$$

(2) *Standard conjecture D_ℓ holds for A for every $\ell \neq p$.*

Proof. Oort's theorem [Oor74, Theorem 4.2] gives an isogeny $E^g \rightarrow A$, where E is a supersingular elliptic curve. Lemma 2.1 reduces algebraicity of cohomology in even degrees and standard conjecture D_ℓ to E^g .

Choose a product polarization. The Rosati correspondence identifies

$$\mathrm{NS}(E^g)_{\mathbf{Q}} \simeq \mathrm{End}^0(E^g)^\dagger.$$

The algebra $\mathrm{End}^0(E)$ is the quaternion algebra over $\mathbf{Q}$ ramified at p and infinity. A Hermitian $g \times g$ matrix over this algebra has one rational parameter on each diagonal entry and four parameters for each unordered off-diagonal pair. Hence

$$\dim_{\mathbf{Q}} \mathrm{NS}(E^g)_{\mathbf{Q}} = g + 4 \binom{g}{2} = g(2g-1) = \dim_{\mathbf{Q}_\ell} H^2(E^g, \mathbf{Q}_\ell)(1).$$

The Kummer sequence injects $\mathrm{NS}(E^g)_{\mathbf{Q}} \otimes \mathbf{Q}_\ell$ into cohomology in degree two, so divisor classes span H^2 . Since the cohomology algebra of an abelian variety is the exterior algebra on H^1 , products of r classes in degree two span H^{2r} . This proves (1). If $z \in \mathrm{CH}^r(E^g)_{\mathbf{Q}}$ is numerically trivial, its class pairs to zero with the products of $g - r$ divisors that span the complementary cohomology group. Poincaré duality gives $\mathrm{cl}_\ell(z) = 0$. $\square$

Lemma 2.8. *Let $A/\mathbf{F}_q$ be a geometrically simple abelian fivefold.*

- (1) *The Tate conjecture holds for A in every codimension and for every $\ell \neq p$.*
- (2) *The base change $A_{\overline{\mathbf{F}}_p}$ satisfies standard conjecture D_ℓ for every $\ell \neq p$.*

Proof. Tankeev's theorem [Tan84, Theorem 1.2] proves the Tate conjecture for geometrically simple abelian varieties of prime dimension over finite fields. Apply it to A over every finite extension of $\mathbf{F}_q$. Tate surjectivity in complementary codimensions, Frobenius semisimplicity, and Poincaré duality then give standard conjecture D_ℓ on the geometric fibre. $\square$

3. ORDINARY ABELIAN VARIETIES OF DIMENSION AT MOST FIVE

Theorem 3.1. *Let $A/\mathbf{F}_q$ be an ordinary abelian variety with $\dim A \leq 5$.*

- (1) *For every $n \geq 1$ and every $\ell \neq p$, the Tate conjecture holds for $A_{\mathbf{F}_{q^n}}$ in every codimension.*
- (2) *The base change $A_{\overline{\mathbf{F}}_p}$ satisfies standard conjecture D_ℓ for every $\ell \neq p$.*

Proof. We realize the Frobenius-invariant classes as Hodge classes on the canonical lift, then specialize algebraic representatives to A . Let $\mathcal{A}/W(\mathbf{F}_q)$ be the canonical lift of A , and let $K = \mathrm{Frac} W(\mathbf{F}_q)$. Frobenius lifts to an endomorphism F of $\mathcal{A}$. Choose an isomorphism $\iota: \overline{K} \simeq \mathbf{C}$, and write $\tilde{A} = \mathcal{A}_{\overline{K}} \otimes_\iota \mathbf{C}$. Deligne's description of the canonical lift [Del69, Section 3, Theorem 7, and the proposition in Section 8] identifies the two Hodge halves of $H^1(\tilde{A}, \mathbf{C})$ with the Frobenius eigenspaces of positive slope and with unit roots, using the p -adic valuation transported by ι .

Fix $0 \leq r \leq d = \dim A$ and $n \geq 1$, and set

$$T_n^r = \ker\left((F^n)^* - q^{rn}: H^{2r}(\tilde{A}, \mathbf{Q}) \longrightarrow H^{2r}(\tilde{A}, \mathbf{Q})\right). \quad (3.1)$$

Frobenius is semisimple. After extending scalars to $\mathbf{C}$, the kernel T_n^r has a basis of wedges of $2r$ eigenvectors in degree one. Ordinarity gives slopes 0 and 1, and the eigenvalue equation fixes the total slope of each wedge at r . Exactly r factors lie in each Hodge half, so every basis vector has Hodge type (r, r) . Thus T_n^r is a rational space of Hodge classes.

The Hodge conjecture in dimension at most five follows from [Mar25, Corollary 1.6.1] and [MZ99, Theorem 0.2]; see also [Mar26, Corollary 1.3]. Choose rational algebraic cycles $Z_1, \dots, Z_t$ on $\tilde{A}$ whose Betti classes form a basis of T_n^r .

Transport the finitely many cycles Z_j through ι^{-1} to $\mathcal{A}_{\overline{K}}$. Their finitely many defining coefficients are algebraic over K , so the cycles descend to a common finite extension K'/K . Their flat closures over its valuation ring specialize to rational cycles $z_1, \dots, z_t$ on the geometric special fibre.

For every $\ell \neq p$ and $0 \leq j \leq d$, let Φ_j denote geometric Galois Frobenius on $H_{\mathrm{et}}^{2j}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(j))$. Proper smooth base change, with the Tate twist, identifies Φ_j with $q^{-j}F^*$. Hence

$$T_n^r \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \simeq H_{\mathrm{et}}^{2r}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))^{\Phi_r^n=1}, \quad (3.2)$$

and the classes of $z_1, \dots, z_t$ form a basis of the invariant space.

To obtain cycles over $\mathbf{F}_{q^n}$, choose N divisible by n such that every z_j is defined over $\mathbf{F}_{q^N}$. Let $m = N/n$, and let $\pi: A_{\mathbf{F}_{q^N}} \rightarrow A_{\mathbf{F}_{q^n}}$ be the projection. Set

$$z_j^{(n)} = \frac{1}{m} \pi_* z_j \in \mathrm{CH}^r(A_{\mathbf{F}_{q^n}})_{\mathbf{Q}}. \quad (3.3)$$

Compatibility with proper pushforward gives

$$\mathrm{cl}_\ell(z_j^{(n)}) = \frac{1}{m} \sum_{a=0}^{m-1} \Phi_r^{na} \mathrm{cl}_\ell(z_j) = \mathrm{cl}_\ell(z_j).$$

Thus (3.2) proves (1).

For (2), let $z \in \mathrm{CH}^r(A_{\overline{\mathbf{F}}_q})_{\mathbf{Q}}$ be numerically trivial and set $a = \mathrm{cl}_\ell(z)$. Choose $e \geq 1$ such that z descends to $A_{\mathbf{F}_{q^e}}$. Semisimplicity and Poincaré duality give the nondegenerate pairing

$$\begin{aligned} & H_{\mathrm{et}}^{2r}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(r))^{\Phi_r^e=1} \\ & \times H_{\mathrm{et}}^{2d-2r}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(d-r))^{\Phi_{d-r}^e=1} \longrightarrow \mathbf{Q}_\ell. \end{aligned} \quad (3.4)$$

For every fixed class b in the complementary degree, (1) gives cycles $y_j \in \mathrm{CH}^{d-r}(A_{\mathbf{F}_{q^e}})_{\mathbf{Q}}$ and coefficients $c_j \in \mathbf{Q}_\ell$ such that $b = \sum_j c_j \mathrm{cl}_\ell(y_j)$. Numerical triviality gives $\langle a, b \rangle = \sum_j c_j \deg(z \cdot y_j) = 0$. Nondegeneracy of (3.4) gives $a = 0$. Compatibility with products and the trace map gives the converse implication, proving (2). $\square$

Corollary 3.2. *Let A/k be an ordinary abelian variety of dimension at most five over an algebraically closed field of characteristic $p > 0$. In every codimension, the kernel of the rational ℓ -adic cycle class map is independent of $\ell \neq p$.*

Proof. Suppose that the kernels at two primes ℓ_0 and ℓ_1 differ. Apply Lemma 2.5 and shrink the base to the open ordinary locus. The resulting closed geometric fibre is ordinary and is defined over $\overline{\mathbf{F}}_p$. The specialized cycle has zero ℓ_0 -adic class, hence is numerically trivial. Standard conjecture D_{ℓ_1} for the ordinary fibre, proved in Theorem 3.1(2), forces its ℓ_1 -adic class to vanish, a contradiction. $\square$

4. THE TENSOR BLOCK WITH FOUR ODD FACTORS

Lemma 4.1. *Let B/k be a geometrically simple abelian surface over a finite field and let $E_1, E_2, E_3/k$ be elliptic curves. Let*

$$P = h^1(B) \otimes h^1(E_1) \otimes h^1(E_2) \otimes h^1(E_3)(2).$$

- (i) *For every $\ell \neq p$, the realization of P has zero geometric Tate subspace.*
- (ii) $\mathrm{Hom}_{\mathrm{NUM}(\overline{k})_{\mathbf{Q}}}(\mathbf{1}, P_k) = 0$.

Proof. We compare the Frobenius field of B with the compositum of the elliptic Frobenius fields. Pass to a finite field $\mathbf{F}_q$ over which all geometric endomorphisms are defined and every supersingular elliptic Frobenius is rational. Oort's theorem [Oor74, Theorem 4.2] implies that B is nonsupersingular. The Honda–Tate dimension and local invariant formulas [Wat69, Chapter 2, p. 527] give

$$F = \mathbf{Q}(\alpha) = \mathrm{End}^0(B), \quad [F : \mathbf{Q}] = 4,$$

for each Frobenius root α of B : a quadratic centre with Schur index two would force every slope to be $1/2$. Zarhin's Corollary 5.2 [Zar15] implies that the CM field F contains no imaginary quadratic subfield.

A nonzero Frobenius invariant in $R_\ell P$ would, by semisimplicity, give elliptic roots ϵ_i satisfying

$$\alpha \epsilon_1 \epsilon_2 \epsilon_3 = q^2, \quad \alpha = \frac{q^2}{\epsilon_1 \epsilon_2 \epsilon_3}.$$

Each ordinary elliptic root lies in an imaginary quadratic field, and each supersingular elliptic root is rational. Their compositum K is multiquadratic, and the displayed equality gives $F \subset K$. Every quartic subfield of K is biquadratic. A biquadratic CM field contains an imaginary quadratic subfield, contradicting Zarhin's Corollary 5.2. The same contradiction over every further finite extension proves (i).

A nonzero numerical class has a cycle representative whose pairing with some complementary cycle is nonzero. Compatibility with products and degree makes its realization nonzero. The representative descends to a finite field, so its realization is a geometric Tate class. Assertion (i) therefore gives (ii). $\square$

For an abelian surface B/k and elliptic curves $E_1, E_2, E_3/k$, let

$$M = h^1(B) \otimes h^1(E_1), \quad N = h^1(E_2) \otimes h^1(E_3), \quad P = M \otimes N.$$

Polarizations of the factors give algebraic dualities

$$M^\vee \simeq M(2), \quad \text{Hom}(M, N) \simeq \text{Hom}(\mathbf{1}, P(2)). \quad (4.1)$$

These isomorphisms commute with realizations and preserve numerical triviality, since numerically trivial morphisms form a tensor ideal. Writing $X = B \times E_1$ and $Y = E_2 \times E_3$, the two Hom spaces are represented by projected cycles in $\text{CH}^3(X \times Y)_{\mathbf{Q}}$ and $\text{CH}^2(X \times Y)_{\mathbf{Q}}$, respectively. The codimension shift on B is supplied by the algebraic Lefschetz isomorphism $h^1(B) \simeq h^3(B)(1)$ [Kün93, Theorem 5.2].

Theorem 4.2. *Let B, E_1, E_2, E_3 be defined over a finite field k , and let Φ be geometric Frobenius.*

(1) *For every $\ell \neq p$, the realization map*

$$\text{Hom}_{\text{CHM}(k)_{\mathbf{Q}}}(M, N) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \longrightarrow \text{Hom}_{\mathbf{Q}_\ell[\Phi]}(R_\ell M, R_\ell N)$$

is surjective.

(2) *Numerical and ℓ -adic homological equivalence agree on*

$$\text{Hom}_{\text{CHM}(\bar{k})_{\mathbf{Q}}}(M_{\bar{k}}, N_{\bar{k}})$$

for every $\ell \neq p$.

Proof. If B is geometrically simple, Lemma 4.1(i) applied to $P(2)$, together with (4.1), makes the target in (1) zero. Every geometric Chow morphism descends to a finite field and has zero realization, hence is numerically trivial. This proves (1) and (2) for geometrically simple B .

Suppose that B is geometrically decomposable. Over a finite extension k'/k , choose an isogeny $B \sim F_1 \times F_2$ with elliptic curves F_1, F_2 . It gives

$$P \simeq \bigoplus_{j=1}^2 h^1(F_j) \otimes h^1(E_1) \otimes h^1(E_2) \otimes h^1(E_3).$$

Each summand is a Chow retract of the fourth cohomological motive of the abelian fourfold $F_j \times E_1 \times E_2 \times E_3$. By Theorem 2.3 and (4.1), every map commuting with Frobenius over k' is realized by a correspondence. To descend a map φ equivariant over k , choose a correspondence γ' with $\mathbf{Q}_\ell$ -coefficients over k' and realization φ . Let $d = [k' : k]$, and let π_M, π_N be the projectors over k defining M, N . Set

$$\gamma = \pi_N \circ \frac{1}{d} \text{Cor}_{k'/k}(\gamma') \circ \pi_M. \quad (4.2)$$

Here $\text{Cor}_{k'/k}$ is proper pushforward along $(X \times Y)_{k'} \rightarrow X \times Y$. Its realization is the sum of the Galois conjugates, so

$$R_\ell(\gamma) = \frac{1}{d} \sum_{a=0}^{d-1} \Phi_N^a \varphi \Phi_M^{-a} = \varphi. \quad (4.3)$$

This proves (1). Proposition 2.6(1) gives D_ℓ on the two geometric fourfolds. Lemma 2.1 passes this equality to their projected cycle spaces, and (4.1) gives (2). $\square$

Theorem 4.3. *Let k be an algebraically closed field of characteristic $p > 0$, let B/k be an abelian surface, and let $E_1, E_2, E_3/k$ be elliptic curves. Set $M = h^1(B) \otimes h^1(E_1)$ and $N = h^1(E_2) \otimes h^1(E_3)$. The kernel of*

$$R_\ell: \operatorname{Hom}_{\operatorname{CHM}(k)_{\mathbf{Q}}}(M, N) \longrightarrow \operatorname{Hom}_{\mathbf{Q}_\ell}(R_\ell M, R_\ell N)$$

is independent of $\ell \neq p$.

Proof. Suppose that the kernels at two primes ℓ_0 and ℓ_1 differ, and spread the corresponding data, including the abelian schemes, projectors in degree one, and relative correspondence, over an integral connected $\mathbf{F}_p$ -scheme of finite type. After shrinking, the two realizations are morphisms of lisse sheaves of constant rank. On a closed geometric fibre, vanishing at ℓ_0 implies numerical triviality. The equality of numerical and homological equivalence in Theorem 4.2(2) forces vanishing at ℓ_1 , contradicting constancy of the positive rank. $\square$

5. A DECOMPOSABLE FIVEFOLD

Theorem 5.1. *Let B be an abelian surface and E_1, E_2, E_3 elliptic curves over a field k of characteristic $p > 0$. Let*

$$A = B \times E_1 \times E_2 \times E_3.$$

- (1) *If k is finite, the Tate conjecture holds for A in every codimension and for every $\ell \neq p$.*
- (2) *If k is finite, $A_{\bar{k}}$ satisfies standard conjecture D_ℓ for every $\ell \neq p$.*
- (3) *If k is algebraically closed, the kernels of the rational cycle class maps on A are independent of $\ell \neq p$ in every codimension.*

Proof. We first treat codimension two by decomposing $h^4(A)$. Deninger–Murre projectors [Sch94, Theorem 5.2(i)] give

$$h^4(A) = \bigoplus_{i_B + i_1 + i_2 + i_3 = 4} h^{i_B}(B) \otimes h^{i_1}(E_1) \otimes h^{i_2}(E_2) \otimes h^{i_3}(E_3). \quad (5.1)$$

If some $i_j \neq 1$, the corresponding elliptic factor contributes $\mathbf{1}$ or $\mathbf{L}$. The summand is then a Tate twist of a retract of the motive of an abelian variety of dimension at most four, so Tate surjectivity, standard conjecture D_ℓ , and independence of cycle class kernels follow from Theorem 2.3, Proposition 2.6, and Lemma 2.1. If every $i_j = 1$, then $i_B = 1$, giving the summand

$$h^1(B) \otimes h^1(E_1) \otimes h^1(E_2) \otimes h^1(E_3). \quad (5.2)$$

With M, N as in Section 4 and $P = M \otimes N$, (4.1) gives

$$\operatorname{Hom}(\mathbf{1}, P(2)) \simeq \operatorname{Hom}(M, N).$$

When k is finite, the duality identifies Tate classes in $R_\ell(P(2))$ with maps from $R_\ell M$ to $R_\ell N$ commuting with Frobenius. It preserves numerical triviality and realizations. Theorem 4.2 therefore proves Tate surjectivity and standard conjecture D_ℓ on (5.2). Theorem 4.3 gives independence of realization kernels over algebraically closed fields of characteristic p .

To pass from codimension two to the other codimensions, use the algebraic Fourier transform and its inverse [Kün93, Lemma 6.1(i), (iii)]. They induce isomorphisms

$$\mathcal{F}_\ell: H_{\text{et}}^{2r}(A_{\bar{k}}, \mathbf{Q}_\ell(r)) \xrightarrow{\sim} H_{\text{et}}^{2(5-r)}(\widehat{A}_{\bar{k}}, \mathbf{Q}_\ell(5-r)).$$

The dual $\widehat{A}$ is again a product of a surface and three elliptic curves. For finite k , Fourier transformation commutes with Frobenius, so surjectivity in codimension two gives surjectivity in codimension three. Tate’s divisor theorem [Tat66, Theorem 4] gives surjectivity in codimension one, and Fourier transformation gives it in codimension four. The fundamental class and the class of the origin treat codimensions zero and five. This proves (1).

For (2) and (3), use the Beauville decomposition. By Lemma 2.2, each component of a numerically trivial cycle is numerically trivial, and every component of nonzero grade has zero class for

all $\ell \neq p$. Fourier transformation sends $\mathrm{CH}^r(A)_{(0)}$ isomorphically to $\mathrm{CH}^{5-r}(\widehat{A})_{(0)}$ and preserves numerical triviality and realizations in both directions. It reduces the parts of grade zero in codimensions three and four to codimension two and to divisors, respectively. For divisors on an abelian variety, numerical and algebraic equivalence agree with rational coefficients; the Kummer sequence identifies the cycle class kernel with the group of algebraically trivial rational divisor classes for every $\ell \neq p$. The fundamental class and degree give (2) and (3) in codimensions zero and five. $\square$

6. REDUCTION TO THE RESIDUAL BLOCK

We reduce the Tate conjecture for fivefolds to one tensor product involving two surfaces and a supersingular elliptic curve. We use the Künneth decomposition, then classify the Frobenius-invariant coordinate lines modulo products of divisors.

A constant extension replaces $\mathbf{F}_q$ by $\mathbf{F}_{q^m}$ and each Frobenius root α by α^m . A coefficient extension is a finite extension of $\mathbf{Q}$ that splits endomorphism algebras and Chow projectors while preserving the finite field and its Frobenius roots.

The Tate conjecture for ordinary, supersingular, and geometrically simple fivefolds follows from Theorem 3.1(1), Lemma 2.7(1), and Lemma 2.8(1), respectively. Choose a finite constant extension $k = \mathbf{F}_q$ defining the simple isogeny factors, polarizations, geometric endomorphisms, and required Chow projectors. Enlarge k so that the group generated by the Frobenius roots contains no non-trivial root of unity. Such a field is sufficiently large in Zarhin's sense [Zar15, Remark 2.6]. All subsequent constant extensions apply to every factor.

For a simple factor X , let Ω_X be the set of distinct normalized roots α^2/q , and let $\Gamma'_X = \langle \Omega_X \rangle$. Its angle rank is $\mathrm{rk} \Gamma'_X$. Largeness makes Γ'_X torsion-free and $\alpha \mapsto \alpha^2/q$ injective on distinct roots. Galois acts transitively on Ω_X . Consequently, angle rank zero forces $\Omega_X = \{1\}$. At angle rank one, every Galois automorphism acts on the infinite cyclic group Γ'_X by a sign, so $\Omega_X = \{\theta, \theta^{-1}\}$. In particular, a simple factor with more than two distinct Frobenius roots has angle rank at least two.

An abelian variety is neat in Zarhin's sense if representatives of the inversion orbits of size two in its set of normalized roots are multiplicatively independent. The root 1 contributes no generator. When all Frobenius roots are distinct and nonsupersingular, neatness is equivalent to maximal angle rank [Zar15, Examples 2.10 and 2.11].

For a polarized abelian variety X , let $T^2(X)$ denote the orthogonal complement of its Néron–Severi motive in $h^2(X)$, and let $P^3(X)$ denote the primitive summand of $h^3(X)$. A symmetric divisor basis and its algebraic Lefschetz duals for the nondegenerate divisor intersection pairing define the Néron–Severi Chow projector; for an elliptic curve, $T^2(X) = 0$. The primitive projector is supplied by the motivic Lefschetz decomposition [Kün93, Theorem 5.1]. Tate's theorem on homomorphisms gives

$$\mathrm{Hom}(X, Y)_{\mathbf{Q}} = 0 \implies \left(R_\ell(h^1(X) \otimes h^1(Y)(1)) \right)^{\mathrm{Frob}_q=1} = 0 \quad (6.1)$$

for geometrically simple factors X and Y in distinct isogeny classes [Tat66, §1, Main Theorem].

For every factor X and every $m \geq 1$, Tate's divisor theorem [Tat66, Theorem 4] and the motivic Lefschetz decomposition [Kün93, Theorem 5.1] give

$$\left(R_\ell(T^2(X))(1) \right)^{\mathrm{Frob}_q^m=1} = 0, \quad h^3(X) \simeq P^3(X) \oplus h^1(X)(-1). \quad (6.2)$$

The vanishing of invariants follows from the orthogonality of $T^2(X)$ to the geometric Néron–Severi space. The Lefschetz decomposition applies for $\dim X \geq 2$, with $P^3(X) = 0$ for a surface. For an elliptic curve, $h^3(X) = 0$.

6.1. The Künneth table in degree four. Write a geometrically decomposable abelian fivefold, up to isogeny, as a product of geometrically simple factors. Theorem 5.1(1) proves the Tate conjecture for products of an abelian surface and three elliptic curves, including products of five elliptic

curves. The remaining factor dimensions are $4 + 1$, $3 + 2$, $3 + 1 + 1$, and $2 + 2 + 1$. A Künneth summand in degree four with an entry in degree zero is supported on a subproduct of dimension at most four, so its Tate classes are algebraic by Theorem 2.3. We therefore consider the summands supported on every factor, listed in Table 1.

TABLE 1. Remaining Künneth summands in degree four supported on every factor

factor dimensions	positive degree tuples	primitive remainder
$4 + 1$	$(2, 2), (3, 1)$	$P^3(X_4) \otimes h^1(E)$
$3 + 2$	$(1, 3), (2, 2), (3, 1)$	$T^2(X_3) \otimes T^2(Y_2); P^3(X_3) \otimes h^1(Y_2)$
$3 + 1 + 1$	$(2, 1, 1), (1, 2, 1), (1, 1, 2)$	$T^2(X_3) \otimes h^1(E_1) \otimes h^1(E_2)$
$2 + 2 + 1$	$(2, 1, 1), (1, 2, 1), (1, 1, 2)$	$T^2(B_1) \otimes h^1(B_2) \otimes h^1(E); T^2(B_2) \otimes h^1(B_1) \otimes h^1(E)$

Frobenius invariants are taken after the Tate twist (2). For an elliptic curve E , $h^2(E) = \mathbf{L}$. In the $4 + 1$ row, the invariant part of $(2, 2)$ is spanned by products of divisors on X_4 with the origin of E . The Lefschetz part of $(3, 1)$ has zero invariant part by (6.1). In the $3 + 2$ row, (6.1) gives zero invariant parts for $(1, 3)$ and the Lefschetz part of $(3, 1)$. In $(2, 2)$, the tensor product of the Néron–Severi summands is spanned by divisor products. Each mixed tensor product of a Néron–Severi summand with T^2 has zero invariant part by (6.2), leaving $T^2(X_3) \otimes T^2(Y_2)$.

In the $3 + 1 + 1$ row, Tate’s divisor theorem on $E_1 \times E_2$ shows that the invariant part of the Néron–Severi term in $(2, 1, 1)$ is spanned by divisor products. Its complement is the block containing $T^2(X_3)$. The other two degree tuples have zero invariant part by (6.1). In the $2 + 2 + 1$ row, the Néron–Severi term with degree two on a surface has zero invariant part, again by (6.1). The invariant part of $(1, 1, 2)$ is spanned by divisors on $B_1 \times B_2$, multiplied by the origin of E ; it is zero when the surfaces lie in distinct isogeny classes.

6.2. CM coordinates and divisor products. Choose a Rosati-stable maximal CM field in the endomorphism algebra of each geometrically simple factor, and take their product inside $\text{End}^0(A)$. Let $L/\mathbf{Q}$ be a finite Galois coefficient field splitting this CM étale algebra. Ancona’s algebraic projectors give

$$h^1(A)_L = \bigoplus_{\sigma \in \Sigma} M_\sigma, \quad h^4(A)_L = \bigoplus_{\substack{I \subset \Sigma \\ |I|=4}} M_I, \quad M_I = \bigotimes_{\sigma \in I} M_\sigma. \quad (6.3)$$

Each summand in (6.3) has rank one. The sum over a complete $\text{Gal}(L/\mathbf{Q})$ -orbit is a rational Chow summand [Anc21, Propositions 6.6 and 6.7]. Fix $\ell \neq p$ and a finite extension $F_\ell/\mathbf{Q}_\ell$ containing the image of an embedding $L \hookrightarrow \overline{\mathbf{Q}_\ell}$. Write V_σ for the realization of M_σ over F_ℓ , and let $V_I = \bigotimes_{\sigma \in I} V_\sigma$. The projectors commute with Frobenius, since the CM endomorphisms are defined over k . A line $V_I(2)$ fixed by Frobenius satisfies

$$\prod_{\sigma \in I} \alpha_\sigma = q^2. \quad (6.4)$$

Here $\alpha_\sigma \in L$ is the Frobenius root on M_σ . Galois conjugation of (6.4) shows that every line in the complete label orbit satisfies the same Tate relation. Complex conjugation pairs the ten elements of Σ , and $\alpha_\sigma \alpha_{\bar{\sigma}} = q$.

Lemma 6.1. *Let $|I| = 4$ and suppose that (6.4) holds. If distinct $\sigma, \tau \in I$ satisfy*

$$(\alpha_\sigma \alpha_\tau)^n = q^n \quad \text{for some } n \geq 1, \quad (6.5)$$

then $V_I(2)$ is generated by a product of two divisor classes with F_ℓ -coefficients on $A_{\mathbf{F}_{q^n}}$.

Proof. Write $I \setminus \{\sigma, \tau\} = \{\nu, \omega\}$. Equations (6.4) and (6.5) give $(\alpha_\nu \alpha_\omega)^n = q^n$. Thus both lines

$$V_\sigma \otimes V_\tau(1), \quad V_\nu \otimes V_\omega(1)$$

are fixed by Frobenius over $\mathbf{F}_{q^n}$. Tate's divisor theorem [Tat66, Theorem 4], extended from $\mathbf{Q}_\ell$ to F_ℓ , supplies a nonzero divisor class in each line. Their cup product is nonzero because the four labels are distinct, and it spans $V_I(2)$. $\square$

In particular, a Frobenius-fixed coordinate containing a complete conjugate pair is generated by divisor products over $\mathbf{F}_q$.

Definition 6.2. A complete rational Galois orbit in (6.3) is admissible if every index set I satisfies:

- (i) Frobenius acts by q^2 on M_I .
- (ii) For all distinct $\sigma, \tau \in I$ and every $m > 0$,

$$(\alpha_\sigma \alpha_\tau)^m \neq q^m. \quad (\text{R0})$$

The Tate relation and condition (R0) are preserved by Galois conjugation. Thus every Frobenius-fixed coordinate either belongs to an admissible orbit or is generated by divisor products, by Lemma 6.1. Condition (R0) excludes a complete conjugate pair. Thus an admissible index set selects one element from four of the five conjugate pairs. Write $o(I) = \{\rho, \bar{\rho}\}$ for the unique omitted pair.

The last column of Table 1 contains five primitive block types, where interchange of the surface factors identifies the two $2 + 2 + 1$ blocks. In the order

$$\begin{aligned} P^3(X_4) \otimes h^1(E), \quad T^2(X_3) \otimes T^2(Y_2), \quad P^3(X_3) \otimes h^1(Y_2), \\ T^2(X_3) \otimes h^1(E_1) \otimes h^1(E_2), \quad T^2(B_1) \otimes h^1(B_2) \otimes h^1(E), \end{aligned}$$

the lower bounds for the primitive subset orbits are

$$8, \quad 6, \quad 4, \quad 6, \quad 4. \quad (6.6)$$

The map $I \mapsto o(I)$ is Galois equivariant. For these five block types, the omitted pair belongs respectively to the simple factors of dimensions 4, 3, 2, 3, 2. The chosen maximal CM field of each factor has degree twice its dimension. Galois acts transitively on the conjugate pairs of embeddings of this field, so the orbit of $o(I)$ has cardinality 4, 3, 2, 3, 2, respectively. The pairs consist of CM labels, even when several labels share a Frobenius root. Complex conjugation fixes $o(I)$ and sends I to $\bar{I} \neq I$ by admissibility. Thus the stabilizer of $o(I)$ has an orbit of size at least two on the subsets with that omitted pair, which proves (6.6).

6.3. The blocks containing a simple threefold.

Proposition 6.3. *No admissible orbit is supported on $P^3(X_3) \otimes h^1(Y_2)(2)$ when X_3 and Y_2 are geometrically simple.*

Proof. A Tate relation would equate a primitive triple product on X_3 with q^2 divided by a root of Y_2 . We compare their Galois orbits. Assume first that the Frobenius centre of X_3 is a sextic CM field. The admissible triple selects one root from each conjugate pair. Identify the eight choices with $V = \mathbf{F}_2^3$. If $H \leq V \rtimes S_3$ is the Galois image, its permutation image P is C_3 or S_3 , and $N = H \cap V$ is a P -stable subspace containing central conjugation $c = (1, 1, 1)$. The only possibilities are

$$N = \langle c \rangle \quad \text{or} \quad N = V. \quad (6.7)$$

Indeed, the plane of vectors with even coordinate sum is the only other nonzero proper P -invariant subspace, and its intersection with $\langle c \rangle$ is trivial. If $N = V$, every orbit has length eight. If $N = \langle c \rangle$, the induced group on $V/\langle c \rangle \simeq \mathbf{F}_2^2$ is isomorphic to P and lies in $\text{AGL}_2(\mathbf{F}_2) \simeq S_4$. An order-three subgroup fixes one point and permutes the other three; its normalizer in S_4 is the stabilizer of that point, of order six. The induced group has orbit partition 1 + 3 for both $P = C_3$ and $P = S_3$. The triple orbit lengths are therefore

$$2, \quad 6, \quad 8. \quad (6.8)$$

The map from these triples to their products is injective on an admissible orbit. At Hamming distance one, equality of products gives $\alpha = q/\alpha$, hence $\alpha^2 = q$. This contradicts $[\mathbf{Q}(\alpha) : \mathbf{Q}] = 6$. At distance two it contradicts (R0). At distance three, writing γ for the triple product, one obtains $\gamma^2 = q^3$. A Tate relation with a root β of Y_2 reads

$$\gamma\beta = q^2; \quad (6.9)$$

equality at Hamming distance three would then give $\beta^2 = q$. This would make Y_2 supersingular, contrary to its geometric simplicity. Thus (6.9) identifies the stabilizers of γ and β . The orbit of a root of a geometrically simple nonsupersingular surface has length four, in contradiction with (6.8).

Now assume that the centre is an imaginary quadratic field F and the Schur index is three. If its two Frobenius roots are $\alpha, \bar{\alpha}$, each with multiplicity three, the eigenvalues occurring in primitive degree three are among

$$\alpha^3, \quad q\alpha, \quad q\bar{\alpha}, \quad \bar{\alpha}^3. \quad (6.10)$$

For each γ in (6.10), (6.9) gives $\beta \in F$: for example, $\beta = q/\alpha$ when $\gamma = q\alpha$, and $\beta = q^2/\alpha^3$ when $\gamma = \alpha^3$. After a sufficiently large constant extension, a geometrically simple surface has a quartic Frobenius field: the alternative of a quadratic centre has Schur index two and is supersingular, by the local invariant formula [Wat69, Chapter 2, p. 527]. Hence $[\mathbf{Q}(\beta) : \mathbf{Q}] = 4$, whereas $\beta \in F$ gives degree at most two. $\square$

Proposition 6.4. *Let X_3 and Y_2 be geometrically simple, and let E_1, E_2 be elliptic curves. Neither of the following blocks supports an admissible orbit:*

$$\begin{aligned} \text{(i)} \quad & T^2(X_3) \otimes T^2(Y_2)(2); \\ \text{(ii)} \quad & T^2(X_3) \otimes h^1(E_1) \otimes h^1(E_2)(2). \end{aligned} \quad (6.11)$$

Proof. Suppose first that the Frobenius centre of X_3 is a sextic CM field. Choose one root α_i from each reciprocal pair and let

$$\theta_i = \alpha_i^2/q \quad (1 \leq i \leq 3).$$

The six normalized roots are distinct, so the angle rank is at least two. At rank three the θ_i are multiplicatively independent. At rank two the relation lattice is saturated of rank one, because its quotient is the torsion-free group Γ'_{X_3} . Write its primitive generator as (v_1, v_2, v_3) . Galois acts on this generator by a sign and permutes the reciprocal pairs transitively. Thus $|v_1| = |v_2| = |v_3|$, and primitivity makes all three absolute values equal to one. Reorienting the reciprocal pairs gives

$$\ker(\mathbf{Z}^3 \longrightarrow \langle \theta_1, \theta_2, \theta_3 \rangle) = \mathbf{Z}(1, 1, 1). \quad (6.12)$$

Encode a primitive pair by $v \in \{0, \pm 1\}^3$ with exactly two nonzero entries. Equality of two pair products gives $\theta^{v-w} = 1$. At angle rank three, multiplicative independence gives $v = w$. At angle rank two, (6.12) gives $v - w = n(1, 1, 1)$. If the supports differ, reduction modulo two gives two nonzero coordinates on the left and either zero or three on the right. If the supports agree, one coordinate of $v - w$ is zero. Thus $n = 0$, so the pair product map is injective. The omitted reciprocal pair in X_3 has an orbit of length three. Complex conjugation fixes that omitted pair and changes the selected primitive pair, so each fibre of the map to omitted pairs has at least two elements. The selected X_3 -pair therefore has at least six conjugates with distinct products. A primitive surface pair and a product of two elliptic roots each have at most four values. A Tate relation would identify their stabilizers with the stabilizer of the X_3 -pair product, which is impossible.

Suppose now that the simple threefold has imaginary quadratic centre and Schur index three. Its Newton polygon is

$$\{(1/3)^3, (2/3)^3\}. \quad (6.13)$$

A pair with mixed orientations has product q , contrary to (R0). A pair with the same orientation has slope $2/3$ or $4/3$. A primitive surface pair has slopes in $\{0, 1, 2\}$ or $\{1/2, 3/2\}$, and a product

of two elliptic roots has slopes in $\{0, 1/2, 1, 3/2, 2\}$. The required complementary slopes $4/3$ and $2/3$ lie outside both sets. $\square$

6.4. Two surfaces and an ordinary elliptic curve.

Lemma 6.5. *Let B/k be a geometrically simple abelian surface and let E/k be an ordinary elliptic curve. Suppose that k is sufficiently large for $B \times E$.*

- (i) $F = \mathbf{Q}(\pi_B) = \text{End}^0(B)$ is a quartic CM field with no imaginary quadratic subfield.
- (ii) If a, b represent the two reciprocal root pairs of B and ϵ is a Frobenius root of E , then

$$a^2/q, \quad b^2/q, \quad \epsilon^2/q$$

are multiplicatively independent. Thus $B \times E$ has angle rank three and is neat.

Proof. A geometrically simple surface is nonsupersingular. The Honda–Tate dimension formula and local invariants [Wat69, Chapter 2, p. 527] give a quartic CM Frobenius field and Schur index one. Indeed, the alternative of a quadratic centre and index two forces every slope to be $1/2$. All geometric endomorphisms are defined over k by largeness. Zarhin’s Corollary 5.2 [Zar15] therefore excludes imaginary quadratic subfields of F , proving (i).

For (ii), let N be the normal closure of F . Its transitive action on the four reciprocal roots is a subgroup of $C_2^2 \rtimes S_2$. The absence of an imaginary quadratic subfield excludes biquadratic F . Thus the Galois group is C_4 or D_4 , both of which contain a four-cycle. Choose such a cycle and orient the roots so that it sends a to b , b to q/a , and q/a to q/b . Lift the cycle to an automorphism r of $N\mathbf{Q}(\epsilon)/\mathbf{Q}$. Let $\theta_a = a^2/q$, $\theta_b = b^2/q$, and $\eta = \epsilon^2/q$. Then r^2 inverts both θ_a and θ_b , while it fixes the quadratic field $\mathbf{Q}(\epsilon)$.

Suppose $\theta_a^m \theta_b^n \eta^k = 1$, with $m, n, k \in \mathbf{Z}$. Multiplying this relation by its r^2 -conjugate gives $\eta^{2k} = 1$. Ordinarity makes η have infinite order, so $k = 0$. Applying r to the remaining relation gives $\theta_b^m \theta_a^{-n} = 1$. Eliminating θ_b yields $\theta_a^{m^2+n^2} = 1$. The element θ_a has infinite order: torsion-freeness would otherwise give $a^2 = q$, contradicting $[\mathbf{Q}(a) : \mathbf{Q}] = 4$. Therefore $m = n = 0$. $\square$

Proposition 6.6. *Let B_1, B_2 be geometrically simple abelian surfaces and let E be an ordinary elliptic curve. No admissible orbit is supported on*

$$T^2(B_1) \otimes h^1(B_2) \otimes h^1(E)(2). \quad (6.14)$$

Proof. Largeness passes to every abelian subproduct: the group generated by its Frobenius roots is a subgroup of the group for the full fivefold. Assume first that B_1 and B_2 are isogenous to a geometrically simple surface B . By Lemma 6.5, the three representatives of the normalized reciprocal pairs of $B \times E$ are multiplicatively independent. Squaring and normalizing a Tate relation in (6.14) would give a multiplicative relation among the normalized roots of $B \times E$. After expressing reciprocal roots as inverses, the exponent of the elliptic normalized representative is 1 or -1 . This contradicts multiplicative independence.

Now suppose that B_1 and B_2 lie in distinct isogeny classes. Lemma 6.5 gives multiplicative independence for $B_1 \times E$. An equality between two of the eight products formed by a primitive B_1 -pair and one root of E gives a relation $\theta^{v-w} = 1$, where $v, w \in \{\pm 1\}^3$ encode the three chosen orientations. Multiplicative independence gives $v = w$, so the product map is injective. Let δ be the product of the selected B_1 -pair and elliptic root, and let β be the selected root of B_2 . The Tate relation

$$\delta\beta = q^2 \quad (6.15)$$

identifies the stabilizers of δ and β . The lower bound (6.6) and the four roots of B_2 give orbit length four for δ and β : injectivity recovers the B_1 -pair and elliptic label from δ , and $\beta = q^2/\delta$ recovers the B_2 -label, so the subset orbit also injects into the value orbit of δ . If an automorphism fixes δ , injectivity forces it to fix the selected elliptic root ϵ . Therefore

$$\mathbf{Q}(\epsilon) \subset \mathbf{Q}(\delta) = \mathbf{Q}(\beta). \quad (6.16)$$

Since $\mathbf{Q}(\epsilon)$ is imaginary quadratic, (6.16) contradicts Lemma 6.5 for $B_2 \times E$. $\square$

6.5. A simple fourfold and an elliptic curve.

Proposition 6.7. *Let X_4 be a geometrically simple abelian fourfold and let E be an elliptic curve. No admissible orbit is supported on $P^3(X_4) \otimes h^1(E)(2)$.*

Proof. After a sufficiently large constant extension, let the Frobenius centre of the geometrically simple fourfold X_4 have degree $2d$, and let e be the Schur index. The Honda–Tate dimension formula gives $de = 4$ [Wat69, Chapter 2, p. 527]. For the d reciprocal pairs of distinct centre roots, choose α_i and let $\theta_i = \alpha_i^2/q$. If the selected triple has a_i occurrences of α_i and b_i occurrences of q/α_i , let $u_i = a_i - b_i$. If γ is the triple product, ϵ an elliptic root, and $\eta = \epsilon^2/q$, the squared Tate relation is

$$\gamma^2/q^3 = \theta^u, \quad \theta^u \eta = 1. \quad (6.17)$$

Let $\mathcal{R} \subset \mathbf{Z}^d$ be the relation lattice of the θ_i . For ordinary E , Galois acts on η through a sign character χ , and (6.17) gives

$$gu - \chi(g)u \in \mathcal{R}. \quad (6.18)$$

For supersingular E , enlarge the constant field so that $\eta = 1$; then $gu \in \mathcal{R}$ for every g .

We now use the three possibilities for (d, e) to exclude (6.17). For $(d, e) = (4, 1)$, admissibility forces the triple to contain roots from three distinct conjugate pairs; hence $u \bmod 2$ has pattern 1110. Label the omitted coordinate by 4. When E is ordinary, transitivity on the conjugate pairs makes the reductions of (6.18) contain $e_j + e_4$ for $j = 1, 2, 3$. These vectors span the subspace of $\mathbf{F}_2^4$ with even coordinate sum. When E is supersingular, the four translates of u are the rows of $J + I$, where J is the matrix whose entries are all one. The matrix $J + I$ is invertible over $\mathbf{F}_2$. Thus the image of $\mathcal{R}$ modulo two has dimension at least three when E is ordinary and four when E is supersingular. Its dimension is at most $\text{rk } \mathcal{R}$, so $\text{rk } \Gamma'_{X_4} = 4 - \text{rk } \mathcal{R}$ is at most one or zero, respectively. At angle rank at most one, X_4 has at most two normalized roots, contradicting $\#\Omega_{X_4} = 8$.

For $(d, e) = (2, 2)$, the centre multiplicities and the odd total degree give $u \bmod 2 = (1, 0)$ or $(0, 1)$. Transitivity on the two centre pairs makes the reduction of $\mathcal{R}$ modulo two contain $(1, 1)$ when E is ordinary, and both basis vectors when E is supersingular. The angle rank is again at most one or zero, respectively, contradicting $\#\Omega_{X_4} = 4$.

Finally, let $(d, e) = (1, 4)$. The centre is imaginary quadratic. If p has a unique place in the centre, conjugation forces slope $1/2$, and the local degree two gives an integral Honda invariant, hence Schur index one. Thus $e = 4$ requires p to split in the centre and forces slopes $1/4, 3/4$. If k of the three selected roots have slope $1/4$, their sum is

$$\frac{9 - 2k}{4} \quad (0 \leq k \leq 3). \quad (6.19)$$

Adding any elliptic slope 0, 1, or $1/2$ gives a sum different from two, contradicting the Tate relation. $\square$

6.6. Newton polygons of the residual block. Let E be supersingular in (6.14). After an even constant extension, write

$$q = Q^2, \quad (6.20)$$

where Frobenius acts on $h^1(E)$ by the scalar Q . Fix a place above p of a common normal closure of the Frobenius fields, and normalize its valuation by $v(q) = 1$. A geometrically simple surface is either ordinary, with slopes 0, 0, 1, 1, or almost ordinary, with slopes 0, $1/2$, $1/2$, 1. The primitive cross products of an ordinary surface have slopes 0, 1, 1, 2; those of an almost ordinary surface have slopes $1/2$, $1/2$, $3/2$, $3/2$.

Let z be the selected primitive pair product on the surface contributing T^2 , and let β be the selected root of the surface contributing h^1 . The Tate relation is

$$z\beta Q = q^2 = Q^4. \quad (6.21)$$

If the surface contributing T^2 were ordinary and the surface contributing h^1 almost ordinary, the orbit of β would have four elements. Equation (6.21) identifies the stabilizers of z and β , so the orbit of z would consist of all four ordinary cross products. Their slopes 0, 1, 1, 2 would give slopes $3/2, 1/2, 1/2, -1/2$ for the corresponding conjugates of β , contradicting the possible Newton slopes of an abelian surface. Equation (6.21) also excludes two ordinary surfaces.

Suppose now that the surface contributing T^2 is almost ordinary. The orbit of β has four elements, and (6.21) identifies its stabilizer with that of z . Thus the complete orbit of the cross products of z contains two roots of slope $1/2$ and two of slope $3/2$. Equation (6.21) therefore gives two roots of slope zero and two of slope one for the other surface, which is ordinary.

In Table 2, F denotes the Frobenius field, e the Schur index of the endomorphism algebra, and ρ_X the angle rank of X . In the threefold and fourfold rows, (F, e) refers to X ; in the surface rows, it refers to each surface. The Tate twist (2) is omitted from the displayed blocks.

TABLE 2. Arithmetic restrictions on admissible orbits

Block	$([F : \mathbf{Q}], e)$	Restriction
$P^3(X_3) \otimes h^1(Y_2)$	(6, 1)	Orbit sizes of triple products: 2, 6, 8; the surface root has orbit size 4 (Proposition 6.3).
$P^3(X_3) \otimes h^1(Y_2)$	(2, 3)	The surface root would lie in a quadratic field; its Frobenius field has degree 4 (Proposition 6.3).
Both blocks in (6.11)	(6, 1)	$\rho_{X_3} = 2$ or 3. The selected pair product has at least 6 conjugates, the complement at most 4 (Proposition 6.4).
Both blocks in (6.11)	(2, 3)	Pair slopes $2/3, 4/3$ require complementary slopes $4/3, 2/3$, excluded by the slopes of the other factors (Proposition 6.4).
$P^3(X_4) \otimes h^1(E)$	(8, 1), (4, 2)	$\rho_{X_4} \geq 2$; the Tate relation forces $\rho_{X_4} \leq 1$ for ordinary E and $\rho_{X_4} = 0$ for supersingular E (Proposition 6.7).
$P^3(X_4) \otimes h^1(E)$	(2, 4)	Triple slopes $3/4, 5/4, 7/4, 9/4$ require elliptic complements $5/4, 3/4, 1/4, -1/4$ (Proposition 6.7).
$T^2(B_1) \otimes h^1(B_2)$ $\otimes h^1(E_{\text{ord}})$	(4, 1)	Multiplicative independence for $B_i \times E$ and the absence of an imaginary quadratic subfield in $\mathbf{Q}(\pi_{B_i})$ exclude the Tate relation (Proposition 6.6).
$T^2(B_1) \otimes h^1(B_2)$ $\otimes h^1(E_{\text{ss}})$	(4, 1)	The Tate relation forces B_1 almost ordinary and B_2 ordinary, giving (6.22) (Section 6.6).

Proposition 6.8. *Let $A/\mathbf{F}_q$ be a product of geometrically simple factors with dimensions listed in Table 1. Assume that all geometric endomorphisms of A are defined over $\mathbf{F}_q$ and that the group generated by its Frobenius roots is torsion-free. Every admissible orbit in a primitive block of Table 1 is supported on*

$$T^2(B_{\text{ao}}) \otimes h^1(B_{\text{ord}}) \otimes h^1(E_{\text{ss}})(2), \quad (6.22)$$

where B_{ao} and B_{ord} are geometrically simple surfaces, respectively almost ordinary and ordinary, and E_{ss} is a supersingular elliptic curve.

Proof. Propositions 6.3 and 6.4 exclude the threefold blocks, and Proposition 6.7 excludes the fourfold block. By Proposition 6.6, an admissible orbit in (6.14) has supersingular elliptic factor. Pass to an even constant extension so that this elliptic Frobenius is the scalar Q of (6.20); condition (R0) preserves admissibility under the extension. The slopes in (6.21) then force the stated Newton polygons of the two surfaces. $\square$

Proposition 6.9. *Let A satisfy the hypotheses of Proposition 6.8, and let $V_I(2)$ be a CM coordinate line fixed by Frobenius in a Künneth summand of degree four with positive degree on every factor. At least one of the following holds.*

- (i) *The line is generated by a product of two divisor classes with F_ℓ -coefficients over a finite extension of $\mathbf{F}_q$.*
- (ii) *The line belongs to an admissible orbit supported on the residual block (6.22).*

Proof. If (6.5) holds for a pair in I , Lemma 6.1 gives (i). Otherwise I contains no complete conjugate pair and satisfies (R0); its complete Galois orbit is admissible. We check that the selected line belongs to a primitive block of Table 1.

Consider a factor X contributing degree two. Its selected pair has Frobenius eigenvalue λ with $(\lambda/q)^m \neq 1$ for every $m \geq 1$. The geometric Néron–Severi space has eigenvalues $q\zeta$, where ζ is a root of unity. Its Frobenius-equivariant projector therefore kills the selected line, which lies in $T^2(X)$. On a factor contributing degree three, the CM polarization pairing couples only conjugate labels [Anc21, Proposition 6.6]. The selected triple has zero contraction and hence lies in $P^3(X)$. A triple on a surface contains a complete conjugate pair, as does degree two on an elliptic curve. Thus (i) applies to both degree patterns. The degree tuples in Table 1 exhaust the possibilities. The classification of admissible orbits in Proposition 6.8 therefore places the line in the residual block, proving (ii). $\square$

6.7. Arithmetic of the residual block. Let $B_1 = B_{\text{ao}}$ and $B_2 = B_{\text{ord}}$. Retain (6.20) and (6.21).

Proposition 6.10. *Assume that the selected residual Tate line belongs to an admissible orbit. Write the reciprocal pairs of Frobenius roots of B_1 as*

$$\{a, q/a\}, \quad \{b, q/b\}, \quad (6.23)$$

where the second pair has slopes $1/2, 1/2$, and let

$$x_0 = a, \quad x_1 = q/a, \quad y_0 = b, \quad y_1 = q/b, \quad z_{ij} = x_i y_j. \quad (6.24)$$

- (R1) *The field $F = \mathbf{Q}(\pi_{B_1}) = \text{End}^0(B_1)$ is quartic CM. Its normal closure N has Galois group $C_2^2 \rtimes S_2 \simeq D_4$, and*

$$F \otimes \mathbf{Q}_p \simeq \mathbf{Q}_p \times F_v \times \mathbf{Q}_p, \quad [F_v : \mathbf{Q}_p] = 2. \quad (6.25)$$

The decomposition group fixes the endpoint roots and interchanges the two roots of slope $1/2$.

- (R2) *The four cross products in (6.24) form one rational Galois orbit. For any of them, say z , the field $K = \mathbf{Q}(z)$ is a primitive quartic CM field whose normal closure is N , with Galois group D_4 , and*

$$K \otimes \mathbf{Q}_p = K_w \times K_{\bar{w}}, \quad [K_w : \mathbf{Q}_p] = [K_{\bar{w}} : \mathbf{Q}_p] = 2, \quad (6.26)$$

with complex conjugation exchanging the two factors. If $D_{p,\infty}$ is the rational quaternion algebra ramified at p and infinity, then

$$K \otimes D_{p,\infty} \simeq M_2(K). \quad (6.27)$$

- (R3) *For every $m > 0$, the four numbers z_{ij}^m are pairwise distinct and form the complete rational Galois orbit of z^m . In particular,*

$$\mathbf{Q}(z^m) = \mathbf{Q}(z) = K. \quad (6.28)$$

(R4) *The projection of the complete rational orbit to the (z, β) -labels is the graph with four elements*

$$\{(z_{ij}, Q^3/z_{ij})\}_{i,j \in \{0,1\}}. \quad (6.29)$$

Each graph pair occurs with both CM coefficient labels of the supersingular elliptic factor, so the complete coordinate orbit has eight elements. Writing $\beta = Q^3/z$ and $\delta = z/Q$, one has

$$z\beta Q = Q^4, \quad \delta = q/\beta = \bar{\beta}, \quad \mathbf{Q}(\delta) = \mathbf{Q}(\beta) = K. \quad (6.30)$$

After a constant extension of degree m , the projected graph in (6.29) becomes

$$\{(z_{ij}^m, Q^{3m}/z_{ij}^m)\}_{i,j}, \quad (6.31)$$

and $\mathbf{Q}(\delta^m) = \mathbf{Q}(\beta^m) = K$.

Proof. The Newton slope multiplicities 1, 2, 1 have greatest common divisor one. Hence the Frobenius polynomial of B_1 has exponent one and is irreducible quartic, with $F = \text{End}^0(B_1)$. The roots of slopes zero and one give two linear factors of $F \otimes \mathbf{Q}_p$. The two roots of slope 1/2 form one quadratic factor, since a linear factor would have Honda invariant 1/2, incompatible with Schur index one. This proves (6.25).

Let N be the normal closure of F , with the place induced by the chosen place above p . The decomposition group has orbit partition 1 + 2 + 1 on the four roots. Its faithful action consists of the identity and the reflection exchanging the two roots of slope 1/2. The action on reciprocal roots embeds $\text{Gal}(N/\mathbf{Q})$ transitively in $C_2^2 \rtimes S_2$. A Galois quartic would act regularly, and every orbit of a subgroup in a regular action has the same cardinality. Thus $F/\mathbf{Q}$ is nongalois, and $\text{Gal}(N/\mathbf{Q})$ is dihedral of order eight. This proves (R1).

The normal subgroup C_2^2 independently exchanges the two reciprocal pairs and acts transitively on the four products z_{ij} . Every quotient of two distinct roots of B_1 has infinite order. Indeed, roots of different Newton slope are distinguished by their p -adic valuations. The only remaining pair is $b, q/b$. If $b/(q/b)$ were torsion, centrality of complex conjugation in the normal closure of a CM field and transitivity would carry this relation to $a/(q/a)$, contradicting the slopes zero and one.

Let $m > 0$, and suppose $z_{ij}^m = z_{i'j'}^m$. If the labels differ in one coordinate, a quotient of two distinct roots of B_1 is torsion, contradicting the infinite order of that quotient. If they differ in both coordinates, then $z_{ij}z_{i'j'} = q^2$, whence $z_{ij}^{2m} = q^{2m}$. The coordinate containing z_{ij} belongs to the admissible Galois orbit of the original coordinate. This contradicts condition (R0) with exponent $2m$. Thus the four powers are distinct and form the Galois orbit of z^m . Consequently $[\mathbf{Q}(z^m) : \mathbf{Q}] = 4$. Together with $\mathbf{Q}(z^m) \subset \mathbf{Q}(z)$, this proves (6.28) and (R3).

We next prove (R2). Taking $m = 1$ gives $[K : \mathbf{Q}] = 4$. Complex conjugation sends z to q^2/z , and condition (R0), with exponent two, excludes a real cross product; hence K is CM. The induced D_4 -action on the four products is faithful: the two generators of C_2^2 flip the two coordinates independently, and the remaining involution exchanges the coordinates. The four products are distinct, so their splitting field is N . Thus N is the normal closure of K , with Galois group D_4 . In particular, K is nongalois; a quartic CM field containing an imaginary quadratic subfield is biquadratic, so K is primitive. The decomposition reflection has precisely the two orbits $\{z_{00}, z_{01}\}$ and $\{z_{10}, z_{11}\}$. These enumerate all places of K above p , each of full local degree two, proving (6.26). Complex conjugation exchanges the two factors. Restriction to either completion multiplies the local invariant 1/2 of $D_{p,\infty}$ by two, giving zero. At finite places away from p the quaternion algebra is already split. Since K has no real place, this proves (6.27) and (R2).

For (R4), (6.21) gives $\beta = Q^3/z$, so each conjugate of z determines a unique matching conjugate of β . The rational orbit projector and Galois equivariance of the labels give (6.29). Let $J \subset \text{End}^0(E)$ be the chosen imaginary quadratic CM field. Complex conjugation is the nontrivial central element of D_4 and generates its commutator subgroup. Every quadratic subfield of N is therefore totally real, so $J \cap N = \mathbf{Q}$. The two surface CM fields lie in N ; hence $\text{Gal}(NJ/\mathbf{Q}) = D_4 \times \text{Gal}(J/\mathbf{Q})$ acts

independently on the four graph pairs and the two elliptic labels. This proves that the complete coordinate orbit has eight elements. Both elliptic coefficient lines have Frobenius eigenvalue Q . Dividing by Q gives (6.30). A constant extension of degree m raises z, β, Q to their m -th powers, yielding (6.31). By (6.28),

$$\mathbf{Q}(\delta^m) = \mathbf{Q}(z^m) = \mathbf{Q}(Q^{3m}/z^m) = \mathbf{Q}(\beta^m) = K.$$

This proves (R4). $\square$

7. THE RESIDUAL CORRESPONDENCE

We construct curve correspondences that generate the third cohomology of an abelian threefold with slopes $0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$, proving Theorem 1.3. A Jacobian factor then realizes the residual motive of Proposition 6.10 and supplies its Tate cycles.

7.1. An isogeny pencil through a prescribed point. We first place a prescribed principally polarized threefold in a Moret–Bailly family. The quotient construction [RS22, Sections 1 and 3] requires a polarization with kernel α_p^2 ; we obtain it from the Dieudonné lattice of the prescribed fibre.

Lemma 7.1. *Let k be an algebraically closed field of characteristic $p > 0$. Let (Y, λ_Y) be a principally polarized abelian threefold with Newton slopes $0, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ and $a(Y) = 1$. The following hold.*

- (i) *There are a polarized abelian threefold (A_0, λ_0) with $\ker \lambda_0 \simeq \alpha_p^2$, and isogenies*

$$q_0 : A_0 \longrightarrow Y, \quad u_0 : Y \longrightarrow A_0, \quad q_0 u_0 = [p]_Y,$$

such that $\deg q_0 = p$ and $\lambda_0 = q_0^ \lambda_Y$.*

- (ii) *The quotients of A_0 by the lines in $\ker \lambda_0$ form a principally polarized abelian scheme*

$$q : A_0 \times \mathbf{P}^1 \longrightarrow \mathcal{A}$$

containing (Y, λ_Y) as a fibre. Its Hodge bundle has degree $p - 1$.

- (iii) *The relative isogeny $f_t = q_t u_0 : Y_{\mathbf{P}^1} \rightarrow \mathcal{A}$ satisfies*

$$\deg f_t = p^6, \quad f_t^* \lambda_t = p^2 \lambda_Y.$$

- (iv) *Given a field of definition $k_0 \subset k$ of (Y, λ_Y) , the data may be chosen over a finite extension of k_0 inside k .*

Proof. Fix a field of definition k_0 as in (iv). It suffices to construct the data over the algebraic closure of k_0 inside k . For the lattice calculation, denote this algebraic closure by k . We construct the two isogenies by changing only the supersingular summand of the Dieudonné lattice. We use the co-variant functor $D(G) = \mathbb{D}(G^\vee)$, where $G^\vee$ is the Cartier dual and $\mathbb{D}$ is the contravariant Dieudonné functor [Dem72, Chapter III, pp. 69–72; Chapter IV, Section 1, p. 73]. Write $D(A) = D(A[p^\infty])$. On these $W(k)$ -modules, F is σ -linear, V is σ^{-1} -linear, and $FV = VF = p$, where σ is the Witt vector Frobenius. The Newton break points $(1, 0)$ and $(5, 2)$ lie on the Hodge polygon. Successive applications of Katz’s decomposition theorem [Kat79, Theorem 1.6.1] split $D(Y)$ integrally into summands of slopes $0, \frac{1}{2}, 1$. The integral projections commute with F , hence with $V = pF^{-1}$ on $D(Y)[1/p]$, so each summand is a Dieudonné submodule. The polarization makes the ordinary summand of height two self-dual and orthogonal to a self-dual supersingular lattice M of height four and dimension two. Let $N = FM + VM$. Then $\text{length}_W(M/N) = a(M) = 1$, and $pM \subset N$. For a lattice $L \subset M[1/p]$, write $L^\vee$ for its dual under the fixed rational alternating pairing.

On $H = M/pM$, the operators F, V are nilpotent of rank two and

$$\ker F = \text{im } V, \quad \ker V = \text{im } F, \quad \dim(\text{im } F \cap \text{im } V) = 1.$$

The equality $F^2 = 0$ would imply $\text{im } F = \text{im } V$, contrary to $a(M) = 1$. The nonzero image spaces of a nilpotent semilinear map over the perfect field k strictly decrease: equality of successive images would make the restriction bijective. Since $\text{rank } F = 2$ and $F^2 \neq 0$, we obtain $\text{rank } F^2 = 1$ and $F^3 = 0$. Likewise, $\text{rank } V^2 = 1$ and $V^3 = 0$. Hence

$$\text{im } F^2 = \text{im } F \cap \text{im } V = \text{im } V^2.$$

Lifting to M gives

$$FN = F^2M + pM = V^2M + pM = VN.$$

Thus N is superspecial. The polarization identities

$$\langle Fx, y \rangle = \sigma \langle x, Vy \rangle, \quad \langle Vx, y \rangle = \sigma^{-1} \langle x, Fy \rangle$$

give the lattice equalities

$$(FL)^\vee = V^{-1}L^\vee, \quad (VL)^\vee = F^{-1}L^\vee.$$

Taking duals of $FN = VN$ and applying $FV = VF = p$ therefore gives $FN^\vee = VN^\vee$. Since $M \subset N^\vee$,

$$N = FM + VM \subset FN^\vee + VN^\vee = FN^\vee.$$

Self-duality of M gives

$$\text{length}_W(N^\vee/M) = \text{length}_W(M/N) = 1, \quad \text{length}_W(N^\vee/N) = 2.$$

For any F, V -stable lattice in $M[1/p]$, the colength of its F -image is the valuation of the determinant of F , namely the sum of its four slopes, which is two. Thus $\text{length}_W(N^\vee/FN^\vee) = 2$ as well. Consequently

$$N = FN^\vee = VN^\vee, \quad N^\vee/N \simeq D(\alpha_p^2).$$

The lattice N now determines the required isogenies. Retain the ordinary lattice and replace M by N . The resulting lattice N_{full} lies between $pD(Y)$ and $D(Y)$. It is realized by a quotient $u_0 : Y \rightarrow A_0$ with kernel in $Y[p]$: in covariant modules the map is $p : D(Y) \rightarrow N_{\text{full}}$. Factoring $[p]$ gives $q_0 : A_0 \rightarrow Y$, whose covariant map is the inclusion $N_{\text{full}} \subset D(Y)$. The colengths are

$$\text{length}_W(N_{\text{full}}/pD(Y)) = 6 - 1 = 5, \quad \text{length}_W(D(Y)/N_{\text{full}}) = 1.$$

Thus u_0 and q_0 have degrees p^5 and p , respectively.

For an isogeny $h : A \rightarrow B$ with $\lambda_A = h^\vee \lambda_B h$, the covariant pairing satisfies

$$\psi_{\lambda_A}(x, y) = \psi_{\lambda_B}(D(h)x, D(h)y).$$

Since $D(q_0)$ is the inclusion $N_{\text{full}} \hookrightarrow D(Y)$, the pairing of $\lambda_0 = q_0^* \lambda_Y$ is the restriction of the pairing on $D(Y)$. Its discriminant module is $N^\vee/N$, so $\ker \lambda_0 \simeq \alpha_p^2$ and $\deg \lambda_0 = p^2$. The map $D(u_0)$ is multiplication by p , and hence $u_0^* \psi_{\lambda_0} = p^2 \psi_{\lambda_Y}$. The subgroup of $Y[p]$ and all the resulting data have finitely many defining coefficients in the chosen algebraic closure of k_0 . They therefore descend to a finite extension of k_0 .

To construct the family of principally polarized quotients, we show that every line in $N^\vee/N$ yields a principal polarization. The linking form on $N^\vee/N$ is the reduction of the original alternating pairing with values in $p^{-1}W/W$. It is alternating for every p . The inverse image in $N^\vee$ of every line $\ell \subset N^\vee/N$ is an integral, self-dual lattice. For $q_\ell : A_0 \rightarrow B = A_0/H_\ell$, the rational homomorphism

$$\mu = (q_\ell^\vee)^{-1} \lambda_0 q_\ell^{-1}$$

therefore induces an integral, perfect Dieudonné pairing on $D(B)$. Since $\deg q_\ell = p$, choose $r \geq 1$ such that $u = p^r \mu \in \text{Hom}(B, B^\vee)$. Integrality gives

$$D(u)D(B) \subset p^r D(B^\vee).$$

Under the identifications

$$D(B[p^r]) = D(B)/p^r D(B), \quad D(B^\vee[p^r]) = D(B^\vee)/p^r D(B^\vee),$$

the restriction $u_r : B[p^r] \rightarrow B^\vee[p^r]$ induces the zero map. Faithfulness for finite group schemes [Dem72, Chapter III, Theorem (a), p. 69, and Section 8] gives $u_r = 0$. The fppf quotient by $B[p^r]$ is $[p^r]_B$, so $u = v \circ [p^r]_B$ for a homomorphism $v : B \rightarrow B^\vee$. In $\text{Hom}(B, B^\vee)_{\mathbf{Q}}$, this equality gives $v = \mu$. Since $q_\ell^* \mu = \lambda_0$ and $\deg \mu = \deg \lambda_0 / (\deg q_\ell)^2 = 1$, polarization descent [vdGM11, Chapter XI, Proposition 11.25(ii)] makes μ a principal polarization. These fibrewise polarizations extend to the quotient abelian scheme. At the generic point of $\mathbf{P}^1$, construct μ over an algebraic closure and descend μ by its rational formula and faithful flatness. At each closed point of $\mathbf{P}^1$, the abelian scheme $\mathcal{A}^\vee$ is the Néron model of its generic fibre [BLR90, Section 1.2, Proposition 8]. Since $\mathcal{A}$ is smooth, the Néron mapping property extends μ uniquely over that local ring. The local extensions glue. The identity $q^* \mu = \lambda_0$ holds everywhere, so [vdGM11, Chapter XI, Proposition 11.25(ii)] makes every fibre a principal polarization. The line M/N recovers (Y, λ_Y) .

It remains to compute the Hodge degree and the polarization multiplier. The universal line subgroup has Lie bundle $L = \mathcal{O}(-1)$. The exact Lie sequence for a height-one quotient [RS22, Theorem 1.2] gives

$$0 \longrightarrow L \longrightarrow \mathcal{O}^{\oplus 3} \longrightarrow \text{Lie}(\mathcal{A}/\mathbf{P}^1) \longrightarrow L^{\otimes p} \longrightarrow 0.$$

Its first cokernel is $\mathcal{O}(1) \oplus \mathcal{O}$, and the remaining extension splits because $H^1(\mathbf{P}^1, \mathcal{O}(p+1) \oplus \mathcal{O}(p)) = 0$. Thus

$$\text{Lie}(\mathcal{A}/\mathbf{P}^1) = \mathcal{O}(1) \oplus \mathcal{O} \oplus \mathcal{O}(-p),$$

and the determinant of the Hodge bundle has degree $p - 1$. Finally $q_t^* \lambda_t = \lambda_0 = q_0^* \lambda_Y$, so

$$(q_t u_0)^* \lambda_t = u_0^* q_0^* \lambda_Y = [p]^* \lambda_Y = p^2 \lambda_Y.$$

The degree is $p \cdot p^5 = p^6$. □

Proposition 7.2. *Let (Y, λ_Y) satisfy the hypotheses of Lemma 7.1, and suppose that it is the Jacobian of a nonhyperelliptic curve of genus three. There is a finite map $T \rightarrow \mathbf{P}^1$ from a smooth projective connected curve such that the following hold.*

- (i) *The pullback of the abelian scheme in Lemma 7.1 is the Jacobian of a stable family $\mathcal{C} \rightarrow T$ of curves of genus three and compact type.*
- (ii) *Let $d = \deg(T \rightarrow \mathbf{P}^1)$, and let $\xi = K_{\mathcal{C}}/4$ on the generic fibre C . Write λ for the Hodge line bundle and δ_1 for the number of separating nodes, counted with their thicknesses. Then*

$$\langle \Delta_\xi, \Delta_\xi \rangle = 21 \deg \lambda - 6\delta_1 \geq 3d(p - 1) > 0.$$

Proof. We realize the pencil as a family of Jacobians and bound its separating nodes in terms of the Hodge degree. The generic fibre is indecomposable and nonhyperelliptic, since these open conditions hold at Y . Over an algebraic closure, an indecomposable principally polarized abelian threefold is a Jacobian [OU73, Theorem 4]. A curve and an isomorphism of principally polarized abelian varieties descend to a finite extension of $k(\mathbf{P}^1)$. Take the normalization of $\mathbf{P}^1$ in the finite extension, and make a further finite base change for stable reduction and a degree-one divisor representing ξ . Since the Jacobian has good reduction everywhere, all stable fibres have compact type [OU73, Lemma 6]. In particular $\delta_0 = 0$, and $\deg \lambda = d(p - 1)$.

To bound δ_1 , let $g : \mathcal{C} \rightarrow T$ be the stable family and consider the multiplication map

$$\mu : \text{Sym}^2 g_* \omega_g \longrightarrow g_* \omega_g^{\otimes 2}$$

is between bundles of rank six. It is generically invertible because the generic canonical curve is a plane quartic. Grothendieck–Riemann–Roch gives

$$[\text{div}(\det \mu)] = 9\lambda - \delta_0 - \delta_1.$$

At the generic separating boundary of the moduli stack of stable curves of genus three, the two components have genera two and one. Products of differentials from distinct components vanish and span a two-dimensional subspace of the kernel of the universal multiplication map. On an étale chart, the local ring at a generic point above this boundary is a discrete valuation ring. If t

is a uniformizer, the multiplication matrix modulo t has rank at most four. Its Smith normal form therefore has at least two diagonal entries divisible by t , so its determinant is divisible by t^2 . Thus the determinant vanishes to order at least two along δ_1 in every characteristic. The regularity of the moduli stack gives the effective divisor

$$\operatorname{div}(\det \mu) - 2\delta_1.$$

Pulling back to T and using $\delta_0 = 0$ yields $\delta_1 \leq 3 \deg \lambda$.

We next express the Gross–Schoen height using the same boundary multiplicities. Pass to the regular semistable resolution of $\mathcal{C}$. A node with local equation $xy = t^m$ contributes an edge of length m ; resolution replaces the edge by a rational chain and preserves both $\deg \lambda$ and the relative dualizing self-intersection. Thus δ_1 counts separating nodes with their thicknesses, including the contributions from ramified base change. For the dual graph of a curve of compact type, a separating edge of length ℓ , with genera i and $g - i$ on its two sides, contributes

$$\epsilon = \left(\frac{4i(g-i)}{g} - 1 \right) \ell, \quad \varphi = \frac{2i(g-i)}{g} \ell.$$

Substituting the formulas for ϵ and φ , together with $\omega_g^2 = 12 \deg \lambda - \delta$, in the canonical Gross–Schoen height formula [Zha10, Theorem 1.3.1 and Proposition 4.4.1] gives, for $g = 3$,

$$\langle \Delta_\xi, \Delta_\xi \rangle = 21 \deg \lambda - 6\delta_1 \geq 3 \deg \lambda = 3d(p-1) > 0.$$

The height over a function field is defined over an arbitrary constant field [Zha10, Section 2.1]. The calculation therefore applies at a geometric generic point of a moduli space, after descending the curves and cycles to a finite extension of the generic field. $\square$

7.2. A correspondence detected by its height.

Lemma 7.3. *Let k be an algebraically closed field of positive characteristic, let T/k be a smooth projective connected curve, and let $K = k(T)$. Let $\mathcal{C} \rightarrow T$ be a stable family of curves of genus three with smooth generic fibre and fibres of compact type, and let $\mathcal{J} \rightarrow T$ be its Jacobian abelian scheme, with principal polarization θ . Suppose that an abelian threefold Y/k , a polarization class $\eta \in \operatorname{CH}^1(Y)_{\mathbf{Q}}$, and an isogeny*

$$a : Y \times T \longrightarrow \mathcal{J}$$

are given, with $a_K^ \theta_K = b\eta_K$ for some $b \in \mathbf{Q}_{>0}$. Let $\xi = K_{\mathcal{C}_K}/4$ in $\operatorname{Pic}(\mathcal{C}_K)_{\mathbf{Q}}$ and $h_{\text{GS}} = \langle \Delta_\xi, \Delta_\xi \rangle$. If $h_{\text{GS}} > 0$, there is a correspondence $\Gamma \in \operatorname{CH}^2(T \times Y)_{\mathbf{Q}}$ satisfying the following.*

(i) *For every $\ell \neq \operatorname{char}(k)$,*

$$\operatorname{cl}_\ell(\Gamma) \in H^1(T, \mathbf{Q}_\ell) \otimes \ker(\eta \cup - : H^3(Y, \mathbf{Q}_\ell) \longrightarrow H^5(Y, \mathbf{Q}_\ell)(1))(2).$$

(ii) *The self-intersection satisfies*

$$\deg(\Gamma \cdot \Gamma) = \frac{\deg(a)}{6} h_{\text{GS}} > 0. \tag{7.1}$$

Proof. We extend a homologically trivial cycle on the generic Jacobian to $T \times Y$ and extract its $H^1(T) \otimes H^3(Y)$ component. Use symmetric representatives for θ and η , so that $a_K^* \theta_K = b\eta_K$ holds in $\operatorname{CH}^1(Y_K)_{\mathbf{Q}}$. Let $C = \mathcal{C}_K$ and define

$$\iota : C \longrightarrow \mathcal{J}_K, \quad x \longmapsto [4x - K_C], \quad C_1 = 4^{-3}(\iota_*[C])_{(1)},$$

where $(\)_{(1)}$ denotes the Beauville component of grade one. After a finite extension admitting $\xi_0 \in \operatorname{Pic}^1(C)$ with $4\xi_0 = K_C$, one has $\iota = [4] \circ j_{\xi_0}$. Since $[4]_*$ acts by 4^3 on this component, C_1 agrees with the degree-one Abel component in [Zha10, Theorem 1.5.6]. Let

$$z = \mathcal{F}_\theta(C_1) \in \operatorname{CH}^2(\mathcal{J}_K)_{\mathbf{Q}}.$$

Zhang's Theorem 1.5.6 gives

$$\theta_K z = 0, \quad \text{cl}_\ell(z) = 0, \quad \langle z, z \rangle = h_{\text{GS}}/6.$$

These identities descend to K : pullback on rational Chow groups is injective under finite extensions, and function-field heights scale by the extension degree. Let $z_Y = a_K^* z$. Since $a_K^* \theta_K = b \eta_K$ with $b > 0$,

$$\eta_K z_Y = b^{-1} a_K^* (\theta_K z) = 0, \quad \text{cl}_\ell(z_Y) = 0.$$

To compute the height on the smooth models, choose a Chow extension $\mathcal{Z} \in \text{CH}^2(\mathcal{J})_{\mathbf{Q}}$ of z . Identify the source of a with $T \times Y$ by interchanging the factors, and let

$$Z = a^* \mathcal{Z} \in \text{CH}^2(T \times Y)_{\mathbf{Q}}.$$

Smooth proper base change shows that the fibre classes of $\mathcal{Z}$ and Z vanish: the corresponding sections of the fourth cohomology sheaves have zero generic value. The extensions restrict to numerically trivial cycles on every fibre of the smooth projective models $\mathcal{J}$ and $T \times Y$. Since these fibres are geometrically integral, Zhang's definition of the height pairing [Zha10, Section 2.1, condition (3), p. 17] gives

$$\deg_{\mathcal{J}}(\mathcal{Z}^2) = \langle z, z \rangle, \quad \deg_{T \times Y}(Z^2) = \langle z_Y, z_Y \rangle. \quad (7.2)$$

The isogeny a is finite locally free. Flat pullback preserves intersection products, and $a_* a^* = \deg(a)$ on Chow groups, including for inseparable a . The projection formula and (7.2) give

$$\begin{aligned} \deg_{T \times Y}(Z^2) &= \deg_{T \times Y} a^*(\mathcal{Z}^2) = \deg(a) \deg_{\mathcal{J}}(\mathcal{Z}^2) \\ &= \deg(a) \langle z, z \rangle = \frac{\deg(a)}{6} h_{\text{GS}} > 0. \end{aligned} \quad (7.3)$$

We now project Z to the required Künneth summand without changing its self-intersection. The vanishing of the fibre class removes the $H^0(T) \otimes H^4(Y)$ Künneth component of $\text{cl}_\ell(Z)$. Write the remaining components as

$$\text{cl}_\ell(Z) = c + v, \quad c \in H^1(T) \otimes H^3(Y)(2), \quad v \in H^2(T) \otimes H^2(Y)(2).$$

Both cv and v^2 vanish for degree reasons on T . Apply the algebraic Chow–Künneth projector π_Y^3 to Z and denote the result by Γ . Its class is c , so $\deg \Gamma^2 = \deg Z^2 > 0$.

Finally, $\eta_K z_Y = 0$ and localization show that $\text{pr}_Y^* \eta \cdot Z$ is represented by vertical cycles. The $H^1(T) \otimes H^5(Y)$ component of a vertical class is zero. Hence $(1 \otimes \eta)c = 0$. The construction of Γ uses Chow projectors, Fourier correspondences, pullbacks, and closures, and is therefore independent of ℓ . $\square$

7.3. Curves generating the third cohomology. Fix $n \geq 3$ prime to p . In the moduli space $\mathcal{A}_{3,1,n}$ over $\overline{\mathbf{F}}_p$, let W be the reduced Newton stratum with slopes $0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$, and let $S = W^{\text{sm}}$. Write $\mathcal{Y} \rightarrow S$ for the universal threefold, with principal polarization θ . For a geometric fibre Y_s , let

$$P_\ell(Y_s) = \ker(\theta_s : H^3(Y_s, \mathbf{Q}_\ell) \rightarrow H^5(Y_s, \mathbf{Q}_\ell)(1)).$$

We use monodromy to generate the generic primitive cohomology and Hecke orbits to reach every isogeny class. The three facts below hold for every prime p . References to [CO24] use the version of 25 February 2024.

- (i) The stratum W is irreducible [CO11, Theorem A]. Its smooth locus S is stable under prime-to- p Hecke correspondences: their two projections are étale and identify the corresponding strict local completions. The scheme S is smooth and irreducible, and its fibres have p -rank one. Chai–Oort's theorem [CO11, Theorem 2.12, p. 1368], with $g = 3$ and $d = 1$, gives the principal congruence subgroup $\Gamma(n)$ of level n as the geometric monodromy image on the prime-to- p Tate module; see also [Cha05, Proposition 4.5.4] and [CO24, Theorem 8.2.1(i),

- p. 538]. For every $\ell \neq p$, the ℓ -component of $\Gamma(n)$ is open in $\mathrm{Sp}_6(\mathbf{Z}_\ell)$, hence Zariski dense in $\mathrm{Sp}_{6, \mathbf{Q}_\ell}$. Thus the connected geometric ℓ -adic monodromy group of $\mathcal{Y} \rightarrow S$ is $\mathrm{Sp}_{6, \mathbf{Q}_\ell}$.
- (ii) The full polarized isogeny class of each point of $W(\overline{\mathbf{F}}_p)$ meets every central leaf in W [CO11, Proposition 4.2].
 - (iii) For each $x \in W(\overline{\mathbf{F}}_p)$, the $\mathrm{Sp}_6(\mathbf{A}_f^{(p)})$ -Hecke orbit of x is Zariski dense in the central leaf containing x [CO24, Theorem 9.1.1, p. 551; proof in §9.4.4, p. 584], applied with $g = 3$, $d = 1$, and $n \geq 3$ prime to p , including $p = 2$. For $p > 2$, [DvH25, Theorem 8.3.2] gives the density of the $\mathrm{GSp}_6(\mathbf{A}_f^{(p)})$ -Hecke orbit in the central leaf.

Proposition 7.4. *Let $K = \overline{\mathbf{F}}_p(S)$, fix an algebraic closure $\overline{K}$, and let Y_η/K be the generic fibre of $\mathcal{Y} \rightarrow S$. There are a finite extension L/K inside $\overline{K}$, a smooth projective geometrically connected curve T_L/L , and a correspondence $\Gamma_L \in \mathrm{CH}^2(T_L \times_L Y_L)_{\mathbf{Q}}$ such that, for every $\ell \neq p$,*

$$(\Gamma_L)_* : H^1(T_{\overline{K}}, \mathbf{Q}_\ell)(-1) \longrightarrow P_\ell(Y_{\overline{\eta}})$$

is surjective, where $Y_L = Y_\eta \times_K L$.

Proof. We first verify that the generic fibre admits the height construction. The stratum W has dimension four and generic a -number one [CO24, §2.5.1.6.4(a), p. 110, and §2.9.5, p. 155]. The locus of decomposable principally polarized threefolds of p -rank one has dimension at most two: an elliptic factor and its complementary surface have p -ranks $(1, 0)$ or $(0, 1)$, and the dimensions of their moduli spaces sum to $1 + 1 = 2$ or $0 + 2 = 2$, respectively. The locus of hyperelliptic Jacobians of genus three and p -rank one has dimension three: for odd p , use [GP05, Theorem 1 and Proposition 2] with its correction [GP10, Proposition 1]; for $p = 2$, use [PZ12, Corollary 1.3]. Thus $Y_{\overline{\eta}}$ is indecomposable and nonhyperelliptic. By [OU73, Theorem 4], it is the Jacobian of a curve of genus three.

The isogeny pencil of Lemma 7.1 has positive Gross–Schoen height by Proposition 7.2. Lemma 7.3 therefore gives a curve correspondence over $\overline{K}$ with positive self-intersection and nonzero image in $P_\ell(Y_{\overline{\eta}})$ for every $\ell \neq p$. The curve, correspondence, and polarization descend to one finite extension L/K , independently of ℓ ; denote them by T_L , Γ_L , and θ_L .

Monodromy makes this nonzero image equal to the primitive summand. Let $G_K = \mathrm{Aut}(\overline{K}/K)$ and $G_L = \mathrm{Aut}(\overline{K}/L)$. Since Γ_L is defined over L , its realization is G_L -equivariant. Its image I_ℓ is therefore G_L -stable. Let G_ℓ and $G_{\ell, L}$ be the Zariski closures of the images of G_K and G_L on $H^1(Y_{\overline{\eta}}, \mathbf{Q}_\ell)$. Since S is normal, G_K surjects onto $\pi_1(S, \overline{\eta})$. By (i), the connected monodromy group G_ℓ° is $\mathrm{Sp}_{6, \mathbf{Q}_\ell}$. The subgroup G_L has finite index in G_K ; the purely inseparable part of L/K preserves the absolute Galois group. Consequently,

$$G_{\ell, L}^\circ = G_\ell^\circ = \mathrm{Sp}_{6, \mathbf{Q}_\ell}.$$

Indeed, finitely many translates of $G_{\ell, L}$ cover G_ℓ , so the two groups have the same identity component. The stabilizer of I_ℓ in $G_{\ell, L}$ is Zariski closed and contains the image of G_L . Thus I_ℓ is stable under Sp_6 . The representation $P_\ell(Y_{\overline{\eta}})$ is the irreducible primitive summand of dimension fourteen in the third exterior power of the standard representation. Since $I_\ell \neq 0$, we obtain $I_\ell = P_\ell(Y_{\overline{\eta}})$ for every $\ell \neq p$. $\square$

Lemma 7.5. *For the data L, T_L, Γ_L of Proposition 7.4, there are a nonempty open subset $U \subset S$, a finite dominant morphism $U' \rightarrow U$ with U' smooth and integral of function field L , a smooth projective curve $f : \mathcal{T} \rightarrow U'$, and a relative correspondence $\tilde{\Gamma} \in \mathrm{CH}^2(\mathcal{T} \times_{U'} \mathcal{Y}_{U'})_{\mathbf{Q}}$ with the following properties.*

- (i) *The generic fibre of $(\mathcal{T}, \tilde{\Gamma})$ is (T_L, Γ_L) .*
- (ii) *For every geometric point $u' \in U'$ and every $\ell \neq p$,*

$$(\tilde{\Gamma}_{u'})_* : H^1(\mathcal{T}_{u'}, \mathbf{Q}_\ell)(-1) \longrightarrow P_\ell(Y_{u'})$$

is surjective.

Proof. Normalize a nonempty open subset of S in L . The normalization is finite because S is of finite type over a field. Spread T_L , Γ_L , and the polarization over a nonempty smooth open of the normalization, with the curve smooth and projective and the cycle components flat. Remove from S the image of the complement of this open; that image is closed because the normalization is finite. This gives a finite surjective map $U' \rightarrow U$. This open subset and finite cover depend only on the geometric data, hence are independent of ℓ .

For each $\ell \neq p$, the relative correspondence defines a morphism of lisse sheaves from $R^1 f_* \mathbf{Q}_\ell(-1)$ to the third cohomology of $\mathcal{Y}_{U'}$. Its composite with cup product by θ is zero at the generic point, hence on U' . The morphism therefore takes values in the primitive summand. Its rank is constant on the connected scheme U' . The generic surjectivity of Proposition 7.4 proves (ii) on the same U' for every $\ell \neq p$. $\square$

Lemma 7.6. *Let $U \subset S$ be a nonempty open subset. Suppose that, for each $u \in U(\overline{\mathbf{F}}_p)$, there are a smooth projective curve T_u and $\Gamma_u \in \mathrm{CH}^2(T_u \times Y_u)_{\mathbf{Q}}$ whose cylinder map surjects onto $P_\ell(Y_u)$ for every $\ell \neq p$. Then every abelian threefold $Y/\overline{\mathbf{F}}_p$ with Newton slopes $0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1$ admits a finite disjoint union T of smooth projective curves and a correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$, defined together over a finite field, such that*

$$\Gamma_* : H^1(T, \mathbf{Q}_\ell)(-1) \longrightarrow H^3(Y, \mathbf{Q}_\ell)$$

is surjective for every $\ell \neq p$.

Proof. Every full polarized isogeny class in $W(\overline{\mathbf{F}}_p)$ meets every central leaf by (ii) and contains a dense Hecke orbit in each leaf by (iii). Its closure is therefore W . Replace Y , up to isogeny, by a principally polarized threefold and add level structure. Its full isogeny class meets U at a point u . Choose an isogeny $a : Y \rightarrow Y_u$ by clearing the denominator of a quasi-isogeny.

To add the Lefschetz summand of $H^3(Y_u)$, choose a smooth ample complete intersection curve C in the dual of Y_u . The Poincaré divisor on $C \times Y_u$, multiplied by θ_u , gives a codimension-two correspondence surjecting onto $\theta_u H^1(Y_u)$ for every $\ell \neq p$, by weak Lefschetz and Poincaré duality. Together with the given primitive correspondence on $T_u \times Y_u$, this gives a correspondence on $(T_u \sqcup C) \times Y_u$ surjecting onto $H^3(Y_u)$ for every $\ell \neq p$. Pullback by $1 \times a$ preserves surjectivity, since a^* is an isomorphism on cohomology. The curves, cycles, and isogeny have finite presentation over $\overline{\mathbf{F}}_p$, so they descend together to a finite field. $\square$

Proof of Theorem 1.3. The generic surjection of Proposition 7.4 spreads over a finite cover $U' \rightarrow U$ of a nonempty open $U \subset S$ by Lemma 7.5. Each $u \in U(\overline{\mathbf{F}}_p)$ has a lift to $U'(\overline{\mathbf{F}}_p)$, so its primitive cohomology is generated by curve correspondences for every $\ell \neq p$. Lemma 7.6 transports these correspondences to every threefold with the stated slopes and adds the Lefschetz summand. $\square$

7.4. A Jacobian factor for the residual motive.

Lemma 7.7. *Let k be a finite field of characteristic p , let J/k be an abelian variety, and let M be a rational Chow motive over k that is finite-dimensional in the sense of Kimura. Fix $\ell_0 \neq p$ and suppose that the following hold.*

- (i) *The realization $R_{\ell_0} M$ is concentrated in odd degree. There is a Chow isomorphism $b : M \xrightarrow{\sim} M^\vee(-1)$ whose realization induces a nondegenerate alternating pairing.*
- (ii) *There is a Chow morphism $s : h^1(J) \rightarrow M$ with surjective ℓ_0 -adic realization.*

Then J has an abelian subvariety C of dimension $\frac{1}{2} \dim_{\mathbf{Q}_{\ell_0}} R_{\ell_0} M$ with a Chow isomorphism $M \simeq h^1(C)$ over k .

Proof. We realize the image of the adjoint as an abelian factor of J , equipped with the restricted polarization. Use a polarization of J and the self-duality b to form the Chow adjoint $j = s^\dagger : M \rightarrow h^1(J)$. Its realization is injective. Let $a = js \in \text{End}(h^1(J))$; then

$$\text{im}(a_{\ell_0}) = j_{\ell_0}(M_{\ell_0}).$$

By [Sch94, Corollary 5.10], write $a = \alpha^*$ for a rational endomorphism α of J . On $H^1(J)$, pullback by α is dual to its action on $V_{\ell_0}J$. After clearing its denominator, let $N = (\ker \alpha)_{\text{red}}^0$ and take the polarization complement $C = N^\perp$. Poincaré complete reducibility over k gives $J \sim N \times C$ and a Rosati self-adjoint projector $e = e_C^*$ with

$$\text{im}(a_{\ell_0}) = \text{Ann}(V_{\ell_0}N) = \text{im}(e_{\ell_0}).$$

Here $V_{\ell_0}N$ is the rational Tate module of N , viewed inside $V_{\ell_0}J$, and the annihilator lies in $H^1(J, \mathbf{Q}_{\ell_0})$. Let $i_C : C \hookrightarrow J$ be the inclusion and λ_J the chosen polarization. Its restriction

$$\lambda_C = i_C^\vee \lambda_J i_C : C \longrightarrow C^\vee$$

is a polarization of the actual abelian factor C . The isogeny $N \times C \rightarrow J$ identifies the two factors as orthogonal subspaces for the polarization pairing. Hence the pairing on $H' = eh^1(J) \simeq h^1(C)$ is nondegenerate in every realization. The morphism

$$f = ej : M \longrightarrow H'$$

is an isomorphism on ℓ_0 -adic realizations, giving the stated dimension of C .

To lift the realization isomorphism to Chow motives, form the adjoint $f^\dagger$ using b and the pairing induced by λ_C . Both pairings are nondegenerate, so the adjoint of f_{ℓ_0} is invertible. Hence $U = f^\dagger f \in \text{End}(M)$ has invertible realization, and

$$\text{Tr}_{\ell_0}(U^r) = -\text{tr}_{\text{cat}}(U^r) \in \mathbf{Q} \quad (r \geq 1),$$

because M_{ℓ_0} has odd degree. Newton's identities give $p(t) = \det(t - U_{\ell_0}) \in \mathbf{Q}[t]$ with $p(0) \neq 0$. The correspondence $p(U)$ is homologically, hence numerically, trivial. Finite-dimensionality of M and Kimura nilpotence [Kah03, Proposition 1.5] make $p(U)$ nilpotent. Writing $p(t) = p(0) + tc(t)$ shows that $Uc(U) = p(U) - p(0)$ is invertible. Since $c(U)$ commutes with U , the endomorphism U is invertible in Chow motives, and $v = U^{-1}f^\dagger$ satisfies $vf = 1_M$. The idempotent $1_{H'} - fv$ has zero realization. Since H' is a summand of an abelian motive, Kimura nilpotence makes the idempotent nilpotent, hence zero. Thus $f : M \xrightarrow{\sim} H'$ is a Chow isomorphism. $\square$

We apply the lemma to the residual motive. Write

$$\begin{aligned} B_1 &= B_{\text{ao}}, & B_2 &= B_{\text{ord}}, & Y &= B_1 \times E, \\ P &= T^2(B_1), & S &= h^1(E), & H &= h^1(B_2), & M &= P \otimes S(1). \end{aligned}$$

Fix a field of definition $k_0 = \mathbf{F}_{q_0}$ for the residual data of Proposition 6.10. Thus $q_0 = Q_0^2$ and Frobenius on E is the scalar Q_0 . Write $z_i^{(0)}, \beta_i^{(0)}$ for the matching roots over k_0 , and $\beta^{(0)}$ for the selected root of B_2 . Choose a finite extension k/k_0 defining the polarizations, Chow projectors, and curve correspondences supplied by Theorem 1.3 for Y . For $n = [k : k_0]$, let

$$q = q_0^n, \quad Q = Q_0^n, \quad z_i = (z_i^{(0)})^n, \quad \beta_i = (\beta_i^{(0)})^n.$$

Take all motives and Frobenius labels over $k = \mathbf{F}_q$.

The eigenvalues on $H^2(B_1)$ are two copies of q and $z_1, \dots, z_4$. Condition (R0) and (R2) exclude roots of unity among z_i/q . Tate's divisor theorem [Tat66, Theorem 4] therefore gives $\dim_{\mathbf{Q}} \text{NS}(B_1)_{\mathbf{Q}} = 2$. Choose symmetric divisors D_1, D_2 representing a basis of $\text{NS}(B_1)_{\mathbf{Q}}$ and let $G_{ab} = \deg(D_a D_b)$. The Hodge index theorem implies that G is invertible, and

$$\pi_P = \pi_2^{B_1} - \sum_{a,b} (G^{-1})_{ab} D_a \times D_b$$

is the self-adjoint Chow projector onto P . Indeed, symmetric divisors have Beauville grade zero, and the composition rule for the correspondences $D_a \times D_b$ gives the projector identities. The cup-product pairing on P and a polarization of E give an algebraic nondegenerate alternating pairing on M , with associated self-duality

$$M \simeq M^\vee(-1). \quad (7.4)$$

Let $\beta = (\beta^{(0)})^n$ and $K = \mathbf{Q}(\beta^{(0)})$. By (R3) and (R4), the powered labels satisfy

$$z_i \beta_i Q = q^2, \quad \frac{z_i}{Q} = \frac{q}{\beta_i} = \overline{\beta_i}, \quad \mathbf{Q}(\beta^m) = K \quad (m \geq 1).$$

The β_i are the four conjugates of β . If $\beta_i = \sigma_i(\beta)$, then $\mathbf{Q}(\beta_i^m) = \sigma_i(K)$ for every $m \geq 1$.

Theorem 7.8. *For the residual data of Proposition 6.10 over k_0 , a finite extension k/k_0 admits an isomorphism*

$$T^2(B_1) \otimes h^1(E)(1) \simeq h^1(B_2)^{\oplus 2} \quad (7.5)$$

in rational Chow motives over k , and hence over every finite extension of k .

Proof. We first realize M as the degree-one motive of an abelian fourfold; its Frobenius polynomial will then identify that fourfold up to isogeny. The motive $M = T^2(B_1) \otimes h^1(E)(1)$ is finite-dimensional because it is a summand of an abelian motive. Its realization is a summand of $H^3(Y)(1)$, hence has odd degree, and its dimension is $4 \cdot 2 = 8$. The self-duality (7.4) induces a nondegenerate alternating pairing.

The Newton slopes of $Y = B_1 \times E$ are $(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1)$. By Theorem 1.3, a finite disjoint union of smooth projective curves T and a correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$ give a surjection

$$\Gamma_* : H^1(T, \mathbf{Q}_\ell) \longrightarrow H^3(Y, \mathbf{Q}_\ell)(1).$$

Applying the projector onto M gives a Chow morphism $s : h^1(T) \rightarrow M$ with surjective realization for every $\ell \neq p$. Let J be the product of the Jacobians of the components of T . Rational degree-one zero-cycles define the curve projectors over k and identify $h^1(T) \simeq h^1(J)$. Lemma 7.7, applied at one prime $\ell_0 \neq p$, gives an abelian fourfold $J_M \subset J$ and a Chow isomorphism $M \simeq h^1(J_M)$ over k .

The Frobenius roots of M are $z_i/Q = \overline{\beta_i}$, each with multiplicity two. The set of conjugates of β is stable under complex conjugation. Thus J_M and B_2^2 have the same Frobenius polynomial and are k -isogenous, by Tate's isogeny theorem [Tat66, Theorem 1(c)]. Their h^1 motives are isomorphic, proving (7.5). $\square$

7.5. Eight residual cycles and their numerical duals.

Corollary 7.9. *Set $X = B_1 \times E \times B_2$ and*

$$R = P \otimes S \otimes H(2) = M \otimes H(1).$$

The following hold.

- (i) *There are cycles $c_1, \dots, c_8 \in \mathrm{CH}^2(X)_{\mathbf{Q}}$ in the summand R and cycles $d_1, \dots, d_8 \in \mathrm{CH}^3(X)_{\mathbf{Q}}$, all defined over k , with*

$$\deg(c_i \cdot d_j) = \delta_{ij}.$$

- (ii) *For every $\ell \neq p$ and $m \geq 1$, the classes of the eight cycles c_i are a basis of $R_\ell^{\mathrm{Frob}_q^m=1}$.*
- (iii) *For every $\ell \neq p$ and $m \geq 1$, realization induces an isomorphism*

$$\mathrm{Hom}_{\mathrm{CHM}(\mathbf{F}_{q^m})_{\mathbf{Q}}}(\mathbf{1}, R_{\mathbf{F}_{q^m}}) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \xrightarrow{\sim} R_\ell^{\mathrm{Frob}_q^m=1}.$$

- (iv) *Rational, numerical, and ℓ -adic homological equivalence agree on the projected Chow group $\mathrm{Hom}(\mathbf{1}, R)$ over every finite extension of k and over $\overline{\mathbf{F}}_p$.*

Proof. Fix a Chow isomorphism $\psi : M \xrightarrow{\sim} H \oplus H$ from Theorem 7.8. A polarization identifies $H^\vee \simeq H(1)$, so rigidity gives

$$\mathrm{Hom}(\mathbf{1}, R) \simeq \mathrm{Hom}(H, M) \simeq \mathrm{End}(H)^{\oplus 2}.$$

Since B_2 is an ordinary geometrically simple surface, $\mathrm{End}_k^0(B_2) = K$. The equality $\mathbf{Q}(\beta^m) = K$ also gives $\mathrm{End}_{\mathbf{F}_{q^m}}^0(B_2) = K$ for every $m \geq 1$. The contravariance of h^1 gives the opposite algebra $K^{\mathrm{op}} = K$.

The trace pairing of K supplies the cycles and their numerical duals. Choose a rational basis $a_1, \dots, a_4$ of K and its trace-dual basis $b_1, \dots, b_4$, so that $\mathrm{Tr}_{K/\mathbf{Q}}(a_i b_j) = \delta_{ij}$. Let $\iota_r : H \rightarrow H \oplus H$ and $p_r : H \oplus H \rightarrow H$ be the inclusions and projections, for $r = 1, 2$, and set

$$u_{ri} = \psi^{-1} \iota_r a_i : H \longrightarrow M, \quad v_{sj} = -b_j p_s \psi : M \longrightarrow H.$$

Define $c_{ri} : \mathbf{1} \rightarrow M \otimes H^\vee$ by applying $u_{ri} \otimes 1$ to the coevaluation of H . Define $d_{sj} : M \otimes H^\vee \rightarrow \mathbf{1}$ by applying $v_{sj} \otimes 1$, the symmetry, and evaluation. Transport c_{ri} and d_{sj} through the polarization isomorphism $M \otimes H^\vee \simeq R$, using the isomorphism for c_{ri} and its inverse for d_{sj} . Since $R = (X, \pi_R, 2)$ and $\dim X = 5$, the groups $\mathrm{Hom}(\mathbf{1}, R)$ and $\mathrm{Hom}(R, \mathbf{1})$ are represented by projected cycles of codimensions two and three, respectively, on X . Their compositions are

$$d_{sj} c_{ri} = \mathrm{tr}_{\mathrm{cat}}(v_{sj} u_{ri}) = \delta_{rs} \mathrm{Tr}_{K/\mathbf{Q}}(b_j a_i) = \delta_{rs} \delta_{ij}.$$

The minus sign in v_{sj} compensates for $\mathrm{tr}_{\mathrm{cat}}(a|H) = -\mathrm{Tr}_{K/\mathbf{Q}}(a)$. Since $\mathrm{End}(\mathbf{1}) = \mathbf{Q}$, the compositions give the required intersection identities.

To show that these cycles span all Frobenius invariants, recall that Frobenius acts semisimply on R_ℓ , a summand of the cohomology of an abelian variety. The motive R has rank 32 and Frobenius eigenvalues

$$\frac{z_i \beta_j Q}{q^2} = \frac{\beta_j}{\beta_i} \quad (1 \leq i, j \leq 4),$$

each with multiplicity two. For every $m \geq 1$, exactly the four diagonal labels are fixed, because the β_i^m are pairwise distinct. Thus the invariant space has dimension eight for every ℓ and m . The intersection identities in (i) show that the eight classes form a basis, proving (ii).

The identification $\mathrm{Hom}(\mathbf{1}, R) \simeq K^{\oplus 2}$ and Tate's theorem on homomorphisms [Tat66, §1, Main Theorem], applied over $\mathbf{F}_{q^m}$, prove (iii). The dual cycles d_{sj} detect every nonzero rational vector in $\mathrm{Hom}(\mathbf{1}, R)$, so rational and numerical equivalence agree on that group. Every geometric cycle is defined over a finite extension of k ; taking the union over finite extensions proves (iv) over $\overline{\mathbf{F}}_p$. $\square$

Remark 7.10. Let $Y = B_1 \times E$, $B = B_2$, and let $\pi_M \in \mathrm{CH}^3(Y \times Y)_{\mathbf{Q}}$ and $\pi_H \in \mathrm{CH}^2(B \times B)_{\mathbf{Q}}$ be the projectors defining $M = (Y, \pi_M, 1)$ and $H = h^1(B)$. Retain a curve correspondence $\Gamma \in \mathrm{CH}^2(T \times Y)_{\mathbf{Q}}$ that surjects onto $H^3(Y)(1)$, where T may be a finite disjoint union of curves. Choose divisor correspondences $D_a \in \mathrm{CH}^1(T \times B)_{\mathbf{Q}}$ representing a rational basis of $\mathrm{Hom}(h^1(T), H)$, and form

$$Z_a = \pi_H \circ (\mathrm{pr}_{13})_* (\mathrm{pr}_{12}^*(t^* \Gamma) \cdot \mathrm{pr}_{23}^* D_a) \circ \pi_M \in \mathrm{CH}^2(Y \times B)_{\mathbf{Q}}. \quad (7.6)$$

In (7.6), the maps pr_{ij} are the projections from $Y \times T \times B$. The resulting cycle has codimension $2 + 1 - \dim T = 2$. The outer projectors extract precisely $P \otimes S \otimes H(2)$ from its cohomology class.

For one $\ell \neq p$, the transpose cylinder embeds M_ℓ in $H^1(T, \mathbf{Q}_\ell)$. Tate's theorem on homomorphisms [Tat66, §1, Main Theorem] shows that the rational basis D_a spans the equivariant Hom space after tensoring with $\mathbf{Q}_\ell$. By semisimplicity, the restrictions of the $(D_a)_\ell$ span $\mathrm{Hom}_{G_k}(M_\ell, H_\ell)$, so at least one Z_a , denoted by Z , has nonzero realization.

Fix a Chow isomorphism $\psi : M \xrightarrow{\sim} H \oplus H$ from Theorem 7.8. The morphism $u : M \rightarrow H$ represented by Z satisfies

$$u\psi^{-1} = (a_1, a_2), \quad a_1, a_2 \in \mathrm{End}(H) = K, \quad (a_1, a_2) \neq (0, 0).$$

Choose r with $a_r \neq 0$, and let

$$v = \psi^{-1} \iota_r a_r^{-1} : H \longrightarrow M.$$

Then $uv = 1_H$. Represent v by its projected correspondence $V \in \mathrm{CH}^3(B \times Y)_{\mathbf{Q}}$, and define

$$W = -\frac{1}{4} {}^t V \in \mathrm{CH}^3(Y \times B)_{\mathbf{Q}}. \quad (7.7)$$

At the level of correspondences, $uv = 1_H$ is the equality $Z \circ V = \pi_H$ in $\mathrm{CH}^2(B \times B)_{\mathbf{Q}}$. The correspondence trace formula gives

$$\deg(Z \cdot W) = -\frac{1}{4} \mathrm{tr}_{\mathrm{cat}}(uv) = -\frac{1}{4} \mathrm{tr}_{\mathrm{cat}}(1_{h^1(B)}) = 1.$$

Since B is a surface, its motive $h^1(B)$ is odd of rank four, and $\mathrm{tr}_{\mathrm{cat}}(1_{h^1(B)}) = -4$.

8. THE TATE CONJECTURE

Let $X/\overline{\mathbf{F}}_q$ be smooth and projective. Choose a model over a finite field and let Φ be its geometric Frobenius. Let

$$\begin{aligned} W_{\ell}^r(X) &= \mathrm{im} \left(\mathrm{CH}^r(X)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell} \longrightarrow H_{\mathrm{\acute{e}t}}^{2r}(X, \mathbf{Q}_{\ell}(r)) \right), \\ \mathcal{T}_{\ell}^r(X) &= \bigcup_{m \geq 1} H_{\mathrm{\acute{e}t}}^{2r}(X, \mathbf{Q}_{\ell}(r))^{\Phi^m=1}. \end{aligned}$$

These spaces are independent of the model. Every cycle is defined over a finite field, so $W_{\ell}^r(X) \subseteq \mathcal{T}_{\ell}^r(X)$.

Lemma 8.1. *Let $X/\mathbf{F}_q$ be smooth and projective, with geometric Frobenius Φ_q , and let*

$$t \in H_{\mathrm{\acute{e}t}}^{2r}(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(r))^{\Phi_q=1}.$$

If t is algebraic over $\overline{\mathbf{F}}_q$, it lies in the image of $\mathrm{CH}^r(X)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell}$.

Proof. Choose $n \geq 1$ and $z \in \mathrm{CH}^r(X_{\mathbf{F}_{q^n}})_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell}$ with $\mathrm{cl}_{\ell}(z) = t$. For the projection $\pi_n : X_{\mathbf{F}_{q^n}} \rightarrow X$, let

$$z_0 = \frac{1}{n} (\pi_n)_* z.$$

After base change to $\overline{\mathbf{F}}_q$, compatibility with proper pushforward gives

$$\mathrm{cl}_{\ell}(z_0) = \frac{1}{n} \sum_{j=0}^{n-1} \Phi_q^j(t) = t. \quad \square$$

Proof of Theorem 1.1. Theorem 2.3 proves the Tate conjecture for $\dim A \leq 4$. Assume $\dim A = 5$, fix $\ell \neq p$, and let $X = A_{\overline{\mathbf{F}}_q}$. We first prove algebraicity over $\overline{\mathbf{F}}_q$ in codimension two: $\mathcal{T}_{\ell}^2(X) = W_{\ell}^2(X)$. An isogeny and its inverse rational Chow correspondence are defined over a finite extension and identify both W_{ℓ}^2 and $\mathcal{T}_{\ell}^2$. We may therefore replace X by a product of geometrically simple factors. Theorem 3.1(1), Lemma 2.7(1), and Lemma 2.8(1) prove the equality for ordinary, supersingular, and geometrically simple fivefolds, respectively. Theorem 5.1(1), applied over finite extensions, treats products of an abelian surface and three elliptic curves. The remaining factor dimensions are $4+1$, $3+2$, $3+1+1$, and $2+2+1$.

Let $t \in \mathcal{T}_{\ell}^2(X)$. Choose a finite extension of $\mathbf{F}_q$ defining the factors, their geometric endomorphisms, and the Chow projectors, over which Frobenius fixes t . Enlarge this field so that the Frobenius roots generate a torsion-free group. Choose the CM coefficient field L in (6.3), an embedding $L \hookrightarrow \overline{\mathbf{Q}}_{\ell}$, and a finite extension $F_{\ell}/\mathbf{Q}_{\ell}$ containing its image.

The Künneth projectors commute with Frobenius. A degree-four summand with an entry in degree zero is supported on a subproduct of dimension at most four. Theorem 2.3, followed by pullback and the Künneth projector, makes its Tate classes algebraic. For a summand supported

on every factor, (6.3) gives a decomposition into Frobenius eigenlines after extending coefficients to F_ℓ . Each nonzero coordinate of $t \otimes 1$ lies in a line $V_I(2)$ fixed by Frobenius. By Proposition 6.9, the line is generated by a product of divisor classes with F_ℓ -coefficients or belongs to an admissible orbit in

$$R = T^2(B_{\text{ao}}) \otimes h^1(B_{\text{ord}}) \otimes h^1(E_{\text{ss}})(2).$$

The divisor products belong to $W_\ell^2(X) \otimes_{\mathbf{Q}_\ell} F_\ell$.

For each residual block containing an admissible coordinate of $t \otimes 1$, Proposition 6.10 verifies the hypotheses of the Chow isomorphism in Theorem 7.8. Choose a further finite extension defining this isomorphism and its correspondences. The Frobenius relations of Proposition 6.10 persist after this extension. Corollary 7.9(ii) then makes the Tate classes of R algebraic. Every coordinate of $t \otimes 1$ therefore belongs to $W_\ell^2(X) \otimes_{\mathbf{Q}_\ell} F_\ell$, so

$$t \otimes 1 \in W_\ell^2(X) \otimes_{\mathbf{Q}_\ell} F_\ell.$$

Since $F_\ell/\mathbf{Q}_\ell$ is faithfully flat, the image of t in $H_{\text{ét}}^4(X, \mathbf{Q}_\ell(2))/W_\ell^2(X)$ is zero. Thus $t \in W_\ell^2(X)$, proving algebraicity in codimension two.

We now obtain the other codimensions and descend to $\mathbf{F}_q$. Tate's divisor theorem [Tat66, Theorem 4] proves algebraicity in codimension one. Choose an ample divisor class ϑ defined over $\mathbf{F}_q$. Hard Lefschetz gives Frobenius-equivariant isomorphisms

$$\begin{aligned} \vartheta : H^4(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(2)) &\xrightarrow{\sim} H^6(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(3)), \\ \vartheta^3 : H^2(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(1)) &\xrightarrow{\sim} H^8(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(4)). \end{aligned}$$

The inverse maps preserve Tate classes. Multiplication by ϑ and ϑ^3 therefore gives algebraic representatives in codimensions three and four. The unit treats codimension zero, and the class of a closed point of positive degree spans $H^{10}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell(5))$. Finally, Lemma 8.1 gives representatives over $\mathbf{F}_q$ for every Φ_q -invariant class, proving surjectivity in all codimensions. $\square$

9. THE HODGE STANDARD CONJECTURE OVER FINITE FIELDS

We prove numerical Hodge positivity for every polarization on an abelian variety of dimension at most five over a finite field. The essential step identifies the residual intersection form with a positive Rosati trace form.

For $d = \dim A$, write

$$Q_{\theta,r}(x, y) = (-1)^r \deg_A(xy\theta^{d-2r}) \quad (9.1)$$

for the form on $P_{\text{num}}^r(A, \theta)$.

For a polarization λ , write b_λ for the degree-one form associated with λ^{-1} ; it takes values in $\mathbf{1}(-1)$. Products use the Koszul sign convention; on an odd realization, $\text{Tr}_\ell = -\text{tr}_{\text{cat}}$. Divisor classes have symmetric representatives.

9.1. The positive scalar in the primitive correspondence. The positive self-intersection of the curve correspondence determines the sign of the induced polarization. For a polarized abelian threefold (Y, η) , let

$$P_\eta(Y) = h_{\text{prim}, \eta}^3(Y)(1), \quad b_Y(x, y) = -\deg_Y(xy).$$

The primitive Chow projector and inverse duality pairing are algebraic [Kün93, Theorems 5.1–5.2 and Corollary 5.3]. The form b_Y is alternating on the fourteen-dimensional realization of $P_\eta(Y)$.

Lemma 9.1. *Let T be a smooth projective curve and let $\Gamma \in \text{CH}^2(T \times Y)_{\mathbf{Q}}$ have class in $H^1(T) \otimes H_{\text{prim}, \eta}^3(Y)(2)$. Project Γ through π_1^T and the primitive Chow projector, and let*

$$f : h^1(T) \longrightarrow P_\eta(Y)$$

be the resulting cylinder. Let $b_T(v, w) = \deg_T(vw)$ and let $f^\dagger$ be the adjoint for b_T, b_Y . For every $\ell \neq p$,

$$\mathrm{Tr}_\ell(ff^\dagger) = \deg(\Gamma^2). \quad (9.2)$$

Proof. The projections preserve the cohomology class and consequently its self-intersection degree. Trivialize the Tate lines for the calculation and write $[\Gamma] = \sum_i a_i \otimes y_i$. Let $Q_T(a, b) = \deg_T(ab)$ and $Q_Y(y, z) = \deg_Y(yz)$. The cylinder and its adjoint are

$$f(v) = \sum_i Q_T(v, a_i) y_i, \quad f^\dagger(w) = - \sum_i Q_Y(y_i, w) a_i.$$

Indeed, these formulas give $b_Y(fv, w) = b_T(v, f^\dagger w)$. The Koszul sign in the product on $T \times Y$ is $(-1)^{3 \cdot 1}$, so

$$\deg(\Gamma^2) = - \sum_{i,j} Q_T(a_i, a_j) Q_Y(y_i, y_j).$$

The trace formula for an endomorphism of rank one gives

$$\mathrm{Tr}_\ell(ff^\dagger) = - \sum_{i,j} Q_T(a_i, a_j) Q_Y(y_i, y_j).$$

Both sides are canonical scalars, independent of the chosen trivializations. At the motivic level this equality reads $\deg(\Gamma^2) = - \mathrm{tr}_{\mathrm{cat}}(ff^\dagger)$. $\square$

Retain the Newton stratum W , its smooth locus S , and the universal threefold from Section 7. Let $K = \overline{\mathbf{F}}_p(S)$ and let $(Y, \theta)/K$ be the generic fibre. Its slopes are

$$0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1. \quad (9.3)$$

The generic primitive correspondence of Proposition 7.4 has positive self-intersection by Lemma 7.3. Thus a finite extension L/K admits a smooth projective curve T/L and a cycle $\Gamma \in \mathrm{CH}^2(T \times Y_L)_{\mathbf{Q}}$ satisfying

$$\mathrm{cl}_\ell(\Gamma) \in H^1(T, \mathbf{Q}_\ell) \otimes H_{\mathrm{prim}, \theta}^3(Y_L, \mathbf{Q}_\ell)(2), \quad (9.4)$$

$$c := \deg(\Gamma^2) > 0. \quad (9.5)$$

The cycle Γ is independent of ℓ , and these identities hold for every $\ell \neq p$.

Proposition 9.2. *After a finite extension L/K , there are an abelian sevenfold C/L and a Chow isomorphism*

$$F : h^1(C) \xrightarrow{\sim} \mathbf{P}_\theta(Y_L)$$

such that

$$F^*b_Y = \lambda b_C, \quad \lambda = \deg(\Gamma^2)/14 \in \mathbf{Q}_{>0}. \quad (9.6)$$

The form b_C is induced by an ample rational polarization on C .

Proof. We use monodromy to compute the cylinder composed with its adjoint, then recover a polarized abelian factor of the Jacobian. Choose L, T, Γ as in (9.4)–(9.5) and make the Chow projections in Lemma 9.1. A rational zero-cycle of degree one defines the curve projectors and the identification $h^1(T) = h^1(J)$, where $J = J(T)$. Let

$$H = h^1(J), \quad P = \mathbf{P}_\theta(Y_L), \quad f : H \longrightarrow P, \quad g = f^\dagger : P \longrightarrow H.$$

All these are Chow morphisms over L .

Fix a prime $\ell \neq p$. The endomorphism $(fg)_\ell$ commutes with $G_L = \mathrm{Aut}(\overline{K}/L)$. The separable part of L/K replaces G_K by a subgroup of finite index; a purely inseparable extension preserves the absolute Galois group. Passing to a subgroup of finite index preserves the connected algebraic monodromy group Sp_6 , whose representation on P_ℓ is the absolutely irreducible primitive summand

of $\wedge^3 \mathbf{Q}_\ell^6$ of dimension fourteen. Schur's lemma therefore gives $(fg)_\ell = \lambda_\ell \text{id}$. The self-intersection trace identity of Lemma 9.1 determines the scalar:

$$\lambda_\ell = \lambda = \frac{\deg(\Gamma^2)}{14} > 0. \quad (9.7)$$

Let $a = gf \in \text{End}(H)$. Equation (9.7) implies $a_\ell^2 = \lambda a_\ell$. The degree-one motive functor is fully faithful up to isogeny [Sch94, Corollary 5.10]; rational homomorphisms of abelian varieties act faithfully on H_ℓ^1 . Hence

$$a^2 = \lambda a \quad \text{in } \text{End}(H).$$

Since $a = f^\dagger f$, the Chow idempotent $e = a/\lambda$ corresponds to a Rosati self-adjoint idempotent of J .

Apply Poincaré complete reducibility and descend the resulting abelian subvariety and rational homomorphisms to a further finite extension of L . This supplies an actual abelian factor $j : C \hookrightarrow J$ with $eH \simeq h^1(C)$. Let λ_J be the canonical principal polarization of J and let

$$\lambda_C = j^\vee \lambda_J j, \quad q = \lambda_C^{-1} j^\vee \lambda_J : J \dashrightarrow C.$$

Direct calculation gives

$$qj = \text{id}_C, \quad q\lambda_J^{-1}q^\vee = \lambda_C^{-1}.$$

Thus jq is the self-adjoint projector on C . The embedding $q^* : h^1(C) \rightarrow H$ identifies its image with eH . The restriction of b_T is the form associated with the inverse of the ample polarization λ_C . If ϑ_C represents this polarization, its cup-product normalization is

$$\frac{1}{\chi(\vartheta_C)} \deg_C \left(xy \frac{\vartheta_C^{\dim C - 1}}{(\dim C - 1)!} \right), \quad \chi(\vartheta_C) = \frac{\deg_C(\vartheta_C^{\dim C})}{(\dim C)!} > 0.$$

Denote this form by b_C .

We now identify this polarized factor with P in Chow motives. Let $H' = eH$, $F = fe : H' \rightarrow P$, and $G = eg : P \rightarrow H'$. The exact identity $gf = \lambda e$ gives $GF = \lambda \text{id}_{H'}$. Thus $r = FG/\lambda$ is a Chow idempotent of P , with realization equal to the identity. The complement $1-r$ is homologically, hence numerically, trivial. The motive P is a summand of an abelian motive and is finite-dimensional. André–Kahn's nilpotence theorem [AK02, Proposition 9.1.14] makes $1-r$ nilpotent. An idempotent which is nilpotent vanishes, so

$$GF = \lambda \text{id}_{H'}, \quad FG = \lambda \text{id}_P. \quad (9.8)$$

Thus $F^{-1} = G/\lambda$ and, since $G = F^\dagger$, $b_Y(Fx, Fy) = \lambda b_C(x, y)$ in Chow motives. The equality $\dim H'_\ell = 14$ gives $\dim C = 7$. $\square$

9.2. Specialization of the polarized primitive motive.

Lemma 9.3. *There are a nonempty open $U \subset S$ and a finite surjective morphism $U' \rightarrow U$, with U' smooth and integral, over which the abelian variety, polarization, isomorphism and pairing identity of Proposition 9.2 spread. At every geometric point of U' , specialization gives an isomorphism of Chow motives satisfying (9.6).*

Proof. We spread the isomorphism together with the rational equivalences expressing its inverse and its compatibility with the pairings. Normalize S in L . The normalization is finite because S is of finite type over a field. On a smooth nonempty open of the normalization, spread C , an ample representative of its rational polarization, F , and G/λ . Shrink the open until C is an abelian scheme and its polarization is relatively ample. The Chow Lefschetz projectors are defined for abelian schemes by [Kün93, Theorems 5.1–5.2].

The inverse and pairing identities are finitely many rational equivalences on generic fibres. Each uses finitely many cycles and rational functions. Spread these data and shrink until the identities

hold in the relative Chow groups. Refined pullback to a geometric fibre preserves the correspondences, compositions and duality pairings. Remove from S the image of the excluded closed subset under the finite normalization map to obtain $U' \rightarrow U$. $\square$

Proposition 9.4. *Let $(Y, \theta)/\overline{\mathbf{F}}_p$ be a principally polarized abelian threefold with slopes (9.3). There exist a polarized abelian sevenfold (C, λ_C) and a Chow isomorphism*

$$F : h^1(C) \xrightarrow{\sim} \mathbf{P}_\theta(Y), \quad F^*b_Y = \kappa b_{\lambda_C}, \quad \kappa \in \mathbf{Q}_{>0}.$$

The data descend with (Y, θ) to a common finite field.

Proof. Add a prime-to- p level structure. The resulting point has dense polarized isogeny class in W : the class meets every central leaf by [CO11, Proposition 4.2] and contains a dense Hecke orbit in each leaf by [CO24, Theorem 9.1.1]. Choose a point $u \in U$ in this isogeny class, where U is the open in Lemma 9.3. A point above u in U' gives a polarized Chow isomorphism for Y_u .

We transport this isomorphism to Y and track the pairing scalar. Choose a polarized quasi-isogeny $\varphi : Y \dashrightarrow Y_u$ with $\varphi^*\theta_u = c_0\theta$, where $c_0 \in \mathbf{Q}_{>0}$. Clearing its denominator gives an isogeny $a = N\varphi$ satisfying

$$a^*\theta_u = N^2c_0\theta.$$

Pullback identifies the primitive Chow summands, by compatibility of the Lefschetz operators with the pulled-back polarization. On these summands the projection formula gives the exact pairing equality

$$b_Y(a^*x, a^*y) = \deg(a)b_{Y_u}(x, y). \quad (9.9)$$

An isogeny is finite locally free. Thus (9.9) also holds for inseparable isogenies, with their scheme-theoretic degree. Composing the polarized isomorphism at u with a^* proves the result, since $\deg(a) > 0$. All data have finite presentation over $\overline{\mathbf{F}}_p$ and descend together to a finite field. $\square$

9.3. Positivity of the residual intersection form. All abelian varieties in this subsection are defined over $\overline{\mathbf{F}}_p$. Let B_1 be a geometrically simple almost ordinary surface and let E be supersingular elliptic. Let

$$Y = B_1 \times E, \quad P_1 = T^2(B_1), \quad M = P_1 \otimes h^1(E)(1).$$

The geometric Néron–Severi pairing of B_1 is nondegenerate. Its inverse matrix, using symmetric divisors, gives a self-adjoint Chow projector onto the Néron–Severi motive. Its complementary projector defines P_1 .

Lemma 9.5. *Equip M with the alternating form*

$$b_M(x \otimes s, y \otimes t) = -\deg_{B_1}(xy) \deg_E(st). \quad (9.10)$$

- (i) *There exist a polarized abelian fourfold (J, λ_J) and a Chow isomorphism $F : h^1(J) \xrightarrow{\sim} M$ with $F^*b_M = \kappa b_{\lambda_J}$ for some $\kappa \in \mathbf{Q}_{>0}$.*
- (ii) *For every product polarization η on Y , the motive M is a self-adjoint summand of $\mathbf{P}_\eta(Y)$ with pairing b_M .*

Proof. We prove (ii) first, then use its self-adjoint projector to obtain (i). For a product polarization $\eta = \eta_1 + \eta_E$, the class η_1 belongs to the Néron–Severi summand of $h^2(B_1)$. The transpose of multiplication by η_1 on P_1 factors through that summand, so $L_{\eta_1}|_{P_1} = 0$ in Chow motives. Also $L_{\eta_E}|_{h^1(E)} = 0$ because $h^3(E) = 0$. Consequently

$$L_\eta M = 0. \quad (9.11)$$

The product projector defining M is self-adjoint and nondegenerate for the middle pairing. Equation (9.11) therefore makes M a self-adjoint Chow summand of $\mathbf{P}_\eta(Y)$. The product pairing on $P_1 \otimes h^1(E)$ has no additional Koszul sign, since the first factor has degree two. Its restriction is (9.10), proving (ii).

Choose an isogeny $a : Y \rightarrow Y_0$ to a principally polarized abelian variety (Y_0, θ_0) ; this exists by [Mum70, p. 234, Corollary 1]. The factors B_1 and E have no nonzero homomorphisms between them, so $a^*\theta_0$ is a product polarization. The variety Y_0 has slopes (9.3). Proposition 9.4 realizes the primitive motive of Y_0 as a positively polarized h^1 motive. Pullback preserves this positivity by (9.9), giving the same description of $P_{a^*\theta_0}(Y)$.

Transport the self-adjoint projector onto M across this polarized Chow isomorphism. Full faithfulness of h^1 identifies the transported projector with a Rosati self-adjoint idempotent on an actual abelian variety. Its image is an abelian subvariety with its restricted ample polarization, as in Proposition 9.2. This gives F and the positive scalar κ in (i). Since $\dim R_\ell M = 8$, the abelian variety J has dimension four. $\square$

Theorem 9.6. *Let B_2 be a geometrically simple ordinary surface and let $X = B_1 \times E \times B_2$. For every product polarization $\theta = \eta_1 + \eta_E + \theta_2$, let*

$$R = M \otimes h^1(B_2)(1) \subset h^4(X)(2)$$

and let V_R be its rational numerical cycle space. Then:

- (i) V_R lies in $P_{\text{num}}^2(X, \theta)$.
- (ii) *The form*

$$(\alpha, \beta) \mapsto \deg_X(\theta\alpha\beta)$$

is positive definite on V_R .

Proof. We express the square of a residual cycle as the trace of a homomorphism composed with its Rosati adjoint. Let $H = h^1(B_2)$ and $b_H(v, w) = \deg_{B_2}(vw\theta_2)$. This is a positive rational multiple of the form associated with the inverse of the polarization on H . Equation (9.11) and $L_{\theta_2}^2 H = 0$ give $L_\theta^2 R = 0$, proving primitiveness.

Let α be a rational cycle projected to R , and write its realization as $\sum_i y_i \otimes v_i$. The Y polarization kills M , so only θ_2 contributes to its square. The Koszul sign is $(-1)^{3 \cdot 1} = -1$. With $Q_Y(y, z) = \deg_Y(yz)$, this gives

$$\begin{aligned} \deg_X(\theta\alpha^2) &= - \sum_{i,j} Q_Y(y_i, y_j) b_H(v_i, v_j) \\ &= \sum_{i,j} b_M(y_i, y_j) b_H(v_i, v_j). \end{aligned} \tag{9.12}$$

Rigidity and $b_H : H \xrightarrow{\sim} H^\vee(-1)$ identify the Chow cycle $\alpha : \mathbf{1} \rightarrow M \otimes H(1)$ with a Chow morphism $u : H \rightarrow M$. Using the identification $v \mapsto b_H(-, v)$, its realization and adjoint are

$$u(w) = \sum_i b_H(w, v_i) y_i, \quad u^\dagger(z) = \sum_i b_M(y_i, z) v_i.$$

Computing the rank-one traces yields

$$\deg_X(\theta\alpha^2) = \text{Tr}_\ell(u^\dagger u). \tag{9.13}$$

Lemma 9.5 realizes M as the degree-one motive of a polarized abelian fourfold, with a positive rational multiple of its polarization form. Scholl's Corollary 5.10 [Sch94] therefore identifies u with a rational homomorphism of polarized abelian varieties in the contravariant direction. Embed this homomorphism as an off-diagonal block in the endomorphism algebra of their product. Rosati positivity [Con, Theorem 3.4] gives $\text{Tr}_\ell(u^\dagger u) > 0$ whenever $u \neq 0$; positive rational scaling of either polarization form preserves this inequality.

If the numerical class of α is nonzero, its realization is nonzero, hence $u \neq 0$. Equation (9.13) proves strict positivity on every nonzero rational numerical class. To pass to real coefficients, use the rationality of the matrix of the form. A negative real direction would admit a negative rational approximation; a degenerate positive semidefinite rational matrix would have a nonzero rational kernel vector. Strict positivity on rational vectors therefore gives positive definiteness over $\mathbf{R}$. $\square$

9.4. Reduction to the residual pairing. We decompose numerical classes into orthogonal subspaces. Products of divisors and classes supported on lower-dimensional subproducts have known positive forms; the remaining subspaces lie in the residual motive just treated.

Abbreviate $N^r(A)_{\mathbf{Q}}$ by $N^r(A)$ and $P_{\text{num}}^r(A, \theta)$ by $N^r(A)_{\text{prim}}$, and work in the category $\text{NUM}(k)_{\mathbf{Q}}$ of numerical motives. We choose a common finite field of definition for the abelian varieties and correspondences over $\overline{\mathbf{F}}_p$ used below.

9.4.1. *Polarized simple factors.*

Lemma 9.7. *Let (A, θ) be a polarized abelian variety over $\overline{\mathbf{F}}_p$. Then:*

(i) *A product of geometrically simple abelian varieties admits an isogeny*

$$f: \prod_{i=1}^t A_i \longrightarrow A$$

such that $f^\theta = \sum_i \text{pr}_i^* \theta_i$ in the rational Néron–Severi group, where each θ_i is ample.*

(ii) *For every $r \leq \dim(A)/2$, pullback identifies the primitive numerical spaces and multiplies their intersection pairing by $\deg f$.*

Proof. Choose a simple abelian subvariety $i: X \hookrightarrow A$ and let $\lambda: A \rightarrow A^\vee$ be the polarization defined by θ . The connected reduced kernel

$$Y = (\ker(i^\vee \lambda: A \longrightarrow X^\vee))_{\text{red}}^0$$

is a complementary abelian subvariety: addition $X \times Y \rightarrow A$ is an isogeny [Mil86, Proposition 12.1]. If $j: Y \hookrightarrow A$ is the inclusion, the equality $i^\vee \lambda j = 0$ makes the pulled-back polarization the product of its restrictions to X and Y . Repeating this construction on Y gives the asserted decomposition, including within each isotypic component.

Pullback by an isogeny is an isomorphism on rational numerical cycles and commutes with multiplication by the polarization. The projection formula gives

$$\deg(f^* x f^* y (f^* \theta)^{\dim A - 2r}) = (\deg f) \deg(xy \theta^{\dim A - 2r}).$$

Thus pullback identifies the primitive spaces and scales their pairings by the positive integer $\deg f$. $\square$

Lemma 9.8. *For a polarized product $\prod_i (A_i, \theta_i)$ as in Lemma 9.7, each algebra $\text{End}^0(A_i)$ contains a maximal CM field stable under the Rosati involution of θ_i . Their product defines a Rosati-stable CM structure on $\prod_i A_i$.*

Proof. Fix a factor $X = A_i$, write $D = \text{End}^0(X)$, and denote its Rosati involution by $\dagger$. Choose a finite field of definition for all geometric endomorphisms. For a geometrically simple factor of dimension at least two, the centre F of D is CM, and $\dagger$ acts on F by complex conjugation. For an ordinary elliptic curve, F is imaginary quadratic and $\dagger$ is complex conjugation. These properties follow from the classification of simple abelian varieties over finite fields; a real Frobenius centre gives a supersingular variety, whose geometrically simple factors are elliptic.

Choose a maximal $\dagger$ -stable commutative subfield $C \subset D$ containing F . Its centralizer $Z_D(C)$ is $\dagger$ -stable. Choose $i \in F^\times$ with $i^\dagger = -i$. For $y \in Z_D(C)$, the decomposition

$$y = \frac{y + y^\dagger}{2} + i \frac{y - y^\dagger}{2i}$$

expresses y using two $\dagger$ -fixed elements of $Z_D(C)$. If $y \in Z_D(C) \setminus C$, one of these elements, say x , lies outside C . The commutative algebra $C[x]$ is a field because D is a division algebra, and it is $\dagger$ -stable. Maximality of C therefore gives $Z_D(C) = C$. Thus C is a maximal subfield of D and $[C : \mathbf{Q}] = 2 \dim X$ [Anc21, Theorem 6.1].

The restriction of $\dagger$ to C is positive. The classification of commutative algebras with positive involution shows that C is totally real with the identity involution or CM with complex conjugation [Mil20, Propositions 1.37 and 1.39]. Since $F \subset C$ is CM, so is C .

For a supersingular elliptic curve, D is the definite quaternion algebra and $\dagger$ is quaternion conjugation. A nonzero trace-zero element $x \in D$ satisfies $x^2 \in \mathbf{Q}_{<0}$, so $\mathbf{Q}(x)$ is a $\dagger$ -stable imaginary quadratic maximal subfield. Applying these constructions to each factor proves the lemma. $\square$

9.4.2. *The tensor pairing.* For a polarized abelian variety A of dimension at least four, let

$$\psi_A = \langle \ , \ \rangle_{1, \text{mot}}, \quad b_A = \psi_A^{\otimes 4} \Big|_{h^4(A)}$$

be the pairings defined in [Anc21, Definition 3.6 and Section 5]. We also write b_A for the induced rational symmetric form on $N^2(A)$. Let $L^2(A) \subset N^2(A)$ be the subspace spanned by products of two divisor classes.

Proposition 9.9. *Let A be a polarized abelian variety of dimension at least four over a finite field, and choose a Rosati-stable CM structure. Then:*

- (i) *On $N^2(A)_{\text{prim}}$, the form b_A is a positive rational multiple of*

$$(x, y) \longmapsto \deg(xy\theta^{\dim A - 4}).$$

Positivity of b_A on $N^2(A)$ is equivalent to the numerical Hodge standard conjecture in codimension two.

- (ii) *The restriction of b_A to $L^2(A)$ is positive definite.*
- (iii) *The decomposition of $h^4(A)$ into rational Galois orbit motives for the CM structure is orthogonal for b_A in $\text{NUM}(k)_{\mathbf{Q}}$.*

Proof. The comparison with the primitive intersection form and positivity on products of divisors are [Anc21, Lemma 5.3 and Proposition 5.4]. The orthogonality of rational CM orbit summands is [Anc21, Proposition 6.7(2)]. $\square$

Lemma 9.10. *Let $A = \prod_{i=1}^t A_i$ be a polarized product of total dimension five. Then:*

- (i) *For a tuple $(n_1, \dots, n_t)$ with $\sum_i n_i = 4$ and at least one zero entry, the restriction of b_A to the numerical classes of $\bigotimes_i h^{n_i}(A_i)(2)$ is positive definite.*
- (ii) *The subspaces for distinct degree tuples are orthogonal.*

Proof. We prove orthogonality first and then compare each form with the corresponding form on a fourfold. The decomposition $h^1(A) = \bigoplus_i h^1(A_i)$ is orthogonal for ψ_A . Taking its fourth tensor power gives orthogonality for distinct Künneth degree tuples.

Let X be the product of the factors contributing positive degree, and write $A = X \times Y$ and $s = \dim X \leq 4$. The indicated summand is a direct summand of the pullback of $h^4(X)$, and its numerical classes identify with the corresponding numerical classes on X . The point projector on Y gives the inverse identification.

For the degree-one form $q_V(u, v) = \deg(uv\theta_V^{\dim V - 1})$, expansion of the product polarization gives

$$q_A|_{h^1(X)} = \binom{4}{5-s} \deg(\theta_Y^{5-s}) q_X.$$

The coefficient is positive, and $\psi_V = -q_V$ in degree one. Thus $\psi_A|_{h^1(X)}$ is a positive rational multiple of ψ_X .

If $s = 4$, the numerical Hodge standard conjecture for X gives positivity of b_X [Anc21, Theorem 3.18 and Proposition 5.4]. If $s < 4$, choose a polarized elliptic curve E and let $Z = X \times E^{4-s}$. Expanding the product polarization on Z compares $\psi_Z|_{h^1(X)}$ with ψ_X by a positive rational scalar. Consequently the restrictions of b_A and b_Z to $h^4(X)$ differ by a positive rational scalar. The known positivity on the fourfold Z proves positivity on the given numerical subspace. $\square$

Lemma 9.11. *Let M_O be a rational CM orbit summand of $h^4(A)$ and set*

$$V_O = \text{Hom}_{\text{NUM}(k)_{\mathbf{Q}}}(\mathbf{1}, M_O(2)).$$

Suppose $V_O \neq 0$. Then:

- (i) *Every CM coordinate of $M_O(2)$ is a geometric Tate line.*
- (ii) *If one coordinate is generated, after extending the realization coefficients, by a product of two divisor classes, then $V_O \subset L^2(A)$ and $b_A|_{V_O}$ is positive definite.*

Proof. By [Anc21, Proposition 6.8(1)], the dimension of V_O equals the rank of M_O , and its realizations are spanned by algebraic classes. This proves (i), after a finite extension defining the classes.

The CM orbit projectors preserve the algebra generated by divisors. Indeed, after splitting the CM structure, its characters on $h^4(A)$ are distinct, so the coordinate projectors are polynomials in the actions of rational endomorphisms. Pullback by these endomorphisms preserves products of divisors. The rational orbit projectors therefore preserve $L^2(A)$.

Project the divisor product in (ii) to M_O . Its realization is nonzero. Homological and numerical equivalence agree on M_O by [Anc21, Proposition 6.8(1)], so the resulting numerical Lefschetz class is nonzero. Hence $p_O L^2(A) \neq 0$ over $\mathbf{Q}$, so V_O contains a nonzero rational Lefschetz class. Ancona's Proposition 6.8(2) [Anc21] gives $V_O \subset L^2(A)$. Proposition 9.9(ii) proves positivity. $\square$

9.4.3. The finite-field theorem.

Theorem 9.12. *Let A be an abelian variety of dimension $g \leq 5$ over a finite field, and let θ be an ample rational divisor class. For every $0 \leq r \leq g/2$, the form*

$$(x, y) \mapsto (-1)^r \deg(xy\theta^{g-2r})$$

is positive definite on $N^r(A_{\overline{\mathbf{F}}_p})_{\text{prim}}$.

Proof. Multiplying θ by a positive integer n clears its denominators, preserves the primitive spaces, and multiplies the form in codimension r by the positive scalar n^{g-2r} . We may therefore assume that θ is an ample divisor class. The Hodge index theorem proves the assertion in dimensions at most three, and [Anc21, Theorem 3.18] proves it in dimension four. Thus assume $g = 5$. The polarized decomposition of Lemma 9.7 reduces the problem to a product $A = \prod_i A_i$ of geometrically simple factors, equipped with a product polarization. Each factor admits a CM field preserved by Rosati by Lemma 9.8. The tensor pairing comparison in Proposition 9.9 reduces codimension two to positivity of b_A on $N^2(A)$.

For a geometrically simple fivefold, all numerical classes are generated by products of divisors [Kos24, Remark 1.3]. Positivity then follows from Proposition 9.9(ii).

Suppose the product has at least two factors. Its possible dimension partitions are

$$4 + 1, \quad 3 + 2, \quad 3 + 1 + 1, \quad 2 + 2 + 1, \quad 2 + 1 + 1 + 1, \quad 1 + 1 + 1 + 1 + 1.$$

Every degree-four summand supported on a proper subproduct has a positive numerical pairing by Lemma 9.10. For the partition $1 + 1 + 1 + 1 + 1$, every degree-four tuple has a zero entry. For $2 + 1 + 1 + 1$, the only tuple supported on every factor is $(1, 1, 1, 1)$; its numerical space vanishes by Lemma 4.1(ii).

It remains to treat tuples with positive degree on every factor for the four partitions

$$4 + 1, \quad 3 + 2, \quad 3 + 1 + 1, \quad 2 + 2 + 1.$$

Decompose each such summand into rational CM orbit motives. Each orbit has a fixed degree tuple because Galois conjugation preserves the individual CM field factors. For an orbit O with $V_O \neq 0$, Lemma 9.11(i) makes every coordinate a geometric Tate line. Choose a common finite extension over which the numerical classes and geometric endomorphisms are defined and the group

generated by Frobenius roots is torsion-free. Proposition 6.9 then gives the following alternatives for each coordinate.

- (i) The coordinate is generated by a product of two divisor classes after extending the realization coefficients.
- (ii) The coordinate belongs to an admissible orbit in

$$R = T^2(B_{\text{ao}}) \otimes h^1(B_{\text{ord}}) \otimes h^1(E_{\text{ss}})(2),$$

where the two geometrically simple surfaces are almost ordinary and ordinary, respectively, and the elliptic curve is supersingular.

The first alternative gives a positive pairing on V_O by Lemma 9.11(ii). The second occurs only in the partition $2 + 2 + 1$.

For this residual product write L_B, L_C, L_E for multiplication by the polarizations on $B = B_{\text{ao}}$, $C = B_{\text{ord}}$, and $E = E_{\text{ss}}$. The definition of $T^2(B)$ gives $L_B T^2(B) = 0$. Dimension gives $L_E h^1(E) = 0$ and $L_C^2 h^1(C) = 0$. Hence

$$(L_B + L_C + L_E)^2 R = 0,$$

so R lies in primitive degree four for the given product polarization. Theorem 9.6 proves positivity of its primitive intersection form. Proposition 9.9(i) gives positivity of b_A on its numerical classes.

The orthogonality of CM orbit summands and degree tuples, proved in Proposition 9.9(iii) and Lemma 9.10(ii), now gives positivity on $N^2(A)$. The positivity of $\deg(\theta^5)$ in codimension zero and the Hodge index theorem in codimension one complete the proof over $\overline{\mathbf{F}}_p$.

Pullback to $\overline{\mathbf{F}}_p$ detects numerical equivalence: a geometric test cycle descends to a finite extension, and its pushforward multiplies the intersection number by the extension degree. Geometric positivity therefore implies positivity over the finite field. $\square$

10. STANDARD CONJECTURE D AND FURTHER CONSEQUENCES

Corollary 10.1. *Let $A/\mathbf{F}_q$ be an abelian variety of dimension at most five.*

- (i) *For every $0 \leq r \leq \dim A$ and $\ell \neq p$, the cycle class map*

$$\text{CH}^r(A)_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{Q}_{\ell} \xrightarrow{\sim} H_{\text{ét}}^{2r}(A_{\overline{\mathbf{F}}_q}, \mathbf{Q}_{\ell}(r))^{\text{Frob}_q=1}$$

is an isomorphism.

- (ii) *Rational and numerical equivalence agree with rational coefficients on A and $A_{\overline{\mathbf{F}}_q}$.*
- (iii) *For every $0 \leq r \leq \dim A$,*

$$-\text{ord}_{s=r} \zeta(A, s) = \dim_{\mathbf{Q}} \text{CH}^r(A)_{\mathbf{Q}}.$$

- (iv) *Quillen's algebraic K -groups satisfy*

$$K_i(A) \otimes_{\mathbf{Z}} \mathbf{Q} = 0 \quad (i > 0).$$

Statements (i)–(iv) hold after every finite extension of $\mathbf{F}_q$.

Proof. The motive $h(A)$ is of abelian type, and Frobenius acts semisimply on its ℓ -adic realization [Kah03, Lemma 1.9]. Together with the Tate conjecture in every codimension (Theorem 1.1), these properties give $A \in B_{\text{tate}}(\mathbf{F}_q)$ in [Kah03, Definition 1]. Kahn's theorem [Kah03, Theorem 1.10], with $X = A$ and $X' = \text{Spec } \mathbf{F}_q$, gives

$$\text{CH}^r(A)_{\mathbf{Q}} \xrightarrow{\sim} N^r(A)_{\mathbf{Q}} \quad (0 \leq r \leq \dim A).$$

The perfect intersection pairing on numerical cycles gives injectivity of the cycle class map after extending coefficients to $\mathbf{Q}_{\ell}$. The Tate conjecture gives surjectivity. This proves (i) and (ii) over $\mathbf{F}_q$.

Let $z \in \mathrm{CH}^r(A_{\overline{\mathbf{F}}_q})_{\mathbf{Q}}$ be numerically trivial. Choose $n \geq 1$ and a cycle z_n on $A_{\mathbf{F}_{q^n}}$ defining z . For every cycle w of complementary codimension on $A_{\mathbf{F}_{q^n}}$,

$$\deg(z_n \cdot w) = \deg(z \cdot w_{\overline{\mathbf{F}}_q}) = 0.$$

Thus z_n is numerically trivial. Kahn's Theorem 1.10 over $\mathbf{F}_{q^n}$ gives $z_n = 0$, hence $z = 0$, proving (ii).

The zeta formula and the vanishing of higher rational K -groups follow from [Kah03, Corollaries 2.1–2.2], respectively. All the hypotheses of Kahn's Theorem 1.10 and Corollaries 2.1–2.2 remain valid after finite extension. $\square$

Corollary 10.2. *Let A be an abelian variety of dimension at most five over a finite field or its algebraic closure. Numerical and homological equivalence agree on $\mathrm{CH}^r(A)_{\mathbf{Q}}$ for every r and every Weil cohomology.*

Proof. Corollary 10.1(ii), applied to A or a model of A over a finite field, identifies rational and numerical equivalence on A . For every Weil cohomology H , the equivalence relations satisfy

$$\sim_{\mathrm{rat}} \subseteq \sim_H \subseteq \sim_{\mathrm{num}}.$$

Thus $\sim_H = \sim_{\mathrm{num}}$. $\square$

Proof of Theorem 1.2(ii). Theorem 9.12 proves numerical Hodge positivity for every polarization. Corollary 10.2 identifies the numerical and homological primitive spaces for every Weil cohomology. Thus I and D hold for every Weil cohomology. The algebraic Künneth and Lefschetz projectors for abelian varieties give A , B , and C [Anc21, Theorem 4.1(1),(4)]. $\square$

11. SPECIALIZATION OF THE STANDARD CONJECTURES

Proof of Theorem 1.2(i). Assume first that k is algebraically closed. We prove positivity on homological primitive classes by specialization, then deduce D_ℓ and numerical positivity. Let $d = \dim A$, fix $\ell \neq p$, and write

$$A_\ell^r(A) = \mathrm{im}(\mathrm{CH}^r(A)_{\mathbf{Q}} \longrightarrow H_{\mathrm{et}}^{2r}(A, \mathbf{Q}_\ell(r))).$$

Take a nonzero cohomologically primitive class $x \in A_\ell^r(A)$, where $0 \leq 2r \leq d$, and choose a rational cycle representing x . Spread the variety, polarization, and cycle over a smooth integral scheme S_0 of finite type over $\mathbf{F}_p$. Shrink S_0 so that the resulting morphism $f : \mathcal{A} \rightarrow S_0$ is an abelian scheme, the polarization is relatively ample, and the cycle components are flat. The cycle class is a section of $R^{2r}f_*\mathbf{Q}_\ell(r)$. A section of a lisse sheaf on a connected scheme is determined by its value at any geometric point. The cycle class is therefore nonzero on every fibre. The section $\theta^{d-2r+1}x$ vanishes at the geometric generic point and hence on every fibre.

Choose a closed point $s \in S_0$. Corollary 10.2 identifies x_s with a nonzero primitive numerical class. By Theorem 9.12,

$$(-1)^r \deg(x_s^2 \theta_s^{d-2r}) > 0.$$

Smooth proper base change preserves cup products and trace, so this intersection number equals $(-1)^r \deg(x^2 \theta^{d-2r})$. The homological primitive form is therefore positive on every nonzero rational class.

To prove D_ℓ , let $z \in \mathrm{CH}^r(A)_{\mathbf{Q}}$ be numerically trivial. The algebraic Lefschetz correspondences give a decomposition

$$\mathrm{cl}_\ell(z) = \sum_j L^j \mathrm{cl}_\ell(z_j),$$

where $z_j \in \mathrm{CH}^{r-j}(A)_{\mathbf{Q}}$ represents the primitive generator obtained by applying the inverse Lefschetz operator to the corresponding component [Kün93, Theorems 5.1–5.2]. Each z_j is numerically trivial because algebraic correspondences preserve numerical triviality. Numerical triviality gives

$$(-1)^{r-j} \deg(z_j^2 \theta^{d-2(r-j)}) = 0.$$

The positivity on homological primitive classes therefore forces $\text{cl}_\ell(z_j) = 0$ for every j , hence $\text{cl}_\ell(z) = 0$. Homological triviality implies numerical triviality by the projection formula. This proves D_ℓ .

By D_ℓ , the numerical and homological primitive spaces coincide. They are finite-dimensional over $\mathbf{Q}$, and the forms have rational matrices. A negative real direction has a negative rational approximation; a degenerate positive semidefinite rational matrix has a nonzero rational kernel vector. Strict positivity on rational vectors therefore proves positive definiteness over $\mathbf{R}$. This gives numerical I and I_ℓ . The algebraic Künneth and Lefschetz projectors give the remaining standard conjectures [Anc21, Theorem 4.1(1),(4)].

For a general field k , apply the result over an algebraic closure. Pullback detects numerical equivalence with rational coefficients: a geometric test cycle descends to a finite extension, and pushforward gives a test cycle over k whose intersection number is multiplied by the positive extension degree. Geometric ℓ -adic cohomology is preserved under this base change. Thus D_ℓ and positivity of both primitive forms descend to k . The choice of $\ell \neq p$ was arbitrary. $\square$

Corollary 11.1. *Let A be an abelian variety of dimension at most five over a field of characteristic $p > 0$. For every r , the kernel of*

$$\text{cl}_\ell : \text{CH}^r(A)_{\mathbf{Q}} \longrightarrow H_{\text{ét}}^{2r}(A_{\bar{k}}, \mathbf{Q}_\ell(r))$$

is the group of numerically trivial rational cycles, independently of $\ell \neq p$.

Proof. Apply Theorem 1.2(i) to each $\ell \neq p$. $\square$

12. POWERS AND PRODUCTS WITH CURVES

12.1. Mixed powers. Fix the residual data (B_1, B_2, E) of Proposition 6.10 over a finite field k_0 . For nonnegative integers a, b, c , let

$$X_{a,b,c} = B_1^a \times B_2^b \times E^c.$$

Thus $\dim X_{a,b,c} = 2a + 2b + c$.

Corollary 12.1. *Every $X_{a,b,c}$ satisfies the Tate conjecture in every codimension, for every $\ell \neq p$, over every finite extension of k_0 .*

Proof. Choose a finite extension k/k_0 over which (7.5) is defined, and let $Y = B_1 \times E$. The exceptional threefolds in Zarhin's theorem [Zar15, Theorem 1.1] are geometrically simple. Since Y is a product, the theorem shows that, after enlarging k , Y is neat: the Tate classes on every Y^m are generated by divisors over k and every finite extension.

To pass from powers of Y to $X_{a,b,c}$, let $M = T^2(B_1) \otimes h^1(E)(1)$ and $H = h^1(B_2)$. The isomorphism $M \simeq H^{\oplus 2}$ supplies Chow morphisms $i : H \rightarrow M$ and $p : M \rightarrow H$ with $pi = \text{id}_H$. Since M is a summand of $h^3(Y)(1)$, the motive $h^1(X_{a,b,c})$ is a direct summand of

$$N = h^1(Y)^{\oplus(a+c)} \oplus h^3(Y)(1)^{\oplus b}.$$

Writing $X = X_{a,b,c}$, the Chow decomposition of an abelian motive gives

$$h^j(X) = \text{Sym}^j h^1(X)$$

[Anc21, Theorem 4.1(2)]. The symmetrizer for the usual symmetry of Chow motives realizes the alternating projector on odd cohomology. Hence $h^j(X)$ is a direct summand of $N^{\otimes j}$.

Fix $r > 0$. Expanding $N^{\otimes 2r}(r)$ and applying the Künneth projectors gives split Chow morphisms

$$h^{2r}(X_{a,b,c})(r) \xrightarrow{I} D_r \xrightarrow{P} h^{2r}(X_{a,b,c})(r), \quad PI = \text{id}, \quad (12.1)$$

where

$$D_r = \bigoplus_{t=0}^{2r} h^{2r+2t}(Y^{2r})(r+t)^{\oplus d_{r,t}}$$

for suitable nonnegative integers $d_{r,t}$. The integer t counts the factors $h^3(Y)(1)$ in a tensor summand.

For a finite extension k'/k and a Frobenius invariant class ξ in the source of I , the Tate conjecture for Y_k^{2r} makes $I\xi$ algebraic. Applying the projectors defining D_r and then P produces a cycle with class ξ . Codimension zero is immediate. For any finite extension k_1/k_0 , apply this conclusion over k_1k and descend by Lemma 8.1. $\square$

Corollary 12.2. *Let $X = X_{a,b,c}$ over a finite extension k/k_0 .*

- (i) *The cycle class map $\mathrm{CH}^r(X)_{\mathbf{Q}} \otimes \mathbf{Q}_{\ell} \rightarrow H^{2r}(X_{\bar{k}}, \mathbf{Q}_{\ell}(r))^{\mathrm{Gal}(\bar{k}/k)}$ is an isomorphism for every r and $\ell \neq p$.*
- (ii) *Rational and numerical equivalence agree on X and $X_{\bar{k}}$.*
- (iii) *Standard conjecture D holds for every Weil cohomology on X and $X_{\bar{k}}$.*
- (iv) *For every r ,*

$$-\mathrm{ord}_{s=r} \zeta(X, s) = \dim_{\mathbf{Q}} \mathrm{CH}^r(X)_{\mathbf{Q}}.$$

- (v) *$K_i(X) \otimes_{\mathbf{Z}} \mathbf{Q} = 0$ for every $i > 0$.*

Proof. Corollary 12.1 and semisimplicity of Frobenius show that $X \in B_{\mathrm{tate}}(k)$. Apply [Kah03, Theorem 1.10 and Corollaries 2.1–2.2]. Descent of a numerically trivial cycle to a finite field, as in the proof of Corollary 10.1(ii), gives equality of rational and numerical equivalence over $\bar{k}$. The inclusions $\sim_{\mathrm{rat}} \subseteq \sim_H \subseteq \sim_{\mathrm{num}}$ give (iii) for every Weil cohomology H . $\square$

Remark 12.3. The residual family contains explicit sixfolds with Tate classes beyond the algebra generated by divisors. Over $\mathbf{F}_4$, take surfaces with Weil polynomials

$$f_1(t) = t^4 - t^3 + 4t^2 - 4t + 16,$$

$$f_2(t) = t^4 + 2t^3 + t^2 + 8t + 16.$$

Their real Weil polynomials are $u^2 - u - 4$ and $u^2 + 2u - 7$. Both have two roots in $(-4, 4)$, so f_1, f_2 are Weil polynomials. The quadratic discriminants $u^2 - 16$ have norms 128 and 17. Each norm is a nonsquare in the corresponding real field, $\mathbf{Q}(\sqrt{17})$ or $\mathbf{Q}(\sqrt{2})$. Hence the discriminants are nonsquares, f_1, f_2 are irreducible, and their Frobenius fields are primitive quartic CM fields with dihedral normal closures. The Newton slopes are respectively $(0, \frac{1}{2}, \frac{1}{2}, 1)$ and $(0, 0, 1, 1)$. For f_1 , the Frobenius field has local degrees 1, 2, 1 at 2: at the unit root u of $u^2 - u - 4$, the element $u^2 - 16$ is a square in $\mathbf{Q}_2$, while its norm is 128 and its valuation at the other prime is seven. Thus its middle slope has local degree two. Honda–Tate theory [Tat71, Theorem 1(i),(ii)] gives surfaces B_1, B_2 with these polynomials. Geometric simplicity follows from the almost ordinary Newton polygon for B_1 and the primitive ordinary CM field for B_2 .

Write the roots of f_1 as $a, 4/a, b, 4/b$. Their four cross products have polynomial

$$G(t) = t^4 + 4t^3 + 4t^2 + 64t + 256 = 16f_2(t/2).$$

Choose $E/\mathbf{F}_4$ with Frobenius 2. For each cross product z , the root $\beta = 8/z$ of f_2 satisfies $z\beta \cdot 2 = 16$. These roots give the residual data of Proposition 6.10.

Let $\delta_1 = ab/2$ and $\delta_2 = 2a/b$, so that $\delta_1\delta_2 = a^2$. After splitting Frobenius over the coefficient field, the roots $a, a, 4/\delta_1, 4/\delta_2$ on $B_1^2 \times B_2$ give a Tate line in degree four, since their product is 16. Every divisor class pairs reciprocal Frobenius roots. The displayed four roots contain no such pair: B_1 and B_2 are geometrically nonisogenous, and $a^2/4$ is of infinite multiplicative order. Hence this Tate line has zero intersection with the algebra generated by divisor classes. Corollary 12.1 shows that this line is contained in the cycle class image after extension of coefficients.

12.2. Polarizations on mixed powers. For a polarized abelian variety (A, λ) of dimension $g > 0$, choose an ample rational class θ representing λ and set

$$b_\lambda(x, y) = \frac{1}{\chi(\theta)} \deg_A \left(xy \frac{\theta^{g-1}}{(g-1)!} \right), \quad \chi(\theta) = \frac{\deg_A(\theta^g)}{g!}. \quad (12.2)$$

This is the form on $h^1(A)$ associated with λ^{-1} . In the convention for motivic polarizations in [Anc21, Definition 3.6], the positive form on $h^1(A)$ is $-b_\lambda$. On the primitive motive $h_{\text{prim}}^3(Y)(1)$, the corresponding form is the cup product. These signs agree with $-b_M$ in (9.10).

Lemma 12.4. *Let $\mathcal{C}$ be a full tensor subcategory of numerical abelian motives, with commutativity modified by cohomological degree and a polarization Π . Suppose $h^1(A) \in \mathcal{C}$ and $-b_{\lambda_0} \in \Pi(h^1(A))$ for one ample rational polarization λ_0 . Then $-b_\lambda \in \Pi(h^1(A))$ for every ample rational polarization λ on A .*

Proof. Let $a = \lambda_0^{-1}\lambda \in \text{End}^0(A)$ and let $\dagger$ be the Rosati involution of λ_0 . We use the same symbol for the induced adjoint on $h^1(A)$. Then $a^\dagger = a$. The form $(x, y) \mapsto \text{Tr}_\ell(xy^\dagger)$ on $\text{End}^0(A) \otimes \mathbf{R}$ is positive definite, where Tr_ℓ is the ordinary trace on $H_\ell^1(A)$, extended from its rational values to $\mathbf{R}$ [Mil86, Theorem 17.3]. Left multiplication by a is self-adjoint for this form, by cyclicity of the trace. Consequently a is semisimple with real eigenvalues.

Let θ_0, θ represent λ_0, λ . The degree formula and Riemann–Roch [Mil86, Proposition 12.9 and Theorem 13.3(a),(b)] give the polynomial identity

$$\det(t + a^* \mid H^1(A, \mathbf{Q}_\ell)) = \frac{\deg(t\lambda_0 + \lambda)}{\deg \lambda_0} = \left(\frac{\deg_A((t\theta_0 + \theta)^g)}{\deg_A(\theta_0^g)} \right)^2. \quad (12.3)$$

All mixed intersection numbers of θ_0 and θ are strictly positive. The right side is therefore nonzero for every real $t \geq 0$. Faithfulness of H^1 shows that all eigenvalues of a are strictly positive.

For $s = (a^{-1})^*$, one has $b_\lambda(x, y) = b_{\lambda_0}(sx, y)$. Polynomial interpolation on the finite positive spectrum of s gives $u \in \mathbf{R}[s]$ with $u^\dagger = u$ and $u^2 = s$. Hence

$$-b_\lambda(x, y) = -b_{\lambda_0}(ux, uy).$$

A polarization on a rational Tate category is defined after extension to $\mathbf{R}$ [Mil02, Section 1]. Thus $-b_\lambda \in \Pi(h^1(A))$. $\square$

Corollary 12.5. *Every $X_{a,b,c}$, over a finite field or its algebraic closure, satisfies all standard conjectures for every Weil cohomology and every ample rational polarization.*

Proof. The result is immediate for $X_{0,0,0} = \text{Spec } k_0$. Assume $\dim X_{a,b,c} > 0$ and work over $\bar{k}_0$. We first construct a polarization on the numerical motives generated by $Y = B_1 \times E$. Let

$$\mathcal{C}_Y = \langle h(Y), \mathbf{1}(1) \rangle_{\text{num}}$$

be the full tensor subcategory of numerical motives generated by $h(Y)$ and $\mathbf{1}(1)$. Equip it with the commutativity constraint modified by cohomological degree, as in [Mil02, Section 3]. Let $\mathcal{S}$ consist of finite disjoint unions of varieties

$$Y^m \times \prod_{j=1}^s \mathbf{P}^{n_j}, \quad m, s, n_j \geq 0.$$

The class $\mathcal{S}$ contains all projective spaces and is closed under products and connected components. It therefore satisfies [Mil02, condition (0.1)].

Corollary 12.2(iii), with $b = 0$, gives equality of numerical and ℓ -adic homological equivalence on every Y^m . The projective bundle formula extends this equality to every member of $\mathcal{S}$. Thus ℓ -adic cohomology is good on $\mathcal{S}$ in the sense of [Mil02, Section 3].

Every numerical class on Y^m is generated by divisors, by [Zar15, Theorem 1.1] and Corollary 12.2(ii). Its Hodge positivity follows from [Anc21, Proposition 5.4]. On a product with

projective spaces, every ample rational divisor is the sum of an ample class on Y^m and positive multiples of the hyperplane classes. The projective bundle formula and the tensor product of the polarized Lefschetz decompositions therefore give Hodge positivity on $\mathcal{S}$. Milne's Proposition 3.2 [Mil02] now supplies a polarization Π on $\text{NUM}(\bar{k}_0; \mathcal{S})_{\mathbf{Q}} = \mathcal{C}_Y$ for which the geometric forms are positive.

We next show that Π contains the geometric form of a polarization on B_2 . Let $M = T^2(B_1) \otimes h^1(E)(1)$. Its self-adjoint primitive projector makes $-b_M$ positive for Π . Lemma 9.5 supplies a polarized abelian fourfold (J, λ_J) and a Chow isomorphism $F : h^1(J) \rightarrow M$ with

$$F^*b_M = \kappa b_{\lambda_J}, \quad \kappa \in \mathbf{Q}_{>0}. \quad (12.4)$$

Take the split inclusion $i : h^1(B_2) \rightarrow M$ from (7.5). Scholl's Corollary 5.10 [Sch94] writes $F^{-1}i = q^*$ for a surjective rational homomorphism $q : J \dashrightarrow B_2$. The rational homomorphism

$$\nu = q\lambda_J^{-1}q^\vee : B_2^\vee \dashrightarrow B_2$$

is an ample rational polarization of $B_2^\vee$: it is the pullback of λ_J^{-1} along $q^\vee$. A positive integral multiple of $q^\vee$ is finite onto its image; dividing its pullback by the square of that integer gives the ample rational form ν . Thus $\lambda_{2,0} = \nu^{-1}$ is an ample rational polarization of B_2 , and

$$i^*b_M = \kappa b_{\lambda_{2,0}}.$$

In particular, the restricted form is nondegenerate and $-b_{\lambda_{2,0}}$ is positive for Π .

The factors $h^1(B_1)$ and $h^1(E)$ inherit positive geometric forms from $h^1(Y)$. A product polarization $\lambda_{X,0}$ on $X = X_{a,b,c}$ therefore makes $-b_{\lambda_{X,0}}$ positive for Π . Lemma 12.4 then makes $-b_{\lambda_\theta}$ positive for every ample rational class θ on X .

Let $d = \dim X$ and let r satisfy $1 \leq 2r \leq d$. Tensor powers, subobjects, and Tate twists give a positive form on the primitive summand of $h^{2r}(X)(r)$. By [Anc21, Lemma 5.3], its restriction to primitive numerical classes is a positive rational multiple of

$$(x, y) \longmapsto (-1)^r \deg_X(xy\theta^{d-2r}).$$

Fullness of $\mathcal{C}_Y$ identifies its morphisms from $\mathbf{1}$ with all numerical classes on this summand. This proves numerical Hodge positivity; codimension zero follows from $\deg_X(\theta^d) > 0$. Corollary 12.2(iii) gives standard conjecture D for every Weil cohomology. The algebraic Lefschetz and Künneth decompositions give A, B, and C [Anc21, Theorem 4.1(1),(4)]. Over a finite field, base change embeds the numerical cycle groups into their geometric counterparts and preserves the intersection forms. $\square$

12.3. Products of threefolds with arbitrary curves. Let k be finite and let Y/k be an abelian threefold with slopes

$$0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1.$$

Proposition 12.6. *There is an abelian variety J_3/k of dimension ten and an isomorphism of rational Chow motives over k*

$$h^3(Y)(1) \simeq h^1(J_3).$$

Proof. Let $M = h^3(Y)(1)$. Theorem 1.3 gives, over a finite extension L/k , an abelian variety J/L and a Chow morphism $s : h^1(J) \rightarrow M_L$ with surjective realization. To construct a surjective correspondence over k , let $A = \text{Res}_{L/k} J$. For $G = \text{Gal}(L/k)$, one has $A_L = \prod_{\sigma \in G} {}^\sigma J$. Let $i_\sigma : {}^\sigma J \rightarrow A_L$ denote the factor inclusions and let $i = i_1$. Corestrict the correspondence $s \circ i^*$ to a morphism $s_0 : h^1(A) \rightarrow M$. After extension to L , this morphism is

$$(s_0)_L = \sum_{\sigma \in G} {}^\sigma s \circ i_\sigma^*.$$

Its restriction to the identity summand of $h^1(A_L) = \bigoplus_{\sigma} h^1({}^\sigma J)$ equals s . Thus its realization is surjective. The motive M is finite-dimensional, has odd realization, and has the alternating Poincaré

self-duality $M \simeq M^\vee(-1)$. Lemma 7.7 over k gives $M \simeq h^1(J_3)$, with $2 \dim J_3 = \dim H^3(Y) = \binom{6}{3} = 20$. $\square$

Corollary 12.7. *Let C/k be a smooth projective geometrically connected curve. For every finite extension k'/k , every r , and every $\ell \neq p$, the cycle class map*

$$\mathrm{CH}^r((C \times Y)_{k'})_{\mathbf{Q}} \otimes \mathbf{Q}_\ell \xrightarrow{\sim} H^{2r}((C \times Y)_{\bar{k}}, \mathbf{Q}_\ell(r))^{\mathrm{Gal}(\bar{k}/k')}$$

is an isomorphism.

Proof. Choose a rational zero-cycle of degree one on C . The associated Chow projectors give the Künneth decomposition

$$\begin{aligned} H^{2r}(C \times Y)(r) &= H^{2r}(Y)(r) \oplus H^{2r-2}(Y)(r-1) \\ &\quad \oplus (H^1(C) \otimes H^{2r-1}(Y))(r). \end{aligned}$$

Terms outside the cohomological range are zero. The first two summands satisfy the Tate conjecture by Theorem 2.3. For the third, let $A_1 = Y$, $A_2 = J_3$, and $A_3 = Y$. Proposition 12.6 and algebraic hard Lefschetz give Chow isomorphisms

$$h^{2r-1}(Y)(r-1) \simeq h^1(A_r), \quad r = 1, 2, 3.$$

Tate's theorem on homomorphisms [Tat66, §1, Main Theorem] represents every invariant class in $H^1(C) \otimes H^1(A_r)(1)$ by a divisor correspondence. Composing these correspondences with the displayed Chow isomorphisms proves surjectivity in every codimension. The motive of $C \times Y$ is of abelian type and has semisimple Frobenius. Kahn's theorem [Kah03, Theorem 1.10] therefore gives injectivity. Each step applies over every finite extension. $\square$

Kahn's Theorem 1.10 and Corollaries 2.1–2.2 [Kah03] also give equality of rational and numerical equivalence, standard conjecture D, the zeta formula, and the vanishing of higher rational K -groups for $C \times Y$, as stated in Corollary 12.2(ii)–(v).

12.4. Constant threefolds over function fields. Write $k = \mathbf{F}_q$ and let $K = k(C)$. Let $\mathrm{CH}^2(Y_K)_{\mathbf{Q}, \mathrm{hom}}$ denote the kernel of the cycle class map to $H^4(Y_{\bar{K}}, \mathbf{Q}_\ell(2))$. For the constant representation $V = H^3(Y_{\bar{k}}, \mathbf{Q}_\ell)$, write $L(V, s)$ for its Hasse–Weil L -function over K .

Corollary 12.8. *With J_3/k as in Proposition 12.6,*

$$\dim_{\mathbf{Q}} \mathrm{CH}^2(Y_K)_{\mathbf{Q}, \mathrm{hom}} = \mathrm{rank} J_3(K) = \mathrm{ord}_{s=2} L(H^3(Y_{\bar{k}}, \mathbf{Q}_\ell), s).$$

In particular, Y_K satisfies the Beilinson–Bloch conjecture in codimension two.

Proof. We first relate the cycle group to the order of $L(V, s)$ by Broe's Proposition 4.20 [Bro26a], with $X = Y$ and $i = 2$. The motive of Y is of abelian type. Corollary 12.7 gives the Tate conjecture in codimension two on $Y \times C$. Choose a finite extension N/k with $C(N) \neq \emptyset$; Tate's divisor theorem [Tat66, Theorem 4] applies to Y_N . Broe's Proposition 4.20 therefore gives

$$\dim_{\mathbf{Q}} \mathrm{CH}^2(Y_K)_{\mathbf{Q}, \mathrm{hom}} = \mathrm{ord}_{s=2} L(V, s). \quad (12.5)$$

To identify this order with the Mordell–Weil rank, use the trace formula for the constant representation V :

$$L(V, s) = \frac{\det(1 - q^{-s} \mathrm{Frob}_q \mid H^1(C_{\bar{k}}, \mathbf{Q}_\ell) \otimes V)}{\det(1 - q^{-s} \mathrm{Frob}_q \mid V) \det(1 - q^{1-s} \mathrm{Frob}_q \mid V)}.$$

At $s = 2$ the two denominator factors are nonzero, by their weights. Semisimplicity of Frobenius and Proposition 12.6 identify the order of the numerator with

$$\dim_{\mathbf{Q}_\ell} (H^1(C_{\bar{k}}, \mathbf{Q}_\ell) \otimes H^1((J_3)_{\bar{k}}, \mathbf{Q}_\ell)(1))^{\mathrm{Frob}_q=1} = \mathrm{rank} \mathrm{Hom}_k(J_C, J_3),$$

where J_C is the Jacobian of C . The equality follows from Tate's theorem on homomorphisms [Tat66, §1, Main Theorem]. Every K -point of the constant abelian variety J_3 extends to a morphism

$C \rightarrow J_3$. Using a rational zero-cycle of degree one on C , the Albanese property of C identifies these morphisms, up to translations and after tensoring with $\mathbf{Q}$, with $\mathrm{Hom}_k(J_C, J_3)_{\mathbf{Q}}$. Since $J_3(k)$ is finite, their rank is $\mathrm{rank} J_3(K)$. Combining this with (12.5) proves the formula. $\square$

ACKNOWLEDGEMENTS

The author used OpenAI Codex for literature searches, exploration of proof strategies, reference verification, proof auditing, and LaTeX formatting.

REFERENCES

- [Agu25] Thomas Agugliaro, *Standard conjecture of Hodge type for powers of abelian varieties*, 2025, Preprint, arXiv:2510.21562v1, version of 24 October 2025.
- [AK02] Yves André and Bruno Kahn, *Nilpotence, radicaux et structures monoïdales*, Rend. Sem. Mat. Univ. Padova **108** (2002), 107–291, With an appendix by Peter O’Sullivan.
- [Anc21] Giuseppe Ancona, *Standard conjectures for abelian fourfolds*, Invent. Math. **223** (2021), no. 1, 149–212, Theorem numbering follows arXiv:1806.03216v3.
- [BLR90] Siegfried Bosch, Werner Lütkebohmert, and Michel Raynaud, *Néron models*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 21, Springer-Verlag, Berlin, 1990.
- [Bro26a] Matt Broe, *On the Beilinson–Bloch conjecture over function fields*, 2026, Preprint, arXiv:2505.00696v2, version of 27 January 2026.
- [Bro26b] ———, *The Tate conjecture for abelian fourfolds over finite fields*, 2026, Preprint, arXiv:2608.28651v1.
- [Cha05] Ching-Li Chai, *Monodromy of Hecke-invariant subvarieties*, Pure Appl. Math. Q. **1** (2005), no. 2, 291–303.
- [CO11] Ching-Li Chai and Frans Oort, *Monodromy and irreducibility of leaves*, Ann. of Math. (2) **173** (2011), no. 3, 1359–1396.
- [CO24] ———, *Hecke orbits*, 2024, Book manuscript, version of 25 February 2024.
- [Con] Brian Conrad, *Polarizations*, Lecture notes, Theorem 3.4.
- [Del69] Pierre Deligne, *Variétés abéliennes ordinaires sur un corps fini*, Invent. Math. **8** (1969), no. 3, 238–243.
- [Dem72] Michel Demazure, *Lectures on p -divisible groups*, Lecture Notes in Mathematics, vol. 302, Springer-Verlag, Berlin–Heidelberg, 1972.
- [DKZB24] Taylor Dupuy, Kiran S. Kedlaya, and David Zureick-Brown, *Angle ranks of abelian varieties*, Math. Ann. **389** (2024), no. 1, 169–185.
- [DvH25] Marco D’Addezio and Pol van Hoften, *Hecke orbits on Shimura varieties of Hodge type*, 2025, Preprint, arXiv:2205.10344v2, version of 14 March 2025.
- [GP05] Darren Glass and Rachel Pries, *Hyperelliptic curves with prescribed p -torsion*, Manuscripta Math. **117** (2005), no. 3, 299–317, Theorem 1 and Proposition 2 refer to the published version; see also the erratum.
- [GP10] ———, *Erratum to: Hyperelliptic curves with prescribed p -torsion*, Manuscripta Math. **133** (2010), no. 3–4, 545–546.
- [Kah03] Bruno Kahn, *Équivalences rationnelle et numérique sur certaines variétés de type abélien sur un corps fini*, Ann. Sci. École Norm. Sup. (4) **36** (2003), no. 6, 977–1002, Published version.
- [Kat79] Nicholas M. Katz, *Slope filtration of F -crystals*, Journées de Géométrie Algébrique de Rennes (Juillet 1978), I, Astérisque, no. 63, Société Mathématique de France, Paris, 1979, pp. 113–163.
- [Kos24] Teruhisa Koshikawa, *The numerical Hodge standard conjecture for the square of a simple abelian variety of prime dimension*, Manuscripta Math. **173** (2024), no. 3–4, 1161–1169.
- [Kün93] Klaus Künnemann, *A Lefschetz decomposition for Chow motives of abelian schemes*, Invent. Math. **113** (1993), no. 1, 85–102.
- [Mar25] Eyal Markman, *Cycles on abelian $2n$ -folds of Weil type from secant sheaves on abelian n -folds*, 2025, Preprint, arXiv:2502.03415v2, version of 8 June 2025.
- [Mar26] ———, *Secant sheaves and Weil classes on abelian varieties*, Proceedings of the International Congress of Mathematicians 2026. Vol. 3: Invited Lectures (Sections 1–4) (Susan Friedlander and Yuri Tschinkel, eds.), Society for Industrial and Applied Mathematics, Philadelphia, PA, 2026, pp. 586–602.
- [Mil86] James S. Milne, *Abelian varieties*, Arithmetic Geometry (Gary Cornell and Joseph H. Silverman, eds.), Springer, New York, 1986, Corrected author text, 2 January 2022, pp. 103–150.
- [Mil02] James S. Milne, *Polarizations and Grothendieck’s standard conjectures*, Ann. of Math. (2) **155** (2002), no. 2, 599–610.
- [Mil20] James S. Milne, *Complex multiplication*, 2020, Course notes, version 0.10, July 14, 2020.
- [Mil22] James S. Milne, *The Tate and standard conjectures for certain abelian varieties*, 2022, Preprint, arXiv:2112.12815v2.
- [Mum70] David Mumford, *Abelian varieties*, Tata Institute of Fundamental Research Studies in Mathematics, vol. 5, Oxford University Press, London, 1970.
- [MZ99] B. J. J. Moonen and Yuri G. Zarhin, *Hodge classes on abelian varieties of low dimension*, Math. Ann. **315** (1999), no. 4, 711–733.
- [Oor74] Frans Oort, *Subvarieties of moduli spaces*, Invent. Math. **24** (1974), no. 2, 95–119.

- [OU73] Frans Oort and Kenji Ueno, *Principally polarized abelian varieties of dimension two or three are Jacobian varieties*, J. Fac. Sci. Univ. Tokyo Sect. IA Math. **20** (1973), no. 3, 377–381, Theorem numbering follows the authors’ manuscript.
- [PZ12] Rachel Pries and Hui June Zhu, *The p -rank stratification of Artin–Schreier curves*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 2, 707–726.
- [RS22] Damian Rössler and Stefan Schröer, *Moret–Bailly families and non-liftable schemes*, Algebr. Geom. **9** (2022), no. 1, 93–121.
- [Sch94] Anthony J. Scholl, *Classical motives*, Motives (Seattle, WA, 1991) (Uwe Jannsen, Steven Kleiman, and Jean-Pierre Serre, eds.), Proc. Sympos. Pure Math., vol. 55, Part 1, Amer. Math. Soc., Providence, RI, 1994, pp. 163–187.
- [Tan84] S. G. Tankeev, *On cycles on abelian varieties of prime dimension over finite or number fields*, Math. USSR-Izv. **22** (1984), no. 2, 329–337.
- [Tat66] John Tate, *Endomorphisms of abelian varieties over finite fields*, Invent. Math. **2** (1966), no. 2, 134–144.
- [Tat71] John Tate, *Classes d’isogénie des variétés abéliennes sur un corps fini (d’après T. Honda)*, Séminaire Bourbaki: Vol. 1968/69, Exposés 347–363 (N. Bourbaki, ed.), Lecture Notes in Mathematics, vol. 179, Springer-Verlag, Berlin and Heidelberg, 1971, Exposé no. 352, pp. 95–110.
- [vdGM11] Gerard van der Geer and Ben Moonen, *Abelian varieties*, 2011, Preliminary book manuscript, Chapter XI: Polarizations and Weil pairings, version of 15 September 2011.
- [Wat69] William C. Waterhouse, *Abelian varieties over finite fields*, Ann. Sci. École Norm. Sup. (4) **2** (1969), no. 4, 521–560.
- [Zar15] Yuri G. Zarhin, *Eigenvalues of Frobenius endomorphisms of abelian varieties of low dimension*, J. Pure Appl. Algebra **219** (2015), no. 6, 2076–2098, Also available as arXiv:1312.0377.
- [Zha10] Shou-Wu Zhang, *Gross–Schoen cycles and dualising sheaves*, Invent. Math. **179** (2010), no. 1, 1–73.